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Hamilton-Principle Models of Oral Biofilm Dysbiosis

GPU-Accelerated Inference, Metabolic Priors, and Spatial Stratification

Masterarbeit

im Studiengang Maschinenbau (Double Degree, LUH/Keio)

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Abstract

A point model of a multi-species biofilm describes how the composition at one material point evolves in time, but has no space. A continuum growth model describes where a biofilm grows and what stresses it carries, but does not distinguish species. This thesis brings the point model into an existing finite element framework, the biofilm element of the Institut für Kontinuumsmechanik in ANSYS, so that composition, growth and stress can be computed together. Both models follow from the extended Hamilton principle: the growth field of Klempt et al. (2024) and the multi-species point model of Klempt et al. (2026).

First, the 15 interaction coefficients of a five-species point model are estimated with transitional Markov chain Monte Carlo from in-vitro time courses under four conditions. Compiled with JAX, the forward model runs about 150 times faster on the GPU than on the CPU; every stage is repeated with three seeds, and the result states which of the 15 coefficients the data determine. A metabolic sign prior carries the same type of model to sparse in-vivo data.

Second, the growth kinematics $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$ with $\mathbf{F}_g = \alpha\mathbf{I}$ is implemented as an ANSYS user material, verified against closed-form solutions, and coupled to the element's biofilm field through the growth equation at every Gauss point. Inside ANSYS the coupled run follows the exact solution of a gradient-free region to 3×10^{-11} in α.

Third, the point model runs, unchanged, at every Gauss point. It receives the nutrient and an antibiotic from fields of the element, interpolated in time, and only its living biomass consumes them; through the nutrient the composition reaches the spreading of the biofilm. With uniform fields the coupled point model reproduces the point model alone to round-off. The antibiotic is shown with synthetic data, since the calibration data contain none: it lowers the living volume fraction of a two-species seed from 0.97 to 0.15, or to 0.04 when the species start mixed, while the growth changes little. The growth converges on refined grids within a few percent; the stress needs a second-order treatment of the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ and converges with a GCI of 1 to 29 %.

The coupling has been verified, but not yet validated against measurements. A coupling in which the composition acts on the stiffness is left to the continuation of this work.

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List of Symbols

Units are those of the models; T^* is the normalised time of Klempt et al. 2024, and lengths in Chapter 4 are in mm.

Growth field and mechanics (Klempt et al. 2024, Chapter 4)

$\mathbf{F}$, $\mathbf{F}_e$, $\mathbf{F}_g$	deformation gradient, its elastic and growth parts, $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$
α	isotropic growth, $\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$; the growth is $\alpha - 1$
ϕ, ϕ_{3D}	biofilm state variable of the field (amount of biofilm), $\phi \in [0, 1]$
c	nutrient concentration of the field
β	diffusion coefficient of ϕ [mm$^2/T^*$]
k_α	growth rate [$1/T^*$]
d, g	nutrient diffusivity [mm$^2/T^*$] and consumption rate [$1/T^*$]
r, k	front growth parameter and half-velocity constant
E, ν, μ	Young's modulus, Poisson's ratio, shear modulus
$E(\phi)$, f	stiffness weighted by ϕ^2, small floor for the void
C_{10}, C_{01}, D_1	Mooney–Rivlin constants
J_e, $\mathbf{b}_e$, $\bar{I}_1$, $\bar{I}_2$	$\det \mathbf{F}_e$, left Cauchy–Green tensor of $\mathbf{F}_e$, its isochoric invariants
Λ	Thiele number, $\Lambda^2 = gL^2/d$

Point model (Klempt et al. 2026, Chapters 2 and 3)

ϕ_i, ϕ_0	volume fraction of species i, void fraction
ψ_i, $\bar{\phi}_i$	viability of species i, living fraction $\bar{\phi}_i = \phi_i\psi_i$
$\mathbf{A}$, A_{ij}	symmetric interaction matrix
$\mathbf{B}$, b_i	antibiotic sensitivity
c^*, α^*	nutrient and antibiotic level
η_i, $\eta_{\phi,i}$	dissipation parameters
γ	Lagrange multiplier of $\sum_{l=0}^{n} \phi_l = 1$
Ψ, Δ, C	free energy, dissipation function, constraint function
K_p	coefficient of the barrier term

Coupling (Chapter 4)

s	time link: the point model advances $s\,\Delta t$ per substep, fixed by the duration of the data
$\phi_{\min}$	threshold of ϕ_{3D} for starting the point model at a Gauss point
$\bar{\phi}_i$	living volume fraction of species i in the point model, $\phi_i\psi_i$
a, α^*	antibiotic field (the element's second nutrient) and its value in the point model
n_{sub}	point-model steps per substep
c_0^*, c_{ref}	nutrient level of the point model at the reference concentration c_{ref}

Calibration (Chapter 3)

$\boldsymbol{\theta}$, n_θ	parameter vector, number of parameters (15)
τ_j	tempering exponent of stage j
c_v	target coefficient of variation of the weights
N_p, K	number of samples, mutation steps per stage
s_p	proposal scale
$\mathbf{S}$, λ_S	sign prior from metabolic models, weight of its penalty
Numerics	
h, Δt	element size, time step
λ	$\beta \Delta t / h^2$, the stability measure of the explicit field update
$\lambda_{\max}$, ρ	stability limit of λ; spectral radius of the discrete Laplacian
N, β^*	number of neighbours and width [mm] of the weight of the NEM fit

1 Introduction

1.1 Clinical Background: Oral Biofilm and Peri-implantitis

Dental implants have become a standard treatment for missing teeth; in the United States the share of adults with an implant rose from 0.7% in 1999–2000 to 5.7% in 2015–2016 [1]. Peri-implantitis, a progressive inflammatory condition of the tissue around an implant driven by biofilm, affects about one in five implant patients (patient-level prevalence 22% and 19.5% in two meta-analyses [2, 3]) and is a major cause of late implant loss [4–6]. It is linked to the multi-species communities that colonise the implant surface: dysbiotic shifts in the oral microbiome, driven by pathobionts such as *Porphyromonas gingivalis* (*Pg*) [7], trigger host immune responses that lead to bone resorption [8, 9].

The oral microbiome comprises more than 700 taxa that interact through mutualism, competition and metabolic cross-feeding [10, 11]. In a healthy host the community is dominated by early colonisers such as *Streptococcus oralis* (*So*) and *Actinomyces naeslundii* (*An*). A perturbation can shift it to a dysbiotic state in which late colonisers, most notably the low-abundance keystone pathogen *Pg* [12], sustain a positive feedback loop with the immune system [13].

A biofilm is also a mechanical body. It grows on a surface, and where it grows unevenly, or against a stiff substrate, it carries stresses that bear on detachment and on the loading of the tissue beneath. Which species are present, and where, therefore matters for the mechanics as well as for the biology. This thesis is about bringing these two descriptions together in a finite element setting.

1.2 Models of Biofilm Composition and Growth

1.2.1 Composition at a material point

The generalised Lotka–Volterra (gLV) system, with roots in classical population ecology [14, 15], is the standard model of microbial community dynamics [16–19]:

$$\dot{\phi}_i = \phi_i\Big(r_i + \sum_j A_{ij}\phi_j\Big), \quad i = 1, \dots, n, \tag{1.1}$$

with growth rates r_i and interaction coefficients A_{ij}. Its n^2 interaction coefficients are poorly determined by typical time-course data [17, 20], and relative-abundance data carry a unit-sum constraint that the model does not respect [21, 22].

The extended Hamilton principle [23] derives the evolution of a biofilm from an energy and a dissipation functional. Applied to a material point with several species, it gives the point model of Klempt et al. [24]: volume fractions ϕ_i and viabilities ψ_i, the living fraction $\bar{\phi}_i = \phi_i\psi_i$, a symmetric interaction matrix A_{ij} and a nutrient level c^*. Only the symmetric part of A_{ij} enters the quadratic energy, so the matrix is taken symmetric, which

reduces the number of interaction coefficients to $n(n+1)/2$, 15 for five species. The model describes how the composition at one point evolves in time; it has no space.

1.2.2 Estimating the interactions

The interaction coefficients have to be estimated from measurements, and with five species and a few time points the problem is poorly posed. Bayesian inference quantifies the remaining uncertainty [25]. Transitional Markov chain Monte Carlo (TMCMC) [26, 27] passes from the prior to the posterior through a sequence of tempered distributions, which copes with the multimodal posteriors of this problem and parallelises over the samples; compiled with JAX [28], the forward model runs fast enough to make it practical. Where data are sparse, the signs of the interactions can be fixed beforehand from metabolic network models [29–31]: a species that feeds another implies $A_{ij} > 0$, competition implies $A_{ij} < 0$.

1.2.3 Growth in space

Continuum models of biofilm growth describe the biofilm as a field. In the model of Klempt et al. [32], also derived from the extended Hamilton principle, a biofilm field $\phi(\mathbf{X}, t)$ spreads towards a diffusing nutrient $c(\mathbf{X}, t)$, and the local growth enters the kinematics through the split $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$ with $\mathbf{F}_g = \alpha\mathbf{I}$. Where the growth is uneven or constrained, the elastic part $\mathbf{F}_e$ carries stress. The authors solve this model with a user element in ANSYS, and an earlier comparison of growth models showed pressure inside a growing biofilm and tension at its edge [33]. The same variational approach has been combined with experiments on *Pg* biofilms on titanium [34] and applied to tissue growth and turnover in atherosclerosis [35]. The field model has one biofilm variable; it does not say which species the biofilm consists of.

1.3 Objectives and Thesis Structure

The point model gives the composition but no space; the field model gives space and stress but no composition. The task of this thesis is to bring the point model into an existing, working finite element framework, so that the composition, the growth and the resulting stresses can be computed in space. The framework is the biofilm element developed at the Institut für Kontinuumsmechanik, which solves the field model of Klempt et al. 2024 in ANSYS.

Figure 1.1 shows how the parts fit together.

I make three contributions:

1. *A calibrated point model* (Chapter 3). The 15 interaction coefficients of a five-species point model are estimated with TMCMC from the in-vitro time courses of Heine et al. [36] under four conditions (commensal and dysbiotic, static and in a HOBIC flow reactor). A final section shows how sign constraints from metabolic models carry the same model to sparse in-vivo data; this part was done during an internship.

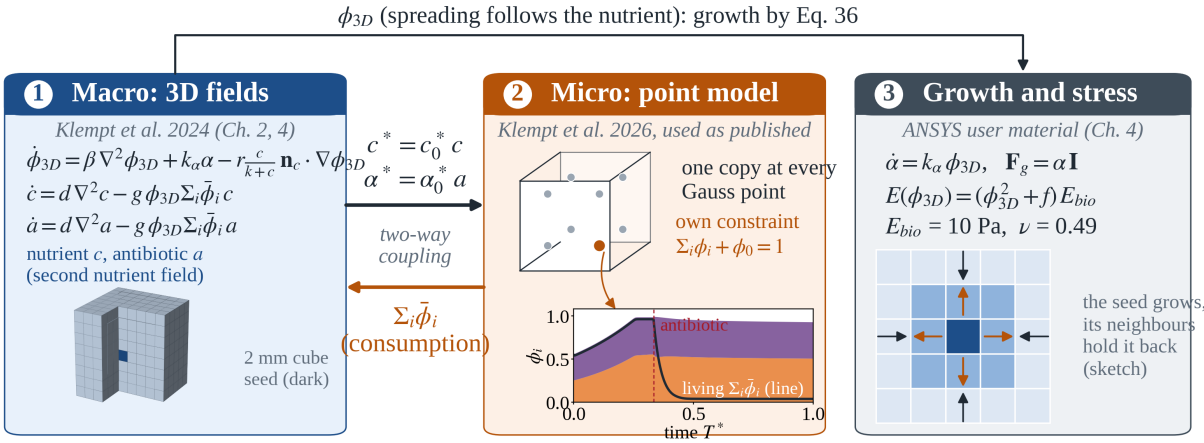


Figure 1.1: Overview of this work. (1) The fields of Klempt et al. 2024 in the IKM element: the biofilm ϕ_{3D}, the nutrient c and, in the element's second nutrient field, an antibiotic a. (2) One copy of the point model of Klempt et al. 2026 at every Gauss point, used as published with its own constraint; it receives c^* and α^* from the fields and returns ϕ_i, ψ_i. Its living volume fraction $\sum_i \bar{\phi}_i$ scales the consumption of c and a, and c enters the spreading of ϕ_{3D}. (3) The growth α and the stresses follow from ϕ_{3D} (Eq. 36 of Klempt et al. 2024). The inset in (2) is computed (synthetic two-species case of Fritsch et al., antibiotic with $\alpha_0^* = 100$ from $T^* = 1/3$); the stress pattern in (3) is a sketch.

2. *A growth law in the finite element framework* (Chapter 4). The growth kinematics $\mathbf{F}_g = \alpha \mathbf{I}$ is implemented as an ANSYS user material, verified against closed-form solutions, and coupled to the element's biofilm field through the growth equation of Klempt et al. 2024 at every Gauss point.

3. *The point model at every Gauss point* (Chapter 4). The point model of Klempt et al. 2026 runs, unchanged, at every Gauss point; it receives the nutrient and an antibiotic from fields of the element, and its living biomass sets the consumption of both. The coupled point model reproduces the point model alone, and the time of both models is linked through the duration of the data. Two-species examples with an antibiotic are verified on refined grids and against the element.

The thesis covers one and two species, for which the papers give numbers to check against. More species and a coupling in which the composition acts back on growth and stiffness are left to the continuation of this work at Keio University.

Chapter 2 sets out the background: the extended Hamilton principle, the field model of Klempt et al. 2024, the point model of Klempt et al. 2026 and Bayesian inference with TMCMC. Chapter 3 calibrates the point model. Chapter 4 brings it into the ANSYS element, with the verification, the results for one and two species and their limits. Chapter 5 summarises the results and outlines the continuation.

2 Theoretical Background

This chapter collects the two models that the thesis connects and the inference method used to calibrate one of them. Both models are derived from the extended Hamilton principle (Section 2.1). The field model of Klempt et al. 2024 describes a single biofilm phase that grows in space and carries stress (Section 2.2); the point model of Klempt et al. 2026 describes the composition of several species at one material point (Section 2.3). Section 2.5 states how the two are combined in this thesis, and Sections 2.6 and 2.7 cover Bayesian inference with TMCMC and the metabolic sign prior.

2.1 Extended Hamilton Principle

The extended Hamilton principle [23] derives the governing equations of a dissipative material from one stationarity condition. For a quasi-static, isothermal process it reads, in the form used by Klempt et al. [32] (their Eq. 19),

$$\mathcal{H} = \int_\tau \big(\mathcal{G} + \mathcal{C} - \mathcal{B} + \mathcal{D}\big)\,\mathrm{d}t \;\to\; \operatorname*{stat}_{\mathbf{u},\boldsymbol{\xi}}, \tag{2.1}$$

with the Gibbs energy $\mathcal{G}$ of the body (stored energy minus the work of external loads), constraint terms $\mathcal{C}$, a non-conservative source $\mathcal{B}$ (growth), and the dissipation $\mathcal{D}$. The displacement $\mathbf{u}$ and the internal variables $\boldsymbol{\xi}$ are varied independently. The dissipation is modelled by a dissipation function, here rate-dependent and quadratic, $\Delta^{\mathrm{diss}} = \frac{\eta}{2}\|\dot{\boldsymbol{\xi}}\|^2$, whose derivative $\partial\Delta^{\mathrm{diss}}/\partial\dot{\boldsymbol{\xi}}$ gives the non-conservative forces. Stationarity with respect to $\mathbf{u}$ gives the balance of linear momentum in weak form; stationarity with respect to each internal variable gives its evolution equation. A model is then fixed by choosing the energy, the source and the dissipation function. For the point model of Chapter 3, which has no displacement field, the principle reduces to its local form, the principle of the minimum of the dissipation potential [37, 38]: $\dot{\Psi} + \Delta + C \to \mathrm{stat}$ with respect to the rates of the internal variables and the Lagrange multiplier. A recent account of this variational framework for dissipative materials is given by Junker, Bode and Hackl [39].

2.2 The Growth Field of Klempt et al. 2024

2.2.1 Variables and kinematics

Klempt et al. [32] describe a biofilm growing into a void that holds a nutrient. The state is given by the displacement $\mathbf{u}(\mathbf{X}, t)$ and three internal variables:

- the biofilm state variable $\phi(\mathbf{X}, t) \in [0, 1]$, 1 in biofilm and 0 in the void;
- the nutrient concentration $c(\mathbf{X}, t)$;
- the local expansion $\alpha(\mathbf{X}, t)$, an internal variable at the Gauss points.

Growth enters the kinematics through the multiplicative split of the deformation gradient [40],

$$\mathbf{F} = \mathbf{F}_e\mathbf{F}_g, \qquad \mathbf{F}_g = \alpha\,\mathbf{I}, \qquad \alpha(t=0) = 1, \tag{2.2}$$

so that $\alpha - 1$ is the isotropic growth strain. The elastic part $\mathbf{F}_e$ carries the stress. The paper treats the biofilm as incompressible, $\det \mathbf{F}_e = 1$, enforced by a pressure p; in this thesis the constraint is replaced by a nearly incompressible material with $\nu = 0.49$ (Chapter 4).

2.2.2 Energy, growth and consumption

The energy density (Eq. 20 of the paper) is

$$\Psi = \phi^2\,\frac{\mu}{2}\big(\mathbf{I} : \mathbf{C}_e - 3\big) + \frac{\bar\beta}{2}\|\nabla_X\phi\|^2 + \frac{\bar d}{2}\|\nabla_X c\|^2 - \bar k_\alpha\,\alpha\,\phi, \tag{2.3}$$

with $\mathbf{C}_e = \mathbf{F}_e^{\mathrm{T}}\mathbf{F}_e$. The first term is a neo-Hookean material whose stiffness is weighted by ϕ^2, so the void carries no stiffness. The two gradient terms regularise ϕ and c, and the last term links the expansion α to the biofilm.

The growth source spreads the biofilm towards the nutrient. It is the product of a Monod function and an orientation function (Eqs. 21–23),

$$h^*_\phi = \tilde r(c)\,o(\nabla_X\phi, \nabla_X c), \qquad \tilde r(c) = \|\nabla_X\phi\|\,\frac{\bar r\,c}{k + c}, \qquad o = \mathbf{n}_{\nabla\phi}\cdot\mathbf{n}_{\nabla c}, \tag{2.4}$$

with the unit vectors $\mathbf{n}_{\nabla\phi} = \nabla_X\phi/\|\nabla_X\phi\|$ and $\mathbf{n}_{\nabla c} = \nabla_X c/\|\nabla_X c\|$. The factor $\|\nabla_X\phi\|$ restricts this growth to the interface between biofilm and void. Consumption is linear in the biofilm as printed (Eq. 24), $g^*_c = \bar g\,\phi$; the authors confirmed that their implementation uses $\bar g\,\phi\,c$ (see below).

2.2.3 Evolution equations

Stationarity of (2.1) with respect to ϕ, c and α gives three evolution equations. The paper neglects the mechanical term $\phi\mu(\mathbf{I} : \mathbf{C}_e - 3)$ in the equation for ϕ, so that growth is driven by the nutrient only and the coupling to the mechanics is one-way. After division by the viscosities, and with the parameters written without the bar, the equations are, as printed (Eqs. 34–36),

$$\dot\phi - \beta\nabla_X^2\phi - k_\alpha\,\alpha + \|\nabla_X\phi\|\,\frac{r\,c}{k + c}\,\mathbf{n}_{\nabla\phi}\cdot\mathbf{n}_{\nabla c} = 0, \tag{2.5}$$

$$\dot c - d\,\nabla_X^2 c + g\,\phi = 0, \tag{2.6}$$

$$\dot\alpha - k_\alpha\,\phi = 0, \tag{2.7}$$

with zero-flux conditions for ϕ and c on the boundary. The authors confirmed that their implementation uses the consumption $g\,\phi\,c$ instead of the printed $g\,\phi$, so that the nutrient equation in the form used for the implementation is

$$\dot c - d\,\nabla_X^2 c + g\,\phi\,c = 0. \tag{2.8}$$

The numbers in Chapter 4 are computed with the printed form (2.6) unless stated otherwise. Time is normalised and written in units of T^*. The parameters of Table 2 of the paper are listed in Table 2.1.

Table 2.1: Parameters of Klempt et al. [32], Table 2. The shear modulus corresponds to $E = 10$ Pa and $\nu = 0.49$.

parameter	symbol	value	unit
shear modulus	μ	3.3557	Pa
nutrient diffusivity	d	10^{10}	$\mu\mathrm{m}^2\, T^{*-1}$
phase-field regularisation	β	2	$\mu\mathrm{m}^2\, T^{*-1}$
growth factor	k_α	10^{-3}	T^{*-1}
half-velocity constant	k	1	–
consumption	g	10^8	T^{*-1}
growth parameter	r	100	T^{*-1}

Three points of the paper are relevant here.

- *The law for α.* Equation (2.7) lets the biofilm expand wherever it is present. The algorithm in Table 1 of the paper instead updates α from the change of ϕ, so that a biofilm that is full from the start does not grow. The text says that α is governed by Eq. 36; I follow Eq. (2.7), as does the element used in Chapter 4.
- *The unit of μ.* The text gives μ in MPa, Table 2 in Pa. I use the table.
- *The printed growth term.* Solved as printed, Eq. (2.5) moves the seeded colony towards the nutrient rather than enlarging it, and the published curves of mean ϕ are not reached. My reproduction of the paper's test cases needs growth on both faces of the colony and a consumption proportional to ϕc (Appendix C). The authors confirmed the consumption $g\,\phi\,c$; their answers are given in the appendix.

2.3 The Point Model of Klempt et al. 2026

2.3.1 State variables

Klempt et al. [24] describe several interacting species at one material point. Species $i = 1, \ldots, n$ has the volume fraction $\phi_i \in [0, 1]$ and the viability $\psi_i \in [0, 1]$, the fraction of living cells in it. The living volume fraction is

$$\bar{\phi}_i := \phi_i\, \psi_i, \qquad i = 1, \ldots, n. \tag{2.9}$$

A further fraction ϕ_0 represents the empty space, and the fractions sum to one,

$$\sum_{l=0}^{n} \phi_l(t) = 1 \quad \forall\, t. \tag{2.10}$$

The authors describe ϕ_0 as a numerical auxiliary variable and caution against reading it as physical void.

2.3.2 Energy, dissipation and evolution equations

The energy contains the nutrient-driven interaction of the living fractions and the effect of an antibiotic on the viabilities,

$$\Psi = -\tfrac{1}{2}\, c^*\, \bar{\boldsymbol{\phi}}^{\mathrm{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}} + \tfrac{1}{2}\, \alpha^*\, \boldsymbol{\psi}^{\mathrm{T}} \mathbf{B}\, \boldsymbol{\psi}, \tag{2.11}$$

with the nutrient level c^*, the antibiotic level α^*, the symmetric interaction matrix $\mathbf{A}$ and $\mathbf{B} = \mathrm{diag}(b_1, \ldots, b_n)$. The constraint (2.10) enters through the constraint function $C = \gamma\Big(\sum_{l=0}^{n} \phi_l - 1\Big)$ with a Lagrange multiplier γ, and the dissipation function is

$$\Delta = \tfrac{1}{2}\, \dot{\bar{\boldsymbol{\phi}}}^{\mathrm{T}} \boldsymbol{\eta}\, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2}\, \dot{\boldsymbol{\phi}}^{\mathrm{T}} \boldsymbol{\eta}\, \dot{\boldsymbol{\phi}}, \qquad \boldsymbol{\eta} = \mathrm{diag}(\eta_1, \ldots, \eta_n). \tag{2.12}$$

The second term is needed because the first alone gives a singular system: $\bar{\phi}_i = \phi_i \psi_i$ couples the two rates.

The stationarity conditions of $\dot{\Psi} + \Delta + C$ with respect to the rates $\dot{\phi}_i$, $\dot{\psi}_i$, $\dot{\phi}_0$ and to γ give

$$0 = -c^*\, \psi_i \left(\mathbf{A}\bar{\boldsymbol{\phi}}\right)_i + \eta_i\Big(\dot{\phi}_i\, \psi_i^2 + \bar{\phi}_i\, \dot{\psi}_i + \dot{\phi}_i\Big) + \gamma, \tag{2.13}$$

$$0 = -c^*\, \phi_i \left(\mathbf{A}\bar{\boldsymbol{\phi}}\right)_i + \alpha^*\, b_i\, \psi_i + \eta_i\Big(\dot{\psi}_i\, \phi_i^2 + \bar{\phi}_i\, \dot{\phi}_i\Big), \tag{2.14}$$

$$0 = \gamma + \dot{\phi}_0, \qquad 0 = \sum_{l=0}^{n} \phi_l - 1. \tag{2.15}$$

The equation for ψ_i has no γ, because ψ_i does not appear in the constraint. The paper prints Eq. 17 with a $+\gamma$; with it, even the two-species examples of the paper are not reproduced; without it, all fourteen examples are (Section 2.3.4).

The system (2.13)–(2.15) has $2n + 2$ unknowns. It is integrated with the implicit Euler method and Newton iterations; a barrier term $K_p/[x^2(1-x)^2]$, which diverges at $x = 0$ and $x = 1$, with $K_p = \eta_i \times 10^{-4}$ keeps ϕ_i, ψ_i and ϕ_0 inside $(0, 1)$ and is omitted above. Numerical treatments that keep such constraints satisfied in constitutive models are compared by Bode et al. [41].

2.3.3 Symmetry and scale

The antisymmetric part of $\mathbf{A}$ does not contribute to the quadratic form (2.11), so $\mathbf{A}$ is taken symmetric; this reduces the number of interaction coefficients from n^2 to $n(n+1)/2$ and keeps the gradient-flow structure that the variational derivation provides [23]. Biologically, symmetry is a reasonable approximation for interactions mediated by metabolites in a shared pool [10]; directional effects such as toxin secretion are excluded.

The nutrient level enters only through the product $c^*\mathbf{A}$, so c^* and the scale of $\mathbf{A}$ cannot be identified separately. In the calibration of Chapter 3 I fix $c^* = 25$ and $\eta_i = 1$, which keeps the coefficients of order one; the paper's examples use $c^* = 100$ and species-dependent η_i. The data of Heine et al. involve no antibiotic, so $\alpha^* = 0$ there.

2.3.4 Check against the paper

The integrator used in this thesis reproduces all fourteen numerical examples of the paper (six two-species cases of its Table 1, four variants of its Appendix A and the four four-species cases), within the accuracy with which values can be read off its figures. Two printed details disagree with the paper's own figures: the $+\gamma$ in Eq. 17 discussed above, and a factor $\frac{1}{2}$ in front of the four-species matrix of Eq. 19 of Klempt et al. 2026; the four-species figures are reproduced only when it is dropped. For two species, case 3 ends at $\bar{\phi}_1 = 0.5945$, $\bar{\phi}_2 = 0.3926$, against 0.60 and 0.40 read from the paper.

2.4 Relation to the Replicator Equation

It is useful to relate the point model to the *replicator equation* of evolutionary game theory [42, 43],

$$\dot{\phi}_i = \phi_i \Big[\big(\mathbf{A}\boldsymbol{\phi}\big)_i - \boldsymbol{\phi}^{\mathrm{T}}\mathbf{A}\boldsymbol{\phi} \Big], \qquad \sum_i \phi_i = 1, \tag{2.16}$$

in which the mean fitness $\boldsymbol{\phi}^{\mathrm{T}}\mathbf{A}\boldsymbol{\phi}$ keeps the composition on the simplex. For a symmetric $\mathbf{A}$, (2.16) is a gradient flow of $W(\boldsymbol{\phi}) = \frac{1}{2}\boldsymbol{\phi}^{\mathrm{T}}\mathbf{A}\boldsymbol{\phi}$ with respect to the Shahshahani metric $g_{ij} = \delta_{ij}/\phi_i$, and W increases monotonically [42, 44]. The point model uses a constant viscosity instead of this ϕ_i-weighted metric; with $\psi_i \equiv 1$ its well-mixed limit is the affine projection $\dot{\phi}_i \propto (\mathbf{A}\boldsymbol{\phi})_i - \langle \mathbf{A}\boldsymbol{\phi} \rangle$. The two have the same fixed points and the same sign structure.

Compared with the generalised Lotka–Volterra model (1.1), the replicator form has a symmetric $\mathbf{A}$ with $n(n+1)/2$ coefficients and keeps $\sum_i \phi_i = 1$ without a separate normalisation. Its interior fixed points satisfy $(\mathbf{A}\boldsymbol{\phi}^*)_i = \boldsymbol{\phi}^{*\mathrm{T}}\mathbf{A}\boldsymbol{\phi}^*$ for every i with $\phi_i^* > 0$. If $\mathbf{A}$ is negative definite on the tangent space $\{\mathbf{x} : \sum_i x_i = 0\}$ of the simplex, W is strictly concave there, so there is at most one interior fixed point; when it exists, it is the maximum of W and attracts every interior trajectory.

2.5 Combining the Two Models

The field model has a single biofilm variable ϕ and says where the biofilm is and how much there is; it does not distinguish species. The point model distinguishes species but has no space. Klempt et al. 2024 drop the mechanical term in the equation for ϕ, so in their model the stress does not act back on growth. I combine the two models as follows:

- the field model, solved by the element, gives the biofilm ϕ_{3D}, the nutrient c and, in its second nutrient field, an antibiotic, and through Eq. (2.7) the growth α and the stress;
- the point model runs, unchanged, at every Gauss point; it receives the nutrient and the antibiotic from the fields and returns ϕ_i and ψ_i;
- only its living biomass, $\sum_i \phi_i\psi_i$, consumes the nutrient and takes up the antibiotic, and the nutrient steers the spreading of ϕ_{3D} through the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$.

How the two times are linked is given in Chapter 4. The stress does not act back, and the composition does not act on the stiffness; these directions are outlined as future work.

2.6 Bayesian Inference and TMCMC

2.6.1 Bayesian parameter identification

Given observed species trajectories $\mathbf{y} = \{\mathbf{y}_t\}_{t=1}^{N_t}$ and a forward model parameterised by the interaction vector $\boldsymbol{\theta} \in \mathbb{R}^{15}$, Bayesian inference seeks the posterior distribution

$$p(\boldsymbol{\theta} \mid \mathbf{y}) \;\propto\; \mathcal{L}(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{2.17}$$

with the likelihood $\mathcal{L}(\mathbf{y} \mid \boldsymbol{\theta})$ and the prior $p(\boldsymbol{\theta})$. I use a Gaussian likelihood with isotropic noise,

$$\mathcal{L}(\mathbf{y} \mid \boldsymbol{\theta}) = \prod_{t,i} \mathcal{N}\Big(y_{t,i} \mid \hat{\phi}_i(t;\boldsymbol{\theta}),\, \sigma^2\Big), \tag{2.18}$$

where $\hat{\phi}_i(t;\boldsymbol{\theta})$ is the solution of the point model at time t.

2.6.2 Transitional Markov chain Monte Carlo

TMCMC [26, 27] constructs a sequence of intermediate distributions between the prior and the posterior through a tempering schedule $0 = \tau_0 < \tau_1 < \cdots < \tau_J = 1$,

$$p_j(\boldsymbol{\theta}) \propto \mathcal{L}(\mathbf{y} \mid \boldsymbol{\theta})^{\tau_j}\, p(\boldsymbol{\theta}). \tag{2.19}$$

At each stage j the samples are weighted by $w_j^{(k)} \propto \mathcal{L}^{\tau_j - \tau_{j-1}}$, resampled, and moved by short Metropolis–Hastings chains. The next exponent τ_j is chosen so that the coefficient of variation of the weights stays at a target value, typically $c_v = 1$.

Computation on the GPU. The cost lies in the forward model, evaluated for every sample at every stage. I compile the point model with JAX [28] and evaluate the whole ensemble, $N_p = 5{,}000$ to $20{,}000$ particles at the final stage, in one batched call (`jax.vmap`). This is about 150 times faster than the CPU baseline (Section 3.2.4).

Narrowed priors. For the multimodal posterior of the dysbiotic condition I run TMCMC twice: the mean and standard deviation of the first posterior define a narrower prior for the second run.

Identifiability. To see which parameters the data determine, I use the ratio

$$r_i \;=\; \frac{\sigma_{\mathrm{post},i}}{\Delta_{\mathrm{prior},i}} \tag{2.20}$$

of the posterior standard deviation of θ_i to its prior width. A value $r_i \ll 1$ means the data constrain θ_i; $r_i \to 1$ means the prior does. Chapter 3 uses this ratio, with the threshold 0.8 relative to a uniform prior, to classify the entries as identified or not.

2.7 Metabolic Sign Priors

Where the data are too sparse to determine the interaction coefficients, their signs can be fixed beforehand from metabolic models. Flux balance analysis (FBA) predicts the metabolic fluxes $\mathbf{v}$ of an organism at steady state [29, 31],

$$\max_{\mathbf{v}} \ \mathbf{w}^{\mathrm{T}}\mathbf{v} \quad \text{s.t.} \quad \mathbf{N}\mathbf{v} = \mathbf{0},\ \mathbf{lb} \le \mathbf{v} \le \mathbf{ub}, \tag{2.21}$$

with the stoichiometric matrix $\mathbf{N}$, an objective $\mathbf{w}$ (usually biomass production) and bounds on the fluxes. Parsimonious FBA [45] additionally minimises the total flux, and MICOM [30] couples the models of several species through shared metabolites. From the predicted exchange between two species follows the sign of their interaction: a species that feeds another implies $A_{ij} > 0$, competition implies $A_{ij} < 0$.

The sign prior $\mathbf{S} \in \{-1, 0, +1\}^{n \times n}$ enters the inference as a penalty on coefficients of the wrong sign,

$$\mathcal{R}(\boldsymbol{\theta}; \mathbf{S}) = \lambda_S \sum_{i,j:\, S_{ij} \neq 0} \max\Big(0,\ -S_{ij}\, A_{ij}\Big)^2, \tag{2.22}$$

with the weight λ_S, and leaves coefficients of the right sign free. Only the sign is used, not the magnitude: predicted interaction strengths of genome-scale models agree poorly with measured ones, while their signs carry useful information [46], and no derivation links metabolic fluxes to the dimensionless coefficients of $\mathbf{A}$ [47, 48]. Section 3.6 applies this prior to in-vivo data.

3 GPU-Accelerated Bayesian Inference for a Five-Species Biofilm

This chapter calibrates the point model of Chapter 2 on the in-vitro time courses of a defined five-species oral biofilm of Heine et al. [36, 49]. The unknowns are the 15 entries of the symmetric interaction matrix $\mathbf{A}$. They are effective parameters of the model, and the main result of the chapter is a statement, entry by entry, of what the data determine and what they leave open. The last section carries the same type of model to sparse in-vivo data with a metabolic sign prior (Section 3.6), and Chapter 4 brings the point model into a finite element framework.

This chapter is based on the manuscript [50]: Nishioka K., Klempt F., Geisler H., Mukherjee R., Heine N., Doll-Nikutta K., Stiesch M., Szafrański S.P., Soleimani M., Junker P., "Identifiability of species interactions in a living material: Bayesian calibration of an extended Hamilton principle multiphase model" (in preparation for *Extreme Mechanics Letters*). All runs of this chapter, with their configurations, convergence records and posterior samples, are archived; an earlier set of calibrations from March and April 2026, whose configurations were not kept, is not used.

3.1 Experimental System

Heine et al. [49] cultivated a defined consortium of five oral taxa, *Streptococcus oralis* (*So*), *Actinomyces naeslundii* (*An*), *Veillonella* (*Vei*), *Fusobacterium nucleatum* (*Fn*) and *Porphyromonas gingivalis* (*Pg*), from early colonisers to the late anaerobic pathogen. The experiment crosses the *community* (commensal or dysbiotic) with the *cultivation* (static or the HOBIC flow reactor), which gives the four conditions used throughout this thesis: Commensal Static (CS), Commensal HOBIC (CH), Dysbiotic Static (DS) and Dysbiotic HOBIC (DH). The two communities differ not only in their ecological state but also in the *Veillonella* species (*V. dispar* and *V. parvula*) and in the *Pg* strain (DSM 20709 and W83).

The composition data are the species fractions of viable cells at days 3, 6, 10, 15 and 21, measured by quantitative real-time PCR after propidium monoazide (PMA) treatment, and the total-count fractions without PMA measured on the same samples. I use the measured replicate values provided by the authors (three to five biological replicates per time point, each in technical triplicate) and the medians over them. Values read off the published box plots deviate from these by up to 0.42 in a species fraction in the dysbiotic conditions, so they are not used. The measured data are available from the authors on request and are not reproduced in this thesis; only summary numbers and figures are shown.

The day-1 composition, the earliest measurement, is the initial state of the model at $t = 0$. Model time is the time since inoculation, $t = s\,d$ with $s = 0.2375/21$; in the flow reactor the inoculated medium is replaced after 24 h, so the growth phase there can also

be taken to start at day 1. The viable and the total-count compositions differ (RMSE between their medians 0.101, 0.087, 0.119 and 0.087 in CS, CH, DS and DH) because the species differ in viability, so each is compared with its own model counterpart.

The fraction of membrane-intact cells from SYTO9/propidium iodide staining is image based and *not* resolved by species; per time point it is the mean over 15–45 confocal images, five per biofilm. It enters the inference as the community live fraction. The pH, measured in the flow reactor, is not used in the inference and serves as a check (Section 3.4).

3.2 Inference Setup

The forward model is the coupled $\boldsymbol{\phi}$–$\boldsymbol{\psi}$ system of Section 2.1, integrated by implicit Euler with Newton's method (2,500 steps over the 21 days) with a barrier term that diverges at 0 and 1 and keeps ϕ_i and ψ_i inside $(0, 1)$. Only the product $c^* a_{ij}$ enters, so $c^* = 25$ fixes the scale (Section 2.4); the assays contain no antibiotic, so $\alpha^* = 0$.

3.2.1 The 15-parameter interaction matrix

Only the symmetric part of $\mathbf{A}$ enters the quadratic free energy, so $\mathbf{A}$ is symmetric with 15 independent entries a_{ij}, $i \le j$. Each coefficient is an effective, undirected interaction of the model: it lumps mechanisms the model does not resolve, such as metabolite exchange, co-aggregation or the pH. Table 3.1 lists the interactions documented for this consortium. All 15 entries are estimated, none is set to zero, and agreement with the table is read as consistency with the literature, not as the measurement of a mechanism.

Pair	Documented mechanism	Status
So–An	Co-aggregation	documented
So–Vei	Lactate cross-feeding	documented
So–Fn	Formate exchange	documented
An–Vei	Co-aggregation	documented
Fn–Pg	Co-aggregation	documented
Vei–Pg	Through the pH	hypothesis

Table 3.1: Interactions documented for the consortium [10, 49]. The model coefficients are symmetric and carry no direction.

3.2.2 Likelihood

The viable fractions y^{v}_{ik} at time t_k are compared with the living volume fractions $\hat{y}^{\mathrm{v}}_{ik} = \phi_i\psi_i / \sum_j \phi_j\psi_j$, the total-count fractions y^{t}_{ik} with $\hat{y}^{\mathrm{t}}_{ik} = \phi_i / \sum_j \phi_j$, and the community live fraction ℓ_k with $\hat{\ell}_k = \sum_i \phi_i\psi_i / \sum_i \phi_i$, through the weighted Gaussian log-likelihood

$$\ln p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_k \sum_{i=1}^{5} w_{ik} \left[\frac{\left(y^{\mathrm{v}}_{ik} - \hat{y}^{\mathrm{v}}_{ik}\right)^2}{(\sigma^{\mathrm{v}}_{ik})^2} + \frac{\left(y^{\mathrm{t}}_{ik} - \hat{y}^{\mathrm{t}}_{ik}\right)^2}{(\sigma^{\mathrm{t}}_{ik})^2} \right] - \frac{\lambda_\psi}{2} \sum_k \frac{(\ell_k - \hat{\ell}_k)^2}{\sigma_\ell^2} + \text{const.} \tag{3.1}$$

The standard deviations σ^{v}_{ik} and σ^{t}_{ik} are estimated from the spread of the measured replicates (interquartile range divided by 1.35, at least 0.05), and σ_ℓ is the standard

deviation of the live fraction between biofilms, averaged over the time points. The weights w_{ik} act on both compositions alike: they up-weight *Pg* ($\lambda_{\mathrm{Pg}} = 5$), whose late increase is the main dynamic feature of the dysbiotic data, and the last two time points ($\lambda_{\mathrm{late}} = 3$), and down-weight species with a mean fraction below 5 % ($\lambda_{\mathrm{rare}} = 0.1$). The live fraction is weighted by $\lambda_\psi = 1.5$ in the commensal conditions, 2.0 in DS and 3.0 in DH. The pH does not enter.

In stages (i) and (ii) below, the viabilities are not evolved but fixed to the values implied by the two compositions and the live fraction, $\psi_i(t_k) = (y^{\mathrm{v}}_{ik}/y^{\mathrm{t}}_{ik})\,\ell_k$, clipped to $[0.01, 0.99]$ (the community value ℓ_k for species below 2 % of the total count), and the live-fraction term is omitted. These fixed values are uncertain (the median width of their 90 % intervals is 0.57), so they serve only to start the inference and as a check, not as data in the likelihood. In stages (iii) and (iv), ψ_i follows Eq. (2.14) and all three terms are used.

3.2.3 Prior and four stages

The prior is uniform on a box $[l_k, u_k]$ for each entry. The first boxes were set from earlier calibrations of a previous version of the model on the same data, and some of them exclude negative values or allow only small ones, so the signs of these entries are partly set by the prior in the first two stages; all of these bounds are widened in stage (iii). Inference proceeds in four stages on the same data (Figure 3.1): (i) the two composition terms with ψ_i fixed to the data, on the box; (ii) as (i), on the box narrowed to the stage-(i) posterior mean ± 4 standard deviations; (iii) the full likelihood with ψ_i evolved, back on the full box, half of the initial particles drawn around the stage-(ii) MAP and half from the prior; (iv) as (iii), on the box narrowed to the stage-(iii) posterior mean ± 4 standard deviations. A ± 4 standard deviation box removes less than 10^{-4} of a Gaussian marginal, so narrowing shortens the tempering without truncating the posterior. If more than 20 % of the posterior mass of an entry lies in the outer 5 % of the box on one side, the bound is moved outwards and the stage is repeated.

3.2.4 TMCMC on the GPU and acceptance criteria

TMCMC [26, 51] passes a particle population through the tempered posteriors $p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta})$, $0 = \beta_0 < \cdots < \beta_K = 1$, choosing each increment so that the coefficient of variation of the importance weights stays at a target value [27]. After resampling, every particle performs n_{mut} Metropolis–Hastings steps, with random-walk proposals scaled by the weighted particle covariance alternating with differential-evolution proposals [52]; proposals outside the box are rejected without solving the model. A run needs 10^5 to 3×10^7 model solutions. I compile the implicit solver, with the Jacobian assembly and the linear solve, into a single XLA kernel with JAX and evaluate all particles at once with `vmap`. On one GPU this is 170 times faster per tempering step than a 12-core CPU at matched settings, and 147 times end to end (Table 3.2), which makes it affordable to repeat every stage with three seeds. General gradient-based samplers (HMC/NUTS) were not used.

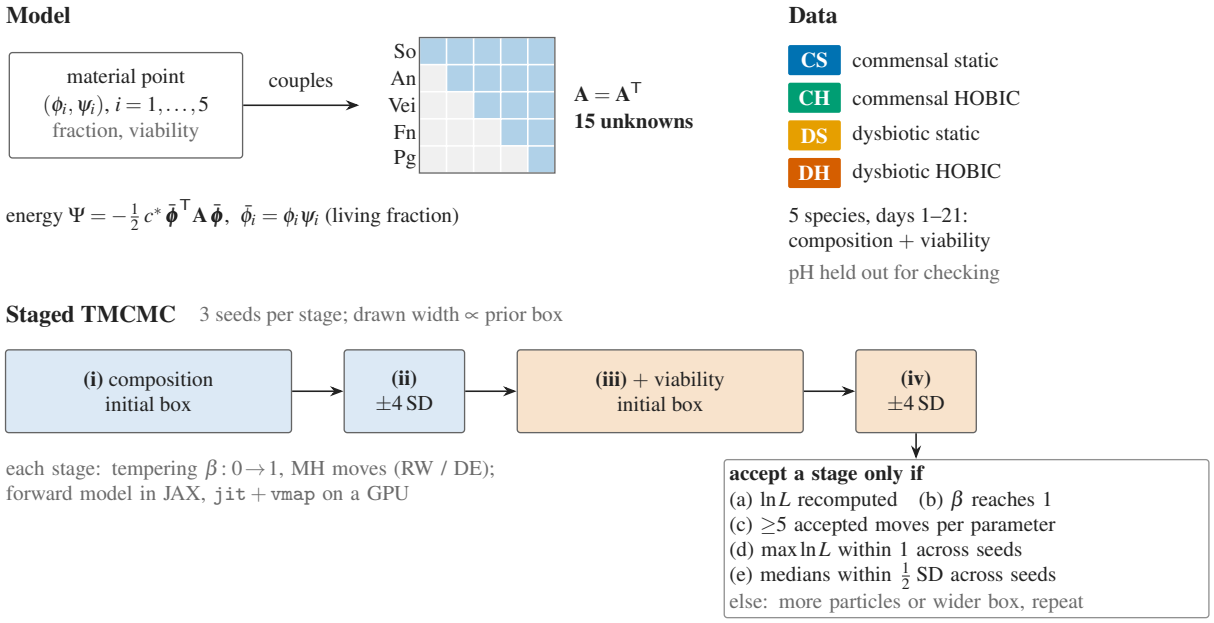


Figure 3.1: Model and inference. A material point carries the volume fractions ϕ_i and viabilities ψ_i of five taxa, coupled through the symmetric matrix $\mathbf{A}$ (15 unknowns). TMCMC with a GPU-compiled forward model infers $\mathbf{A}$ in four stages, first with the viabilities fixed to the data, then with them evolved; every stage is accepted only after the checks (a)–(e) of Section 3.2.4.

Metric	**CPU (12 cores)**	**GPU (RTX 4090)**	**Speedup**
Matched benchmark (DH, $N_p = 500$, 9 tempering steps):			
Per step	2,400 s	14 s	170×
Total	21,600 s	147 s	147×
Final stage of this chapter (one seed, GPU):			
CS, CH, $N_p = 5{,}000$		7.1–8.7 h	
DS, $N_p = 18{,}000$		17.1–18.9 h	
DH (stage iii), $N_p = 20{,}000$		21.1–22.7 h	

Table 3.2: Computational cost. The matched benchmark isolates the GPU speedup; the run times are wall-clock times per seed, with $n_{\text{mut}} = 150$ (120 in DH).

Every stage is run with three independent seeds and accepted only if (a) the stored log-likelihood of every particle agrees with a recomputation; (b) the tempering reaches $\beta = 1$; (c) each particle is moved at least twice per tempering step and at least five times per free parameter over the run; (d) the maximum log-likelihood differs by at most 1 across seeds; and (e) every posterior median differs across seeds by at most half a posterior standard deviation. An entry whose pooled posterior standard deviation is at least 0.8 times that of a uniform distribution on the prior box is reported as *not identified* and excluded from (e); for an entry with several posterior modes, (e) is applied within each mode. All other entries are called *identified*: the data constrain them beyond their prior box. Stages (ii)–(iv) that fail are repeated with more particles or mutation steps and are not interpreted. Stage (i) is exploratory: it sets the box of stage (ii) and gives the evidence estimates, which are read only as indications.

3.3 Results

3.3.1 Fit and convergence

Stages (ii)–(iv) of CS and CH and stages (iii)–(iv) of DS meet the criteria (a)–(e) with three seeds each; DS stage (ii) fails (d), because one seed found a higher mode (spread of the maximum log-likelihood 1.70). In DH, stage (iii) meets them, but at stage (iv) the seeds settle in different posterior modes: criterion (e) fails, with the medians of a_{22} differing by 0.79 posterior standard deviations. The DH results are therefore taken from stage (iii), and this split is discussed as a limitation in Section 3.5. At the final stage the MAP reproduces the viable composition with an RMSE of 0.087 (CS), 0.059 (CH), 0.058 (DS) and 0.073 (DH), the total-count composition with 0.039, 0.078, 0.080 and 0.159, and the community live fraction with 0.23, 0.12, 0.09 and 0.07 (Figure 3.2, Table 3.3). Accepting a stage often needed more than one attempt: posterior mass accumulated at a bound of the initial box in every condition and was released by widening that bound, and the dysbiotic conditions needed more particles. Without the criteria, several of these runs would have been read as converged.

		RMSE of the MAP				
Cond.	Stage	viable	total count	live fraction	$\ln \mathcal{L}$ (MAP)	$\ln \hat{Z}$
CS	(iv)	0.087	0.039	0.23	−32.3	−43.6 to −43.8
CH	(iv)	0.059	0.078	0.12	−25.9	−40.5 to −41.2
DS	(iv)	0.058	0.080	0.09	−96.3	−113.6 to −114.3
DH	(iii)	0.073	0.159	0.07	−51.4	−80.7 to −81.8

Table 3.3: Fit at the final stage. The MAP is the best particle refined by a local optimisation (for DH the best particle); $\ln \mathcal{L}$ without the constant term; $\ln \hat{Z}$: range over the three seeds. Neither is compared across conditions, because the weights and the prior boxes differ; $\ln \hat{Z} \leq \max \ln \mathcal{L}$ holds in every run.

The fit is poorest in CS, in the live fraction (0.23). In this condition the composition

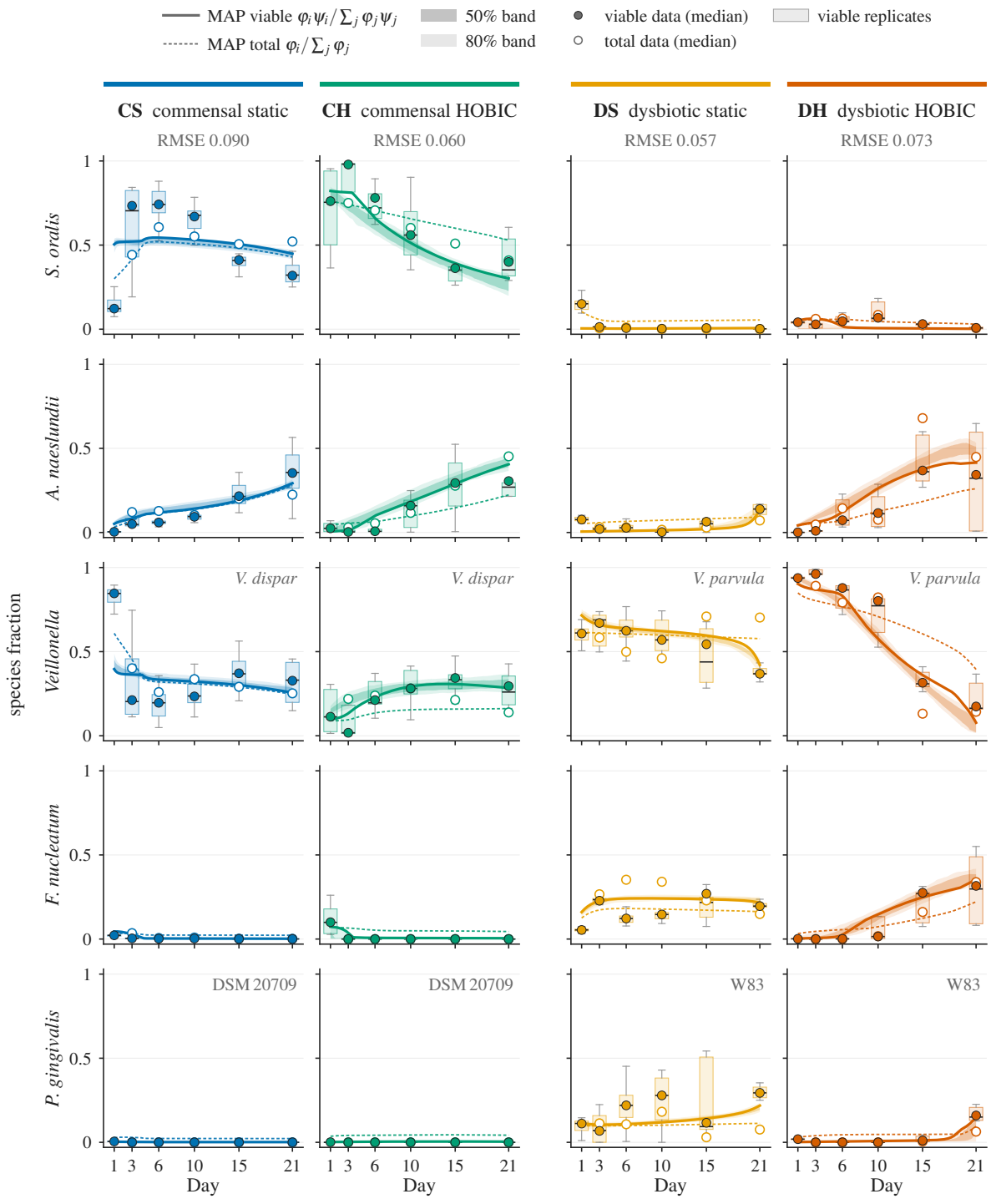


Figure 3.2: Species fractions over 21 days in the four conditions (columns) for the five taxa (rows). Lines: MAP of the final stage, viable composition $\phi_i\psi_i/\sum_j \phi_j\psi_j$ (solid) and total-count composition $\phi_i/\sum_j \phi_j$ (dotted); bands: central 50 % and 80 % posterior predictive intervals of the viable composition, pooled over three seeds; filled dots: median of the viable data; open dots: median of the total-count data; boxes: viable replicates (interquartile range). The RMSE of the viable composition is given above each column for the best particle; the refined MAP of Table 3.3 is slightly lower. DH from stage (iii).

and the live fraction cannot both be matched: raising the weight λ_ψ of the live fraction makes the composition worse, and even a fit to the composition alone does not bring the error of the viable composition below 0.078. This is a limit of the model for CS, not of the sampler, and I report the compromise rather than re-weight the channels for this condition (Figure 3.3).

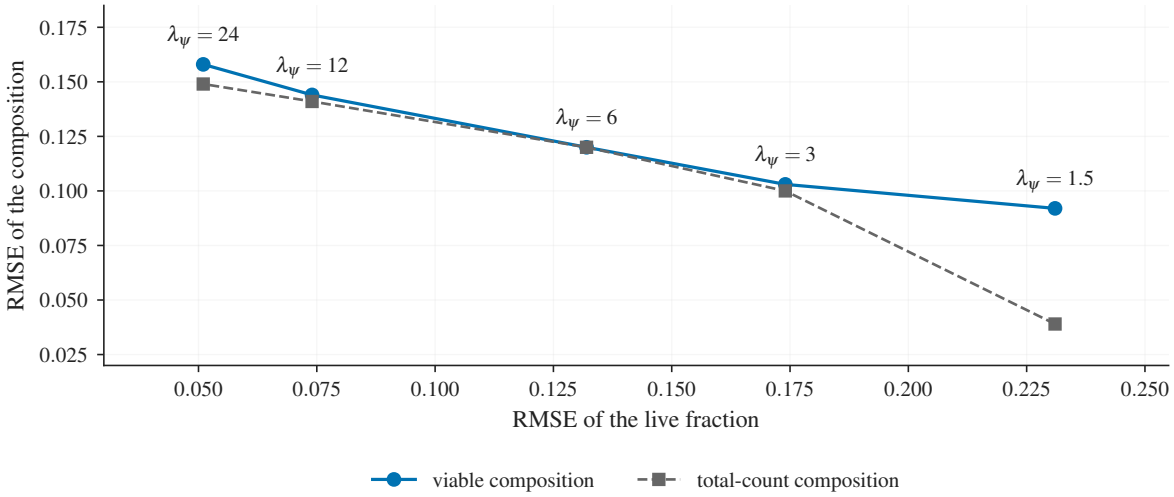


Figure 3.3: CS: error of the composition against the error of the live fraction for different weights λ_ψ of the live fraction (labels). A better live fraction costs composition, and the composition error does not fall below about 0.08.

3.3.2 What the data determine

Figure 3.4 and Table 3.4 classify every entry of **A** at the final stage; Figure 3.5 shows the marginal posteriors. An entry is identified when its posterior standard deviation is below 0.8 times that of a uniform distribution on the final box; otherwise it is determined in sign only (s.o.) when its 90 % interval excludes zero, and not identified (n.i.) when it does not. The number of identified entries is 5, 6, 9 and 8 in CS, CH, DS and DH. Some entries are bounded on one side only: a_{33} in CS lies below -5.3, and in DS a_{44} and a_{13} have only an upper and a_{25} only a lower bound.

In the commensal conditions only a few entries are identified: five in CS (a_{11}, a_{22}, a_{33}, a_{13}, a_{23}) and six in CH (a_{11}, a_{12}, a_{22}, a_{33}, a_{13}, a_{23}), all among the early colonisers *So*, *An* and *Vei*; a_{12} in CS is determined in sign only. Where both conditions determine the sign of an entry, it agrees: the self-interactions of *So* and *Vei* are negative (a_{11}, 90 % intervals $[-7.0, -1.7]$ in CS and $[-1.4, -0.4]$ in CH; a_{33}, $[-14.6, -5.3]$ and $[-5.8, -0.6]$), and the *So*–*Vei* coefficient is positive (a_{13}, $[+4.7, +11.3]$ and $[+0.3, +2.4]$), in line with the lactate cross-feeding between streptococci and *Veillonella* [10], although a_{13} is an effective coefficient and not a measurement of that mechanism. Every entry involving *Fn* or *Pg* spreads over its whole box. This is expected: *Fn* and *Pg* stay below 1 % of the commensal communities from day 3 on, so the data carry little information on their interactions.

In the dysbiotic conditions, where *Fn* and *Pg* reach 10–27 % of the community, more

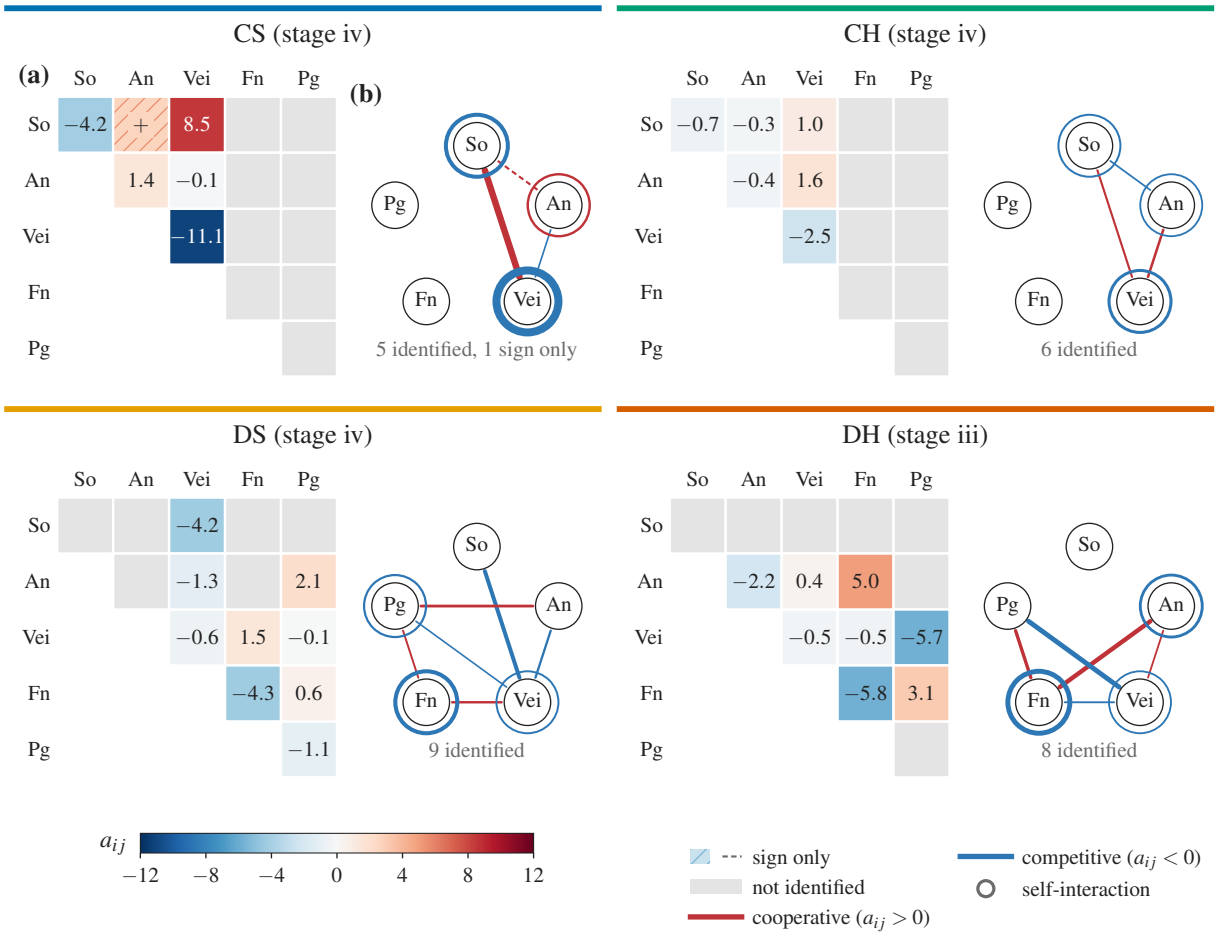


Figure 3.4: What the data determine, at the final stage with three seeds pooled; one block per condition. (a) Left in each block: upper triangle of **A**. Coloured with value: identified; colour and number give the posterior median; hatched: sign only (90 % interval excludes zero); grey: not identified. (b) Right in each block: the same entries as a network: red, $a_{ij} > 0$ (cooperative); blue, $a_{ij} < 0$ (competitive); line width proportional to the absolute median; dashed, sign only; rings, identified self-interactions. DH from stage (iii).

entries are identified, now including interactions of the late colonisers: nine in DS and eight in DH. The self-interaction of *Fn* is negative in both (a_{44}), and the *An*–*Fn* coefficient is positive in DH (a_{24}, $[+1.3, +8.8]$). The *Vei*–*Pg* coefficient a_{35} is negative in DH ($[-20.9, -0.6]$) and close to zero in DS ($[-0.34, +0.24]$). The *So*–*Vei* coefficient, positive in both commensal conditions, is negative in DS (a_{13}, $[-4.9, -2.4]$). The commensal–dysbiotic comparison mixes the ecological state with the strain differences and the cultivation; the DS–DH contrast compares the same strains, although the two conditions also differ in the live-fraction weight and in their prior boxes.

Entry	CS	CH	DS	DH
a_{11}	−4.20 [−7.0, −1.7]	−0.72 [−1.4, −0.4]	n.i.	n.i.
a_{12}	s.o. [+0.0, +1.9]	−0.31 [−0.8, +0.1]	n.i.	n.i.
a_{22}	+1.44 [+0.5, +2.4]	−0.42 [−0.9, +0.5]	n.i.	−2.23 [−6.4, +0.2]
a_{33}	−11.07 [−14.6, −5.3]	−2.50 [−5.8, −0.6]	−0.64 [−0.9, −0.4]	−0.46 [−0.6, −0.4]
a_{34}	n.i.	n.i.	+1.45 [+0.9, +1.9]	−0.51 [−1.3, +0.2]
a_{44}	n.i.	n.i.	−4.29 [−4.9, −2.8]	−5.77 [−9.6, −0.2]
a_{13}	+8.45 [+4.7, +11.3]	+0.96 [+0.3, +2.4]	−4.22 [−4.9, −2.4]	n.i.
a_{14}	n.i.	n.i.	n.i.	n.i.
a_{23}	−0.12 [−1.4, +1.7]	+1.64 [+0.4, +2.7]	−1.29 [−2.0, −0.5]	+0.39 [−0.1, +1.0]
a_{24}	n.i.	n.i.	n.i.	+4.98 [+1.3, +8.8]
a_{55}	n.i.	n.i.	−1.08 [−3.7, +0.2]	n.i.
a_{15}	n.i.	n.i.	n.i.	n.i.
a_{25}	n.i.	n.i.	+2.11 [+1.1, +2.5]	n.i.
a_{35}	n.i.	n.i.	−0.06 [−0.3, +0.2]	−5.72 [−20.9, −0.6]
a_{45}	n.i.	n.i.	+0.63 [−0.3, +2.1]	+3.11 [−2.4, +9.2]
identified / s.o. / n.i.	5/1/9	6/0/9	9/0/6	8/0/7

Table 3.4: Posterior median and 90 % interval of the identified and sign-only entries at the final stage, three seeds pooled; s.o.: sign only; n.i.: not identified. Species 1–5: *So*, *An*, *Vei*, *Fn*, *Pg*. DH from stage (iii).

The classification is a result in its own right: it states which conclusions about the interaction network the data can support, and it comes from the same runs at no additional cost. Because the boxes were narrowed in stages (ii) and (iv), the classification is made against the box of the final stage; judged against a narrowed box, an entry would look identified more easily. Figure 3.6 shows the MAP of every seed: the seeds agree in the identified entries and scatter in the others.

3.3.3 The *Fn*–*Pg* coefficient

The coefficient a_{45} links the bridging organism *Fn* to the pathogen *Pg*. Comparing the full model with a model in which $a_{45} = 0$ gives $\ln B = -0.06$ (CS), $+0.03$ (CH), $+7.5$ (DS) and $+2.1$ (DH). In the commensal conditions the data cannot tell the two models apart; in the dysbiotic conditions they favour a non-zero a_{45}. The uniform prior penalises the full model by the log-width of the a_{45} box, so $\ln B$ is read only within a condition. At the final stage a_{45} is identified in both dysbiotic conditions, but its 90 % intervals include zero ($[-0.3, +2.1]$ in DS, $[-2.4, +9.2]$ in DH): the coefficient is constrained by the data, but its sign is not determined.

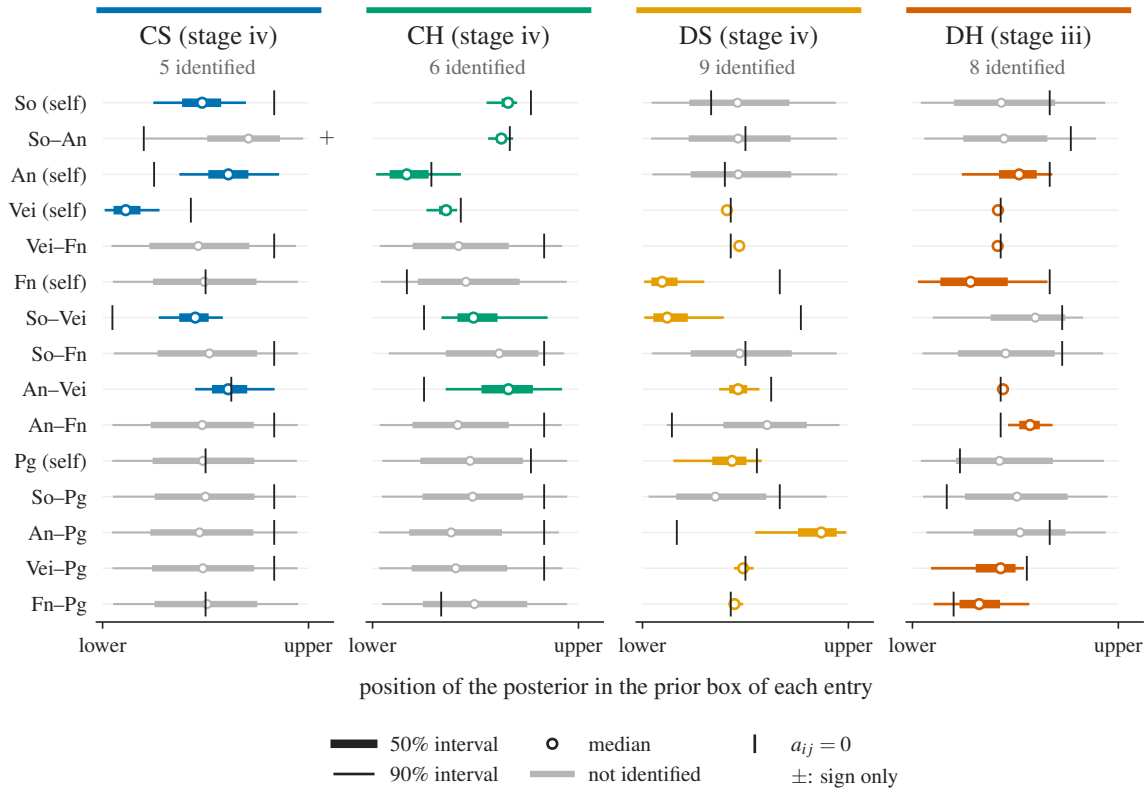


Figure 3.5: Marginal posteriors of the 15 entries at the final stage, three seeds pooled, drawn by their position in the prior box (thick: 50 %, thin: 90 % interval; dot: median; tick: $a_{ij} = 0$; grey: not identified). DH from stage (iii).

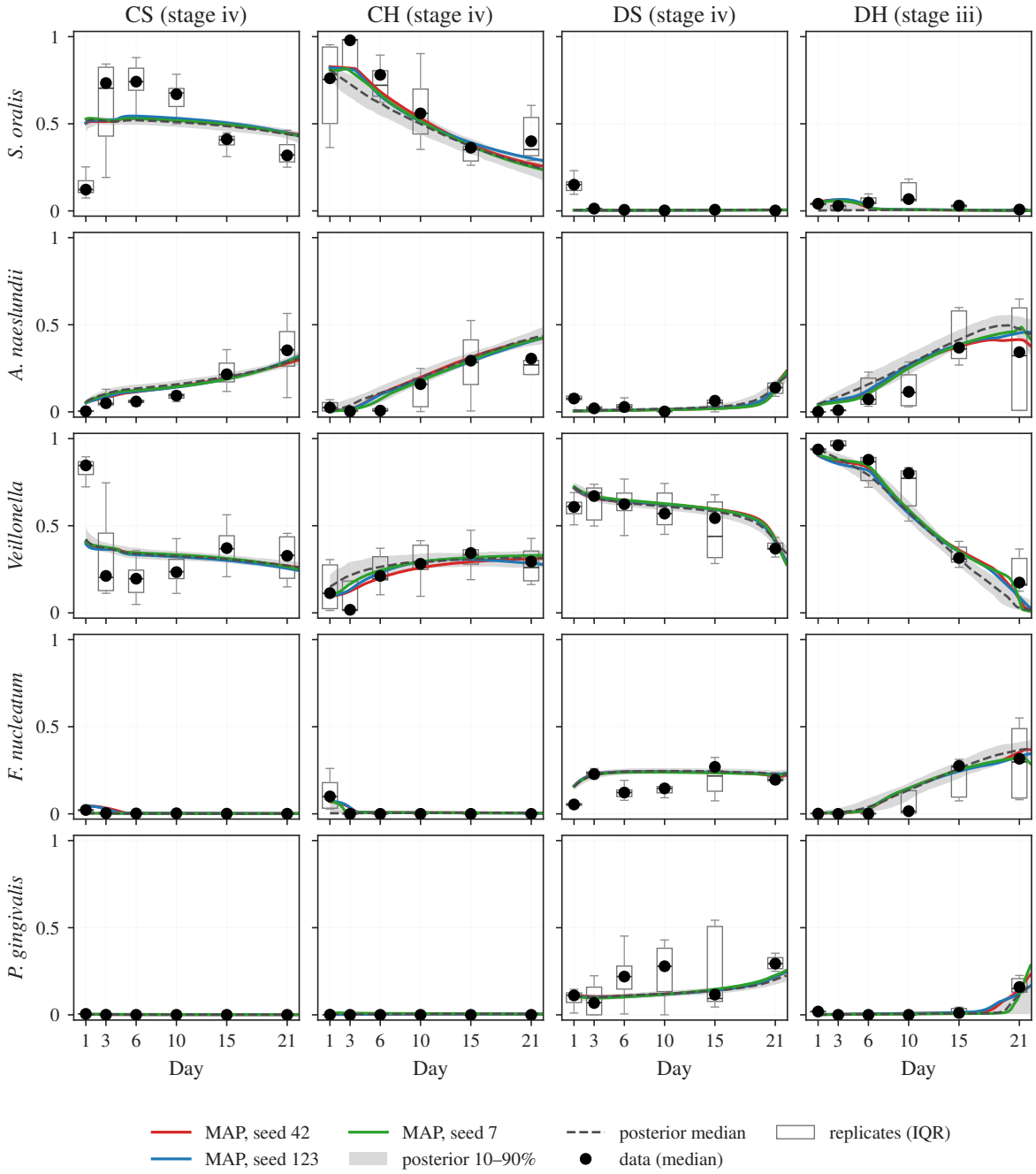


Figure 3.6: Viable species fractions computed with the best particle of each of the three seeds at the final stage (coloured lines), the 10–90 % band (grey) and median (dashed) of the posterior prediction pooled over the seeds, and the measured viable composition (dots: median; boxes: interquartile range of the replicates; day 1 is the initial condition). The maximum log-likelihoods of the three seeds lie within 0.6 of each other (−32.9 to −33.1 in CS, −26.5 to −27.1 in CH, −96.8 to −96.9 in DS, −51.4 to −51.8 in DH), and their composition RMSEs are 0.090–0.093, 0.060–0.066, 0.057 and 0.073–0.076. DH from stage (iii).

3.3.4 Recovery test

To test the classification, I generated synthetic data with known coefficients at the measured time points and repeated the inference. The classification of the 15 entries as identified or not was recovered for 14 (CS), 15 (CH), 14 (DS) and 15 (DH) entries. The intervals are less reliable than the classification: in CH the true value lies inside the 90 % interval for only 2 of the 6 identified entries. The posterior intervals of this chapter are therefore read as a lower bound on the uncertainty (Figure 3.7).

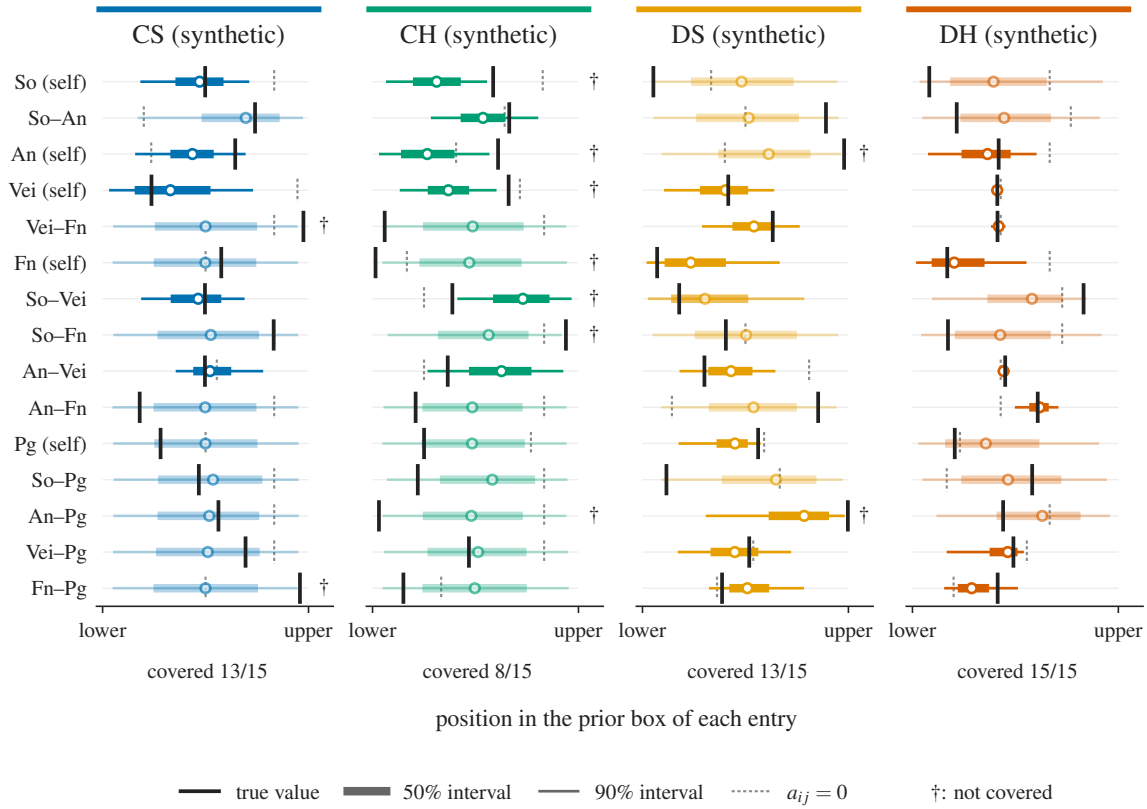


Figure 3.7: Recovery test: posterior of each entry for synthetic data generated with known coefficients (vertical bar: true value; thick and thin: 50 % and 90 % interval), drawn by the position in the prior box. Below each panel: number of entries whose true value lies inside the 90 % interval.

3.4 Checks Against Data Not Used in the Inference

The pH measured in the flow reactor was not used in the inference (Figure 3.8b,c). A linear regression of the measured pH on the measured species fractions (intercept and four fractions, *Vei* as reference) is fitted once on the measured data, the continuous pH records of CH and DH (mean of five reactors, about 14,000 values per condition) with the composition interpolated between the sampling days ($R^2 = 0.88$), and then applied, unchanged, to the posterior predictive composition, so the check tests the predicted composition, not a model of pH. In DH the pH predicted in this way follows the measured rise from 6.4 to 7.0 ($R^2 = 0.71$, RMSE 0.097; 85 % of the measured values lie inside the

90 % band). In CH it matches the level of the measured pH (RMSE 0.082) but not its variation ($R^2 < 0$; 44 % inside the band).

The calibrated model can also be used in silico. Removing *Fn* from the dysbiotic flow-reactor community reduces the late increase of *Pg* to about a quarter: the ratio of the *Pg* fraction on day 21 to day 15 falls from about 12 to 3.0 (Figure 3.8a). Setting only $a_{45} = 0$ removes most of the increase, so the prediction rests mainly on the *Fn*–*Pg* coefficient, and it can be tested directly by a co-culture without *Fn*. The size of the increase itself is not determined by the data: the MAPs of the three seeds give ratios of 9.6 to 18.0 (measured 14.9), and the 90 % interval spans 0.6 to 32. Under static cultivation the model gives a smaller late increase than measured (1.4–1.5 against 2.5); without *Fn* the ratio is 1.0, and $a_{45} = 0$ leaves it unchanged.

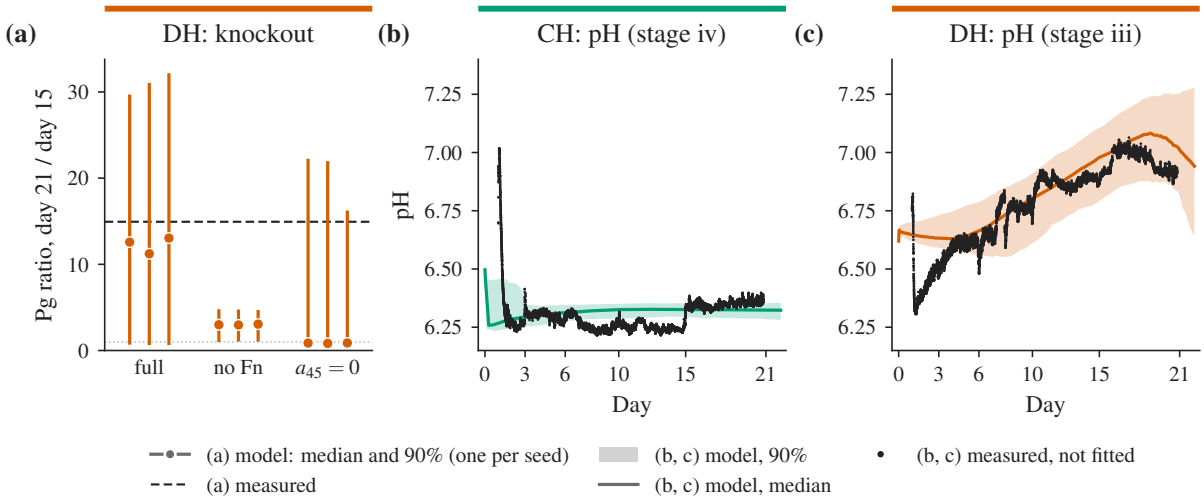


Figure 3.8: Checks against data and predictions not used in the inference. (a) DH: ratio of the *Pg* fraction on day 21 to day 15 for the calibrated model, with *Fn* removed, and with $a_{45} = 0$ (posterior median and 90 % interval, one bar per seed; dashed: measured). (b,c) pH predicted from the posterior composition through a regression fitted once on the measured pH and composition (band: central 90 % interval), against the measured pH. DH from stage (iii).

3.5 Discussion

What the calibration supports. The data determine a small part of the matrix: in the commensal conditions the interactions among the early colonisers, and in the dysbiotic conditions also some interactions of *Fn* and *Pg*. The signs that are determined agree with the documented pathways where both exist, which is consistency, not proof of a mechanism. The repeated stages with three seeds and the acceptance criteria were necessary: several runs that looked converged did not meet them.

Limitations. The likelihood treats the five fractions as independent although they sum to one, so the posterior intervals are too narrow (Section 3.3.4). The community live

fraction does not determine the species-specific viabilities ψ_i (in the model they span 0.03–0.96), so the living fractions of individual species are not predictions of the calibrated model. The interaction coefficients are effective quantities of this model and data set, and the recovery test shows that their intervals understate the uncertainty. The commensal–dysbiotic comparison is confounded by the strain differences. In CS the model cannot match the composition and the live fraction at the same time. In DH the final stage split across seeds into different posterior modes, so DH is reported from stage (iii), and the size of the late *Pg* increase is not determined by the data. The calibration covers a material point; the spatial structure and the mechanics are not part of the inference. Because the likelihood uses fractions only, the absolute amount of biomass is a prediction of the model, and its posterior spread has to be carried into the coupling with the finite element model of Chapter 4. The five-species runs of Chapter 4 use the final CH MAP of this chapter.

3.6 From In-Vitro to In-Vivo Data: a Metabolic Sign Prior

The calibration above rests on rich in-vitro data: five species, six time points, four controlled conditions. Clinical data are much sparser. This section shows, in condensed form, how the same type of model can still be identified there when the signs of the interactions are fixed beforehand from metabolic models. The work was done during an internship at NIFE together with S. P. Szafrański and is summarised here because it completes the calibration side of the thesis; the full analysis is available as an internal report.

Data. Dieckow et al. [53] characterised early biofilm formation on implant abutments in 12 patients. The analysis uses 10 of them, each sampled at three weekly time points. The 16S rRNA amplicon reads were aggregated into 10 class-level guilds and normalised to the simplex, $\sum_i \phi_i = 1$. With 10 guilds and three time points, the first of which is the initial condition, there are 20 observations per patient against up to 55 entries of a symmetric interaction matrix, so the inverse problem cannot be solved without additional information.

Sign prior. The prior of Section 2.7 is assembled in three layers: experimentally documented production and use of metabolites from eHOMD [54] and Szafrański et al. [55] (34 constrained guild pairs); parsimonious FBA of one AGORA2 model per guild [31] (35 pairs); and community FBA with MICOM [30] (72 pairs). The directed signals are symmetrised for the symmetric model, and the penalty acts only on coefficients of the wrong sign.

Model and test. The replicator form of Section 2.4 is fitted with one interaction matrix $\mathbf{A}$ shared by all patients and a growth vector $\mathbf{b}^{(p)}$ per patient. It is tested by leave-one-patient-out cross-validation: $\mathbf{A}$ is fitted on nine patients and frozen, the held-out patient's $\mathbf{b}^{(p)}$ is fitted from the first week only, and the second and third weeks are predicted.

Table 3.5 lists the prediction errors, and Figures 3.9 and 3.10 show the predictions of the best model.

model	prior	sign agreement	LOO-RMSE
replicator, symmetric $\mathbf{A}$	none	–	0.0595
replicator, symmetric $\mathbf{A}$	eHOMD + pFBA	94%	0.0516
replicator, symmetric $\mathbf{A}$	+ MICOM	100%	0.0502
gLV, asymmetric	none	–	0.0588
gLV, asymmetric	eHOMD + pFBA	97%	0.0490
Bayesian gLV (MDSINE2 [56])	sparsity	–	0.0552

Table 3.5: Leave-one-patient-out prediction of the in-vivo guild compositions (10 patients of Dieckow et al.). Sign agreement: share of the constrained pairs whose fitted sign matches the prior. The persistence null model has a Bray–Curtis dissimilarity of 0.281; the best model reaches 0.147.

Results.

- The sign prior improves the symmetric model step by step, from 0.0595 without prior to 0.0502 with all three layers. The asymmetric gLV with the first two layers is slightly better (0.0490); its directed coefficients add freedom that the symmetric model does not have. The Bayesian gLV collapses to $\mathbf{A} \approx 0$ with three time points.
- Only the sign carries over from the metabolic models. A prior that ties A_{ij} to the magnitude of the predicted fluxes recovers the correct sign for only 4 to 11 of 70 pairs. Fitted without any prior, the symmetric model still recovers 11 of the 14 cross-feeding signs predicted by AGORA2 (78.6 %, against 37.7 % under a permutation null, $p = 4 \times 10^{-4}$); the asymmetric gLV shows no such agreement ($p = 0.96$).

Limits. With 10 patients the gain of the community-FBA layer over the first two is only marginally significant ($p \approx 0.07$). The symmetric matrix cannot represent directed effects, which is where the asymmetric gLV gains its edge. For the rest of this thesis the result that matters is a narrower one: the signs of the interaction matrix are the part of the model that is supported by independent evidence, in vitro and in vivo.

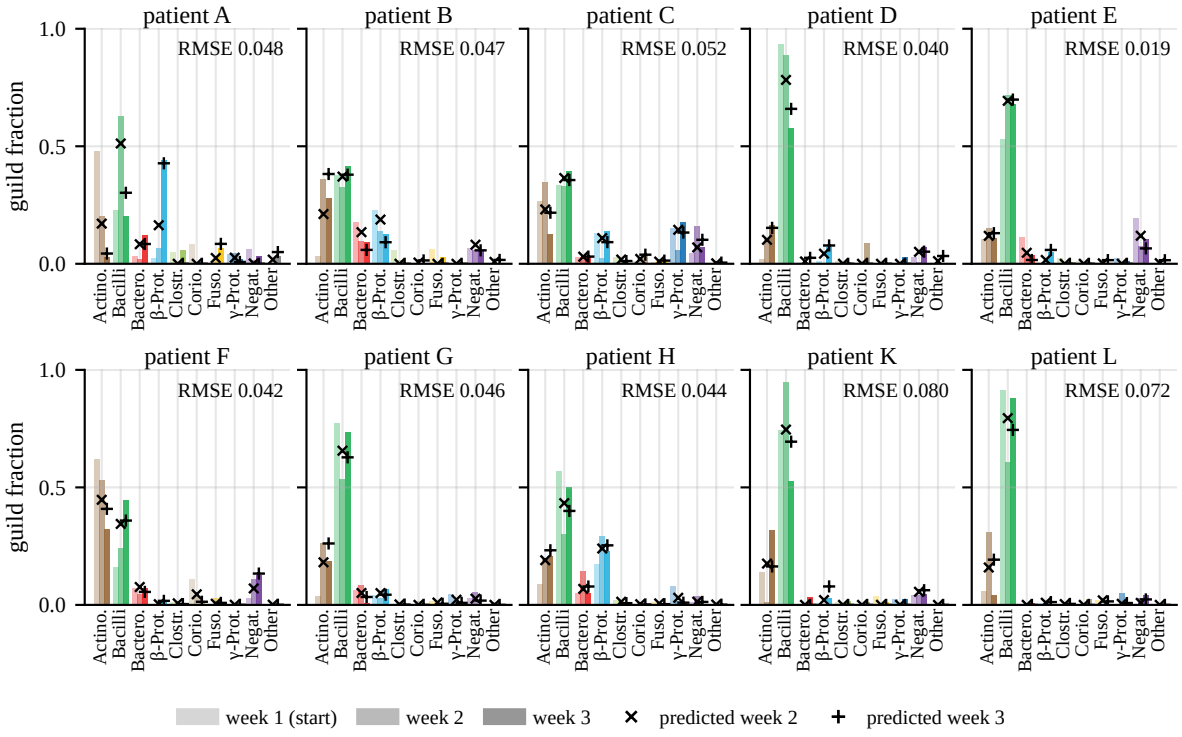


Figure 3.9: Leave-one-patient-out prediction for the ten patients with the best model of Table 3.5 (gLV with the eHOMD and pFBA sign prior). Bars: observed guild fractions in weeks 1 to 3 (light to dark); markers: weeks 2 and 3 predicted with $\mathbf{A}$ fitted on the other nine patients and the patient's growth vector fitted on week 1 only. RMSE per patient.

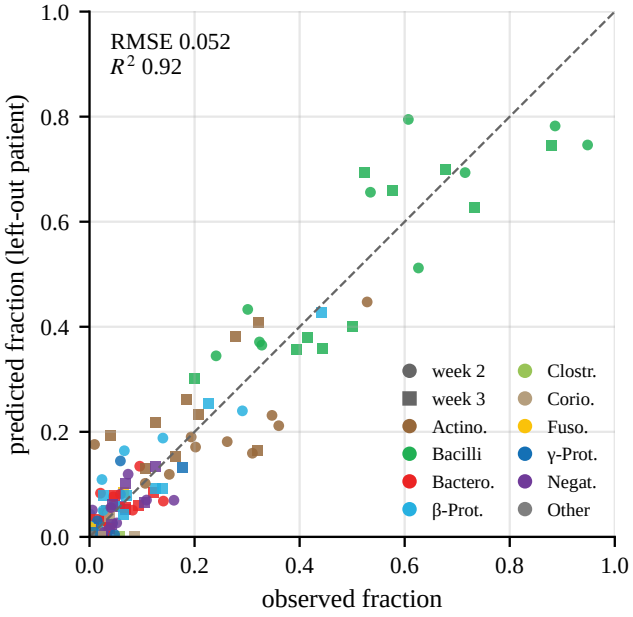


Figure 3.10: Predicted against observed guild fractions of Figure 3.9, all patients, guilds and both predicted weeks (200 values). The pooled RMSE (0.052) is slightly above the mean of the per-patient values (0.049).

4 Coupling biofilm growth into an existing FEM element

4.1 Scope and contribution

This chapter contains the second and third contributions of the thesis (Section 1.3): the growth law $\mathbf{F}_g = \alpha\mathbf{I}$ as an ANSYS user material, verified and coupled to the biofilm field of the IKM element (§4.2–4.5), and the point model of Klempt et al. [24] at every Gauss point (§4.6). The calibrated five-species model of Chapter 3 is shown once in the element (§4.7.2).

4.2 The constitutive model

Growth enters multiplicatively, $\mathbf{F} = \mathbf{F}_e\mathbf{F}_v\mathbf{F}_g$ with $\mathbf{F}_g = \alpha\mathbf{I}$ and $\alpha(0) = 1$, as in Klempt et al. [32]; $\alpha - 1$ is the growth, and $\alpha = 1$ means no growth. The elastic response is compressible Mooney–Rivlin, reducing to neo-Hookean at $C_{01} = 0$. A viscous branch, a deviatoric Newtonian dashpot carrying $\mathbf{F}_v$, is implemented but switched off ($\eta = 0$) in every result of this chapter, as in the paper.

Written out in the order the implementation evaluates it, a step takes the total deformation gradient $\mathbf{F}$ and the viscous state $\mathbf{F}_v^n$ and returns the Cauchy stress and $\mathbf{F}_v^{n+1}$:

1. the mechanical part removes the growth, $\mathbf{F}_{\mathrm{mech}} = \mathbf{F}\,\mathbf{F}_g^{-1}$, and a trial elastic gradient follows from the viscous state at the start of the step, $\mathbf{F}_e^{\mathrm{tr}} = \mathbf{F}_{\mathrm{mech}}(\mathbf{F}_v^n)^{-1}$;

2. with $\mathbf{b}_e = \mathbf{F}_e\mathbf{F}_e^{\mathsf{T}}$ and $J_e = \det\mathbf{F}_e$, the deviatoric Kirchhoff stress follows from $W = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{1}{D_1}(J_e - 1)^2$, with the isochoric invariants $\bar{I}_1$, $\bar{I}_2$ of $\mathbf{b}_e$;

3. the viscous state advances by one step of the deviatoric flow rule,

$$\mathbf{F}_v^{n+1} = \Big(\mathbf{I} + \frac{\Delta t}{2\,\eta\,J_e}\,\boldsymbol{\tau}_{\mathrm{dev}}\Big)\mathbf{F}_v^n, \tag{4.1}$$

   with $\boldsymbol{\tau}_{\mathrm{dev}}$ taken at the trial state;

4. the stress is recomputed at $\mathbf{F}_v^{n+1}$, the volumetric term $\frac{2}{D_1}(J_e - 1)$ is added, and the result is returned in the solver's Voigt ordering.

A body that can grow freely therefore stays stress free; stress appears only where growth is constrained, by boundary conditions or by neighbouring material that grows less.

Stiffness. Klempt et al. weight the elastic energy with ϕ^2, so that the void, $\phi = 0$, has no stiffness, and give $\mu = 3.3557$ Pa ($E = 10$ Pa, $\nu = 0.49$) in their Table 2. The routine supports this form,

$$E(\phi) = (\phi^2 + f)\,E_{\mathrm{bio}}, \qquad \nu = \nu_{\mathrm{bio}}, \tag{4.2}$$

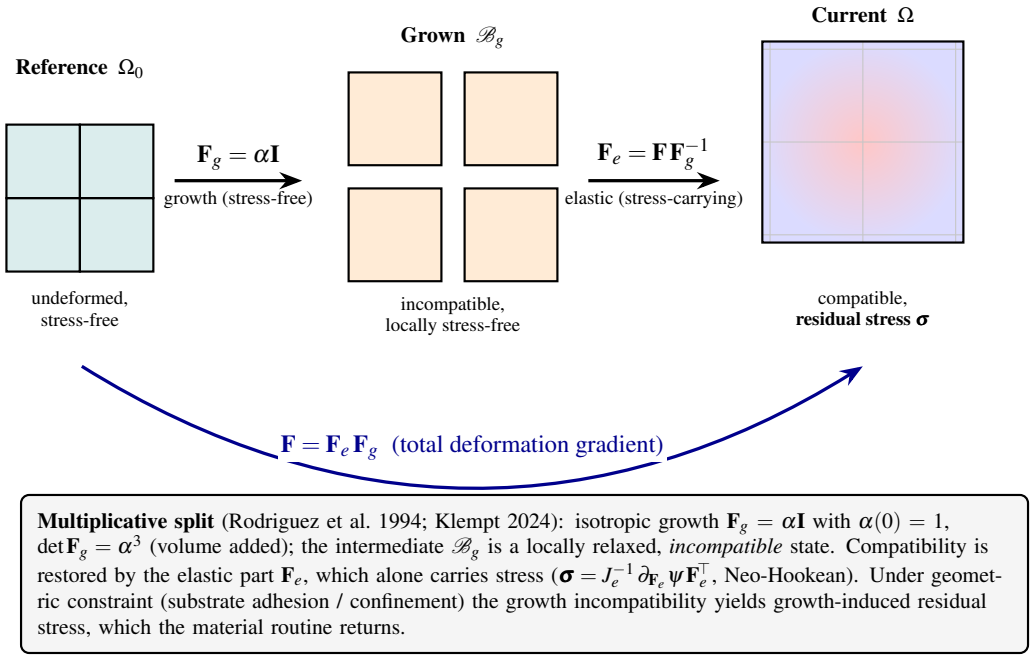


Figure 4.1: The configurations the decomposition introduces. Growth carries the reference state to an intermediate one that is stress free but generally incompatible; the elastic part restores compatibility, and this is where the residual stress comes from.

where f is a small floor that keeps the void solvable; it is not in the paper, whose void has no stiffness. It is tested to reproduce the plain law fed the same E at every ϕ and to scale the stress by ϕ^2. The paper's material is incompressible, with $\det\mathbf{F}_e = 1$ enforced by a Lagrange multiplier, and $\nu = 0.49$ enters there only through μ; here the constraint is approximated by $\nu = 0.49$. The element, SOLID185 in its default $\bar{\mathbf{B}}$ formulation, integrates over the $2 \times 2 \times 2$ Gauss points of each element, where it also stores its fields, and uses one dilatation per element, the same idea as the H1P0 element of the paper. A Python solution of the same problem shows that with this formulation the mean stress changes by less than 10 % between $\nu = 0.49$ and 0.499, while full integration would overestimate it several times (§4.7).

4.3 Implementation

The law is implemented as an ANSYS `USERMAT`.[1] It is delivered to the partner framework as a routine with the same argument shape as the framework's own elastic law: per-phase (E, ν) blended by the local biofilm fraction, returning Cauchy stress and consistent tangent. The delivered routine is an adapter around the verified core. This was tested: fed (E, ν), it reproduces the core fed the (C_{10}, C_{01}, D_1) those map to, at zero tolerance. It uses no solver includes and no common blocks, so it does not depend on the ANSYS release.

The consistent tangent is obtained by perturbing $\mathbf{F}$ rather than in closed form. A

[1] `ansys_usermat/usermat_biofilm.f` in the accompanying repository.

tangent that is slightly wrong does not give a wrong answer but a Newton iteration that converges slowly or not at all, which is hard to diagnose inside someone else's framework. I therefore compared it with forward-mode automatic differentiation over the range the routine is expected to see: stiffness over three decades, Poisson ratios up to 0.49, growth up to $\alpha - 1 = 0.35$, both material paths, elastic and viscous. Over 648 combinations the relative difference has median 1.2×10^{-7} and worst case 1.1×10^{-6}, with no part of the range degrading.

4.4 Verification of the growth law

The checks of this section are verification: they test whether the implementation solves the equations it states. Validation against a measured stress is not attempted (Section 4.4.3).

4.4.1 Growth kinematics against closed forms

Two growth cases have closed forms. Traction-free isotropic growth expands the body to $\mathbf{F} = \alpha\mathbf{I}$ with zero stress. Fully constrained growth gives an isotropic elastic deformation, so the deviatoric stress vanishes and only

$$\boldsymbol{\sigma} = \frac{2}{D_1}\,(J_e - 1)\,\mathbf{I}, \qquad J_e = \alpha^{-3} \tag{4.3}$$

remains, independently of the shear constants and of the viscosity.

In ANSYS 2022 R2, one SOLID185 element at $\alpha = 1.05$, with $C_{10} = 2 \times 10^{-4}$ MPa, $C_{01} = 0$ and $D_1 = 5 \times 10^{3}$ MPa^{-1} (shear and bulk modulus both 4×10^{-4} MPa, $\nu = 0.125$; this deck sets no unit system, so stresses come out in the unit of C_{10}, read here as MPa as in the framework's decks), returns a largest stress component of 1.9×10^{-11} MPa under free growth, seven decades below the constrained answer. Under full constraint it returns $\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = -1.0193 \times 10^{-4}$ MPa, with shear components at machine noise. The free case is needed as well: full constraint can hide a sign error in $\mathbf{F}_e = \mathbf{F}\mathbf{F}_g^{-1}$, because the element cannot move, and only the free case exposes it.

The constrained value is not equation (4.3) alone. It equals that term plus a spherical term that I predicted in closed form from the implementation (§4.4.3), to every digit ANSYS prints, with the extra term making up 47 % of the reported stress for this material ($\nu = 0.125$) (Figure 4.2).

4.4.2 The delivered routine on a larger mesh

A two-layer curved shell of 12 240 elements, a non-growing substrate under a growth layer, was solved twice in ANSYS: once with the verified core and once with the delivered routine. The von Mises fields agree to every digit reported (minimum 4.3523×10^{-9}, maximum 1.1862×10^{-5}, mean 9.1443×10^{-6}, all in MPa), with no solver errors. This is a consistency check between two builds, not a comparison with a closed form.

The same model showed a meshing error that I corrected. The two layers had been meshed separately and joined by merging coincident nodes, and on the curved sweeps only 63 of 5 673 interface nodes coincided, three axial lines. The growth layer was held along

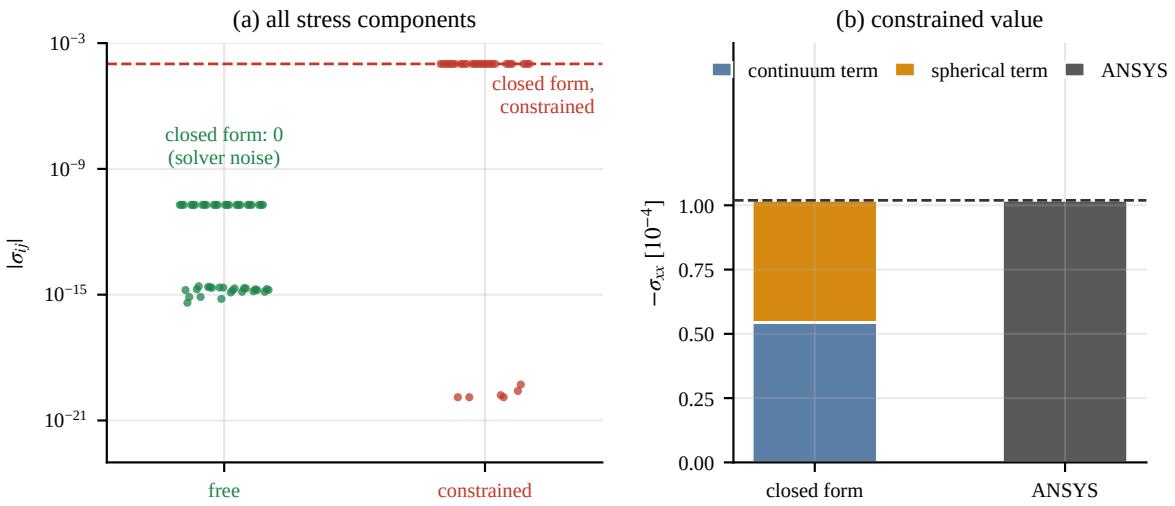


Figure 4.2: One SOLID185 element, $\alpha = 1.05$, the same material, differing only in the boundary conditions. (a) Every stress component of both solves (8 nodes, 6 components each); the closed form is zero for the free case, so the values there are solver noise, and the 16 constrained components that are exactly zero cannot be shown on the log axis. (b) The constrained value decomposed into the continuum term and the spherical term of §4.4.3: their sum, -1.019276×10^{-4}, agrees with the solver's -1.0193×10^{-4} to every printed digit; the spherical term is 47 % of it and does not enter the von Mises stress.

those lines and free between them; the "two-lobe" bulge first read as a buckling mode was the layer lifting between them. With one shared interface mesh the bulge is near uniform and converges at close to second order: the θ-profile error against the finest mesh falls from 2.7 % to 0.58 % over 2.4k, 19k and 150k elements, each step halving the element size, an observed order of 1.9, and every growth value up to $\alpha - 1 = 0.2$ converges (Figure 4.3).

The finite-difference tangent holds up inside the solver as well (Figure 4.4): every substep converges in one to four equilibrium iterations, and the automatic time stepping enlarges the increment at every opportunity, from 0.5 to 1.6875 of a pseudo-time that runs to 5 (the material has no viscosity here, so time has no physical unit), without a single cut-back.

4.4.3 What these checks establish, and what they do not

The isochoric split in the implementation applies $J^{-2/3}$ to the subtracted trace but not to the tensor beside it. The coded and the correct forms agree only at $J = 1$, and growth moves J_e away from 1 by construction. The difference is purely spherical for any deformation, so it is a pressure error: the deviatoric stress, and with it the von Mises stress reported in this work, is unaffected at a given deformation. It is the spherical term of Figure 4.2.

Two of the usual checks cannot see such an error. The free-growth case lands exactly on $J_e = 1$, where both forms agree, and reference values generated by calling the implementation record its behaviour rather than test it. Only the comparison with an independently derived closed form finds it, which is why equation (4.3) is derived from the

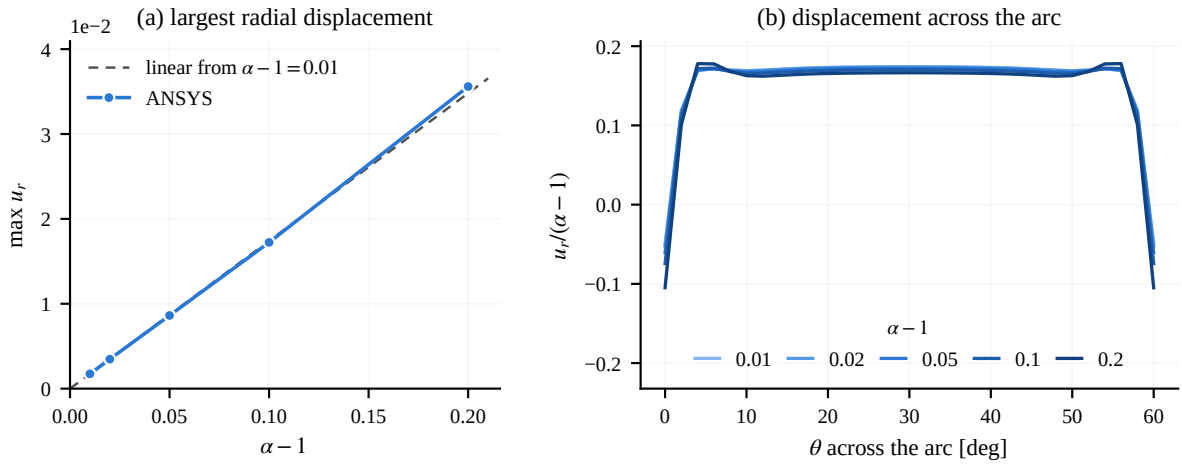


Figure 4.3: The bonded two-layer shell (2.4k elements) under prescribed growth $\alpha-1 = 0.01$ to 0.2; all runs reach the end time. (a) Largest radial displacement against $\alpha - 1$, with the straight line through the $\alpha - 1 = 0.01$ run. (b) The radial displacement across the arc divided by $\alpha - 1$: in the middle of the arc the curves differ from the $\alpha - 1 = 0.01$ curve by 1 % at $\alpha - 1 = 0.05$ and 4 % at 0.2; the larger differences are at the ends of the arc.

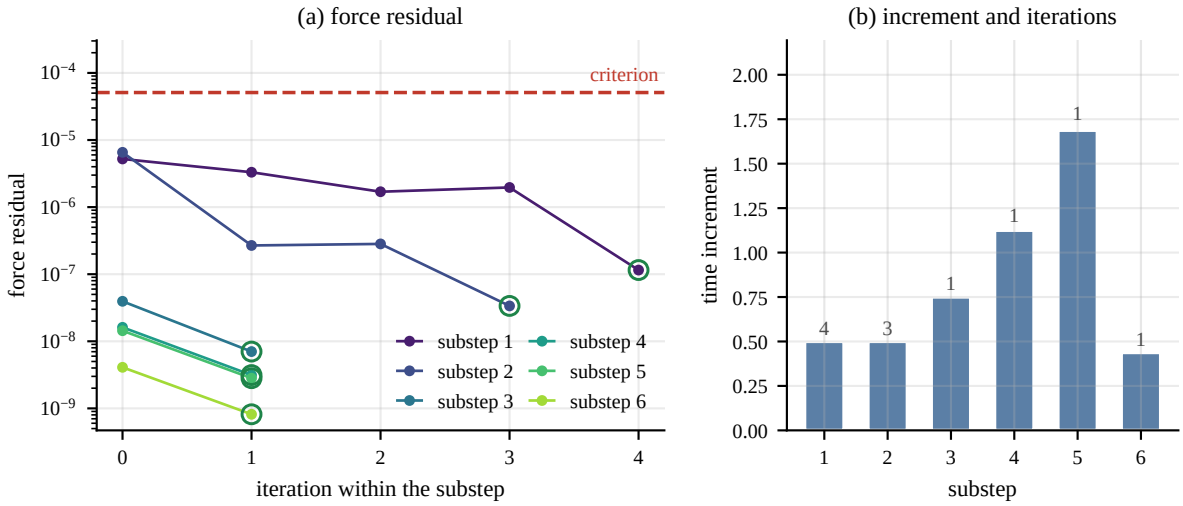


Figure 4.4: Convergence of the 12 240-element solve, read from the solver's listing: the load step finished in 6 substeps. (a) Force residual against the equilibrium iteration, one curve per substep (circle: converged), with the ANSYS criterion dashed. (b) The time increment of each substep and, above the bars, the iterations it needed. The automatic time stepping only enlarged the increment; the last one is short because the load step ends there.

continuum statement and not from the code. A wrong pressure can still reach the von Mises stress indirectly, through the displacement field of an equilibrium solve; this route is not measured here. It shrinks as $\nu \to 0.5$: at the paper's $\nu = 0.49$ the extra term is 1.3 % of the constrained pressure, against 31 % at $\nu = 0.3$.

4.5 Coupling to the element's biofilm field

4.5.1 The framework and the growth equation

The partner framework solves the mechanics in ANSYS and its transport fields in its own routines, alternating substep by substep (Figure 4.6). The biofilm field follows Klempt et al. [32], Eq. 34 and 36:

$$\dot{\phi} = \beta\,\Delta\phi + k_\alpha\,\alpha - r\,\frac{c}{k+c}\,\nabla\phi\cdot\frac{\nabla c}{\|\nabla c\|}, \qquad \dot{\alpha} = k_\alpha\,\phi, \tag{4.4}$$

with diffusion, a local source, and a front term $-r\frac{c}{k+c}\,\mathbf{n}_c\cdot\nabla\phi_{3D}$ that moves the biofilm boundary up the nutrient gradient. The element computes ϕ at its Gauss points; I leave it unchanged. Its Laplacian and the gradients of ϕ and c are approximated from the values at the neighbouring Gauss points, by a second-order Taylor expansion fitted in the weighted least-squares sense, the Neighbored Element Method [57, 58]. The field is updated explicitly, which limits the time step. Rudolf et al. [58] give $\Delta t \le h^2/(2\beta)$ for the element size h, which is the limit of the three-point stencil in one dimension; the seven-point stencil in three dimensions has $\lambda = \beta\Delta t/h^2 \le 1/6$. A Bloch analysis of the NEM stencil on the lattice of the Gauss points (Appendix D) gives the limit of this element: 0.18 for the unweighted fit, because the lattice is not uniform, and lower on coarse meshes, because the width $\beta^* = 0.2$ mm of the Gaussian weight is fixed in millimetres. On 16^3 and 24^3 the limit also depends on which points of the last, equidistant shell enter the fit of 30 neighbours, and lies between 0.122 and 0.140 on 16^3 and between 0.122 and 0.162 on 24^3; on 8^3 it is 0.062. Runs to $T^* = 5$ confirm this: on 8^3 they stay smooth at $\lambda = 0.048$ and stop at 0.080, on 16^3 at 0.136 and 0.144, and on 24^3 at 0.130 and 0.158 (Figure 4.5). Runs to $T^* = 1.1$ are too short to show an instability near the limit. The results of this chapter use $\lambda \le 0.04$ on 8^3 and $\lambda \le 0.072$ on 24^3, below the band, and $\lambda \le 0.128$ on 16^3, inside the band but below 0.136, where the runs to $T^* = 5$ on 16^3 still stay smooth. The time-step series below has two further cases on 24^3: at $\Delta t = 0.05$ ($\lambda = 0.144$, inside the band) the seed growth follows the first-order trend of the other steps, and at $\Delta t = 0.1$ ($\lambda = 0.288$) the run gave no result, so the 24^3 series starts at $\Delta t = 0.05$.

What I add sits in the material routine, at every Gauss point: take ϕ from the field, integrate the second equation of (4.4),

$$\alpha_{n+1} = \alpha_n + k_\alpha\,\phi\,\Delta t, \tag{4.5}$$

and pass α to the growth law. I use the paper's $k_\alpha = 10^{-3}$ per unit of normalised time T^*, in the field and in (4.5) alike, and the deck time is T^*.

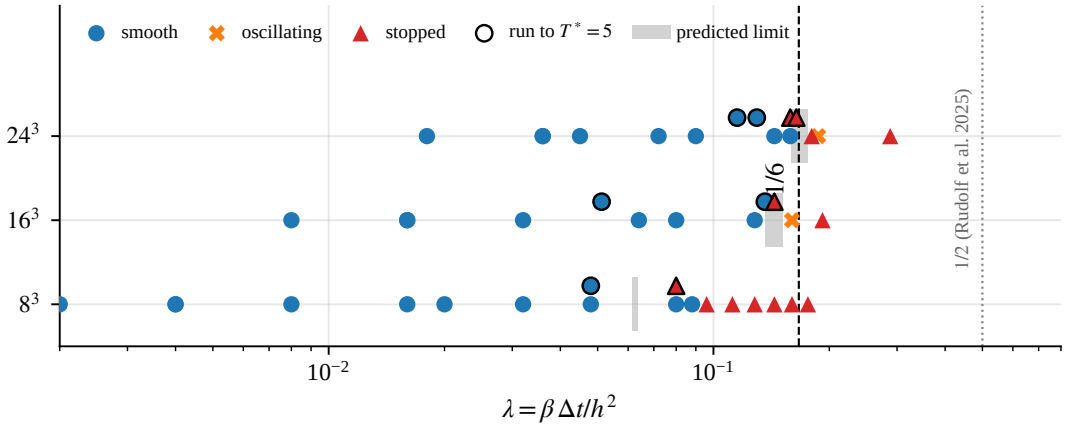


Figure 4.5: Runs of the element against $\lambda = \beta \Delta t / h^2$. Smooth: the run reaches the end time with $\alpha - 1 \geq 0$ and no oscillation between neighbouring elements; oscillating: it reaches the end time, but $\alpha - 1$ alternates in sign; stopped: elements collapse before the end time. End time $T^* = 1.1$; black edge: the two 8^3 runs to $T^* = 5$. Grey: the limit of the NEM stencil from the Bloch analysis of Appendix D (30 to 32 neighbours). Dashed: 1/6, the limit of the seven-point stencil in three dimensions; dotted: 1/2, the condition given by Rudolf et al. [58].

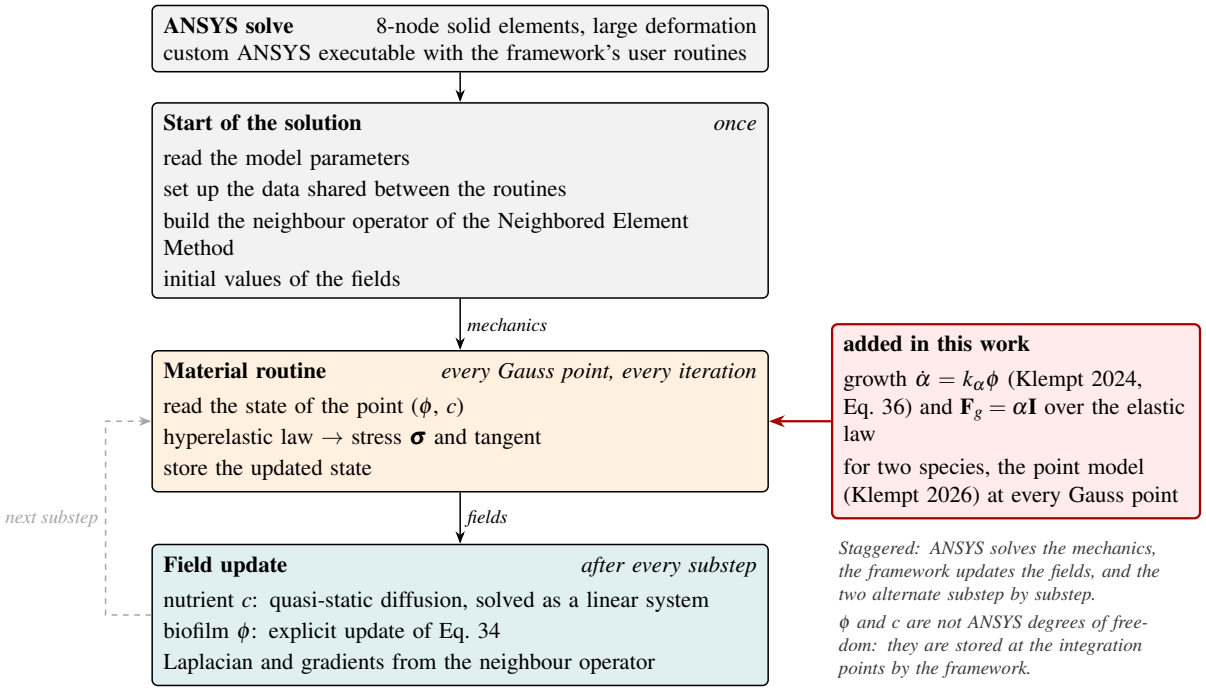


Figure 4.6: Solution loop of the partner framework, traced from its source, and the point at which the growth law of this chapter attaches.

The paper prints a second form of this law (Section 2.2). Table 1, step 3a, of Klempt et al. 2024 integrates α from the change of ϕ, $(\alpha_{n+1}-\alpha_n)/((1+\alpha_{n+1})\Delta t) = k_\alpha(\phi_{n+1}-\phi_n)/\Delta t$ (as printed, with k_α/η_α read as k_α), while its text states that α follows Eq. 36. The two differ in where growth happens. On the reproduced field of test case 4.1, at $T^* = 1$, biofilm present from the start reaches $\alpha - 1 = 9.9 \times 10^{-4}$ with Eq. 36 and stays at zero with Table 1, and newly colonised biofilm reaches 5.4×10^{-4} and 8.5×10^{-4} (Figure 4.7). A seed that is full from the start, as in the runs of this chapter, grows only under Eq. 36. I follow the text, Eq. 36, as does the partner's element.

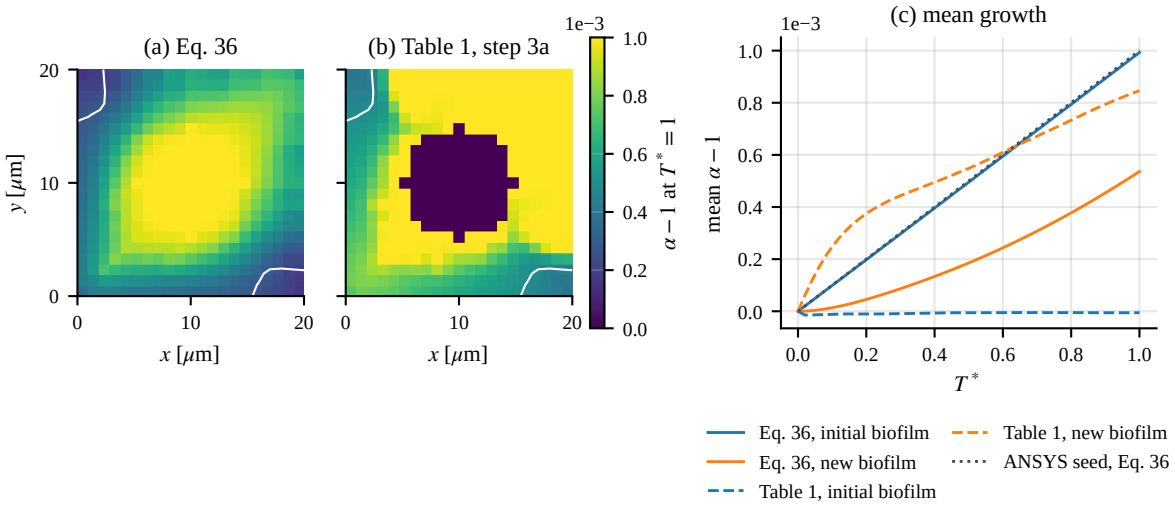


Figure 4.7: The two growth laws of Klempt et al. 2024 on the reproduced field of test case 4.1 (Python), $k_\alpha = 10^{-3}$. (a) Equation 36 and (b) Table 1, step 3a: $\alpha - 1$ at $T^* = 1$ on the mid-plane $z = 10$ μm (white: $\phi = 0.5$). (c) Mean over biofilm present from the start and over newly colonised biofilm, with the ANSYS seed ($\phi = 1$) under equation 36.

4.5.2 Growth counted once per increment

ANSYS calls the material routine several times per substep, once per equilibrium iteration and once more for output, and hands back the converged state each time. Equation (4.5) must therefore start from the stored state variable and must not accumulate over iterations. A trace of every call (46 calls over 11 substeps in the first run) confirms that growth is counted once per increment, that each increment starts from the previous one's end value, and that α follows (4.5) exactly.

4.5.3 Four properties of the framework found along the way

1. *The element's own growth variable is a mean.* The element already integrates $\dot{\alpha} = k_\alpha \phi$ for each of its two species fields and uses their mean as the growth of the material. With one species the unused field stays at $\alpha = 1$, so the growth is halved. Whether the mean is intended is a question for the framework's author.

2. *Start values.* The void fraction is $1 - \sum \phi_i$, so start values must satisfy $\phi_1 + \phi_2 \leq 1$. The framework's example input starts both species at 1 in the seed; for one species I

start the second at 0.

3. *The first result set is not in equilibrium.* While the element sets up its operators it returns zero stress and tangent on its first calls, so the solver converges the first substep on a zero residual. The output written for that substep is not an equilibrium state and is excluded from every result below.

4. *The front term $-r\frac{c}{k+c}\,\mathbf{n}_c\cdot\nabla\phi_{3D}$ was switched off, and differs from the paper.* An input-reading slip sets the orientation weight of species 1 to zero (the weights of both species are read into the same variable, and the last one read is zero), so the front term of (4.4) never acts and the biofilm only diffuses. The term is also multiplied by $|\Delta\phi|$ where the paper has $\|\nabla\phi\|$. With the slip fixed, $|\Delta\phi|$ makes the explicit update unstable, and $\|\nabla\phi\|$ with a central gradient lets ϕ exceed 1 (cell Péclet number about 30). Where the nutrient is uniform the direction $\nabla c/\|\nabla c\|$ is undefined. The one-species checks of §4.7 do not depend on this term, so the example model there shows no spreading. For the two-species examples of Section 4.7.1 the term is on, with the slip fixed, the paper's $\|\nabla\phi\|$, the direction regularised as in the authors' implementation (F. Klempt, personal communication, October 2026) and a stabilised discretisation.

4.6 The point model at every Gauss point

The point model. Klempt et al. [24] describe each species i by its volume fraction ϕ_i and its living fraction ψ_i, with the void $\phi_0 = 1 - \sum_i \phi_i$:

$$\begin{aligned}(\eta_{\phi,i} + \eta_i\psi_i^2)\,\dot{\phi}_i + \eta_i\phi_i\psi_i\,\dot{\psi}_i &= c^*\psi_i\textstyle\sum_j A_{ij}\phi_j\psi_j - \gamma - (\text{barrier}),\\ \eta_i\,(\phi_i\psi_i\,\dot{\phi}_i + \phi_i^2\,\dot{\psi}_i) &= c^*\phi_i\textstyle\sum_j A_{ij}\phi_j\psi_j - \alpha^* b_i\,\psi_i - (\text{barrier}),\end{aligned} \tag{4.6}$$

where A_{ij} are the interactions, b_i the sensitivity to an antibiotic of concentration α^*, c^* the nutrient, γ the multiplier for $\sum_i \phi_i + \phi_0 = 1$, and the barrier terms keep ϕ_i, ψ_i in $(0, 1)$; η_i and $\eta_{\phi,i}$ are the dissipation parameters of the two terms of (2.12), equal to each other in Chapter 3. My implementation reproduces all fourteen numerical examples of the paper. Two printed details had to be read differently to do so: Eq. 17 as printed carries a $+\gamma$ that the figures do not show, and the four-species interaction matrix of Eq. 19 reproduces the figures without its prefactor $\frac{1}{2}$. The same model reproduces the figures of Fritsch et al. [59] within 0.013.

The point model as published. Klempt et al. formulate their model on a material point, with ϕ_0 as an auxiliary variable that enforces $\sum_i \phi_i + \phi_0 = 1$ and is not to be read as a physical void [24]. I therefore do not change the point model: its equations (4.6), its parameters and its constraint stay as published. It is called at every Gauss point with given inputs and returns its state, which is carried from one increment to the next as

state variables of the Gauss point. The field keeps its own amount ϕ_{3D}. Throughout, ϕ_{3D} denotes the field of Klempt et al. 2024 and ϕ_i the point model.

From the field to the point model. The element solves the nutrient c of Klempt et al. 2024, held at given sources and diffusing through the domain. As ϕ_{3D}, the nutrient is a field of the integration points themselves (the neighboured element method solves it on the Gauss points and the surface points), so the material routine receives the value at its own Gauss point directly, without an interpolation from nodes. The point model needs c^* at its own, much smaller time steps. Between the increments t_n and t_{n+1} I interpolate linearly,

$$c^*(t) = c_0^*\left[\tfrac{t_{n+1}-t}{\Delta t}\, c_n + \tfrac{t-t_n}{\Delta t}\, c_{n+1}\right], \qquad t_n \le t \le t_{n+1}, \tag{4.7}$$

with c_n, c_{n+1} the nutrient at the Gauss point and c_0^* the nutrient level of the point model's own examples. Klempt et al. 2026 already use a time-dependent c^* (their four-species case 3), so the point model accepts such an input without change. The field of Klempt et al. 2024 has no antibiotic. The element solves a second nutrient for its second species, which I do not use; I use this field for an antibiotic a, with its own sources and the same diffusion, and pass $\alpha^*(t) = \alpha_0^*\, a$ to the point model, interpolated as in (4.7). Because the element advances its fields after the material routine, the pair (c_n, c_{n+1}) is the one of the previous increment. A short first load step, one internal step of the element's field solver (0.0025 in T^*), solves the fields before the point model starts, so the point model begins with the solved nutrient (in the seed $c = 0.79$) and the living biomass of its start state already enters the first consumption.

From the point model to the field. Only living biomass consumes nutrient. The consumption of Eq. 35 of Klempt et al. 2024 is printed as $g\phi$; it is $g\phi_{3D}c$, and the printed form is a misprint. In general the uptake depends on the nutrient through a Monod-type factor $f(c) = c/(K_c + c)$; for $K_c \gg c$ this is c/K_c, and with $1/K_c$ absorbed in g it is the linear form $g\phi_{3D}c$ used here. The time derivative of Eq. 35 is kept. With the diffusivity of Table 2 of Klempt et al. 2024, $d = 10^{10}$ μm$^2/T^*$ on their 20 μm cube, converted like β to $d = 10^8$ mm$^2/T^*$ on the 2 mm cube, the nutrient adjusts within about $4 \cdot 10^{-8}\, T^*$, so the fields are solved quasi-steadily; with the time derivative and this d the seed values change by less than 0.005 (Python). The consumption follows the same table: $g = 10^8\, T^{*-1}$, so that $g\ell^2/d = 1$ with the half edge $\ell = 1$ mm. The antibiotic, carried by the element's second nutrient field, has the same diffusivity and uptake as the nutrient. Only a diffusion slower by about eight orders of magnitude would matter: with $d = 1$ mm$^2/T^*$, the value of the element's example input, the antibiotic reaches the seed more slowly and kills markedly less of it (in an earlier set-up with $\alpha_0^* = 50$, a living volume fraction of 0.65 instead of 0.20

in the mixed seed). I scale it by the living volume fraction of the point model,

$$\dot{c} = d\,\nabla^2 c - g\,\phi_{3D}\Big(\sum_{i=1}^{N} \bar{\phi}_i\Big)c, \qquad \bar{\phi}_i = \phi_i\psi_i, \tag{4.8}$$

in the notation of Klempt et al. 2026. The point model normalises its volume, so $\bar{\phi}_i \in [0,1]$ is the living volume fraction of species i at the point, and $\phi_{3D}\sum_i \bar{\phi}_i \le \phi_{3D}$ is the living volume fraction in the field. A dead biofilm consumes nothing; the void ϕ_0 has no living fraction and is not part of the sum. The antibiotic is taken up in the same way, $\dot{a} = d_a\nabla^2 a - g_a\phi_{3D}\big(\sum_i \bar{\phi}_i\big)a$. The growth law (4.5) is unchanged: the inflation α measures how much biomass has been produced, alive or dead. The spreading of ϕ_{3D} follows Eq. 34 of Klempt et al. 2024 with its front term $-r\frac{c}{k+c}\,\mathbf{n}_c\cdot\nabla\phi_{3D}$, which depends on c; this is the path by which the point model reaches the growth and the stress. An antibiotic acts on the stress only through it: it lowers ψ_i, so less nutrient is consumed and more reaches the front.

Time. Klempt et al. 2024 normalise time with a reference time chosen for the problem at hand. I choose $T^* = 1$ as the duration covered by the point model's data, so that both models run on the same time: the point model advances $s\,\Delta T^*$ per increment, with s the length of that window in its own units, not a fitted value. For the synthetic data of Fritsch et al. [59], case I (1000 steps of 10^{-4}), $s = 0.1$. With synthetic four-species data and this choice, a point model that keeps its own constraint reproduces the point model alone to 7×10^{-7}; scaled to the field amount instead, the error is 0.019, and a value of s fitted to the data comes out a third too low because it absorbs that error.

Start values. The point model starts at a Gauss point once ϕ_{3D} there reaches $\phi_{\min}$. Where a seed is placed, the start values are those of the seed (one species, or a mixture). A point that the spreading biofilm reaches later takes the living composition of its colonised neighbours: their amounts are divided in the ratio of their living volume fractions $\bar{\phi}_i = \phi_i\psi_i$, and the new point starts alive ($\psi_i = 0.999$, $\gamma = 0$). I chose this rule because the biofilm spreads through the growth of its living cells. The papers do not fix it; the alternative of starting from the seed composition at the nearer seed is compared in Section 4.7. The point model has no transport between points, so the composition moves with the biofilm only through this rule.

Assumptions. The coupling rests on choices that neither paper fixes, each with its sensitivity in Section 4.7: the linear interpolation of c^* and α^* in time; $\sum_i \bar{\phi}_i$ taken from the previous increment; the start values of newly reached points; the threshold $\phi_{\min}$; and, for the examples, the uptake of the antibiotic. The time link is no longer one of them.

The complete step. Algorithm 1 collects one increment of the coupled run. ANSYS solves the equilibrium $\operatorname{div}\boldsymbol{\sigma} = \mathbf{0}$ by Newton's method; the material routine turns the deformation at a Gauss point into a stress after removing the growth and advances the point model. Its state does not enter the stress, so it does not affect the Newton iteration.

Algorithm 1 One increment $t_n \to t_{n+1}$ of the coupled run

Require: at every Gauss point α_n, the point-model state $(\phi_i, \psi_i, \gamma)_n$ and the pair (c, a) seen in the previous increment; the element's fields ϕ_{3D}, c, a of t_n
Ensure: displacements, stresses and state of t_{n+1}
repeat {Newton iteration of ANSYS}
  for all Gauss points **do**
    $\alpha_{n+1} \leftarrow \alpha_n + k_\alpha\, \phi_{3D}\, \Delta t$ {Eq. (4.5)}
    $\mathbf{F}_e \leftarrow \mathbf{F}\,\mathbf{F}_g^{-1}$, $\mathbf{F}_g = \alpha_{n+1}\mathbf{I}$; stress and tangent from $W(\mathbf{F}_e)$ and $E(\phi_{3D})$
    if $\phi_{3D} \geq \phi_{\min}$ or the point model has started **then**
      if not yet advanced in this increment **then**
        (c_n, c_{n+1}), (a_n, a_{n+1}) from the previous and the current value at the Gauss point
        n_{sub} steps of $s\Delta t/n_{\text{sub}}$ of Eq. (4.6) with c^*, α^* from (4.7)
        store $(\phi_i, \psi_i, \gamma)_{n+1}$ and $\sum_i \bar{\phi}_i$
      end if
    end if
  end for
until the residual meets the convergence criterion
the element advances ϕ_{3D} (Eq. 34 with the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$), c and a (Eq. (4.8) with $\sum_i \bar{\phi}_i$)

4.7 Results

The one-species runs use ANSYS 2022 R2 on the framework's example model ($8^3 = 512$ elements) or refinements of it, one process, and the growth parameters of Klempt et al. 2024, Table 2: $k_\alpha = 10^{-3}\,T^{*-1}$, one species, the second field started at zero. The model is a 2 mm cube instead of the paper's 20 μm one. For the diffusion coefficient of (4.4) I use $\beta = 0.02$ mm$^2/T^*$, the value $\beta = 2$ of Klempt et al. 2024, Table 2, converted to the 2 mm cube. The conversion is an assumption: the paper gives no length and time scale, and its first author describes β as chosen for sensible results and dependent on the mesh. The framework's example input, $\beta = 10^{-4}$ mm$^2/T^*$, is reported as a sensitivity case. The remaining coefficients of the field and of the nutrient are those of the framework's example input (`/UNITS,MPA`, lengths in mm), not values from the paper: $r = 100$ mm$/T^*$, $k = 0.1$, $d = 1$ mm$^2/T^*$ and $g = 0.5\,T^{*-1}$; with the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ off (§4.5), r and k do not act. The one-species runs end at $T^* = 1.1$ and the two-species runs at $T^* = 1$, both with $\Delta t = 0.025$; halving the step changes the mean stress in the seed by about 1 %. The two checks against exact solutions below use the example input and $\Delta t = 0.1$, because they need a region without gradients. The two-species examples of Section 4.7.1 differ: they use the constants of the papers, the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ and the agreed

coupling, and their set-up is given there.

Exact solution inside ANSYS. Where the field has no gradients, (4.4) reduces to $\dot{\phi} = k_\alpha \alpha$, $\dot{\alpha} = k_\alpha \phi$, with the solution $\phi = \sinh(k_\alpha t)$, $\alpha = \cosh(k_\alpha t)$. ANSYS follows the exact solution of its own time stepping to 3×10^{-11} in α (round-off) and 1.1×10^{-7} in ϕ, about 10^{-4} relative, because the element still sees a small gradient from its neighbours (Figure 4.8).

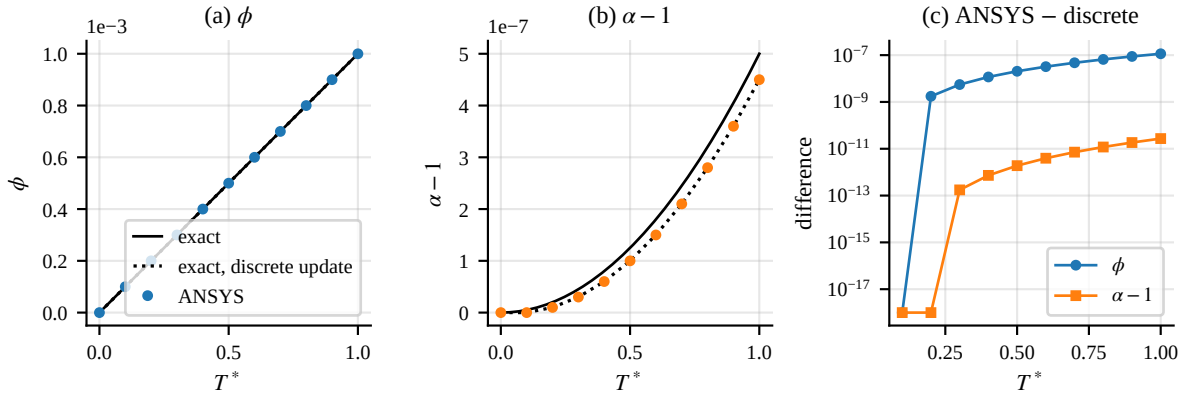


Figure 4.8: A gradient-free region of the coupled ANSYS run (dots): (a) ϕ and (b) $\alpha - 1$ against $\phi = \sinh(k_\alpha t)$, $\alpha - 1 = \cosh(k_\alpha t) - 1$ (solid) and the exact solution of the explicit update (dotted); (c) the difference to the latter.

The whole model. The element's own growth variable and equation (4.5) give two ways of computing the stress. Over all 512 elements the two von Mises fields have the same pattern, cosine similarity 1.0000. The element's variable gives 0.455 times the stress of (4.5), which is $\frac{1}{2} \times \frac{10}{11}$: the mean over both species fields, one of them unused, and a lag of one substep (Figure 4.9).

Stresses with the published stiffness. With the stiffness of Klempt et al. 2024, equation (4.2) with $E = 10$ Pa, $\nu = 0.49$ and a void floor $f = 10^{-3}$, $\alpha - 1$ in a monitored element inside the seed reaches 8.9×10^{-4} at $T^* = 1.1$ on 16^3 elements (Figure 4.10). This is less than $k_\alpha \phi t = 1.1 \times 10^{-3}$, because ϕ diffuses out of the seed. Klempt et al. [32, 33] report pressure inside the biofilm and a ring of tension at its edge. Grouped by distance from the seed, the mean stress is -2.5×10^{-4} Pa in the seed, $+6.0 \times 10^{-5}$ Pa in the first layer of elements around it and $+1.3 \times 10^{-6}$ Pa in the second; every element of these two layers is in tension. The von Mises stress of this element is 4.7×10^{-4} Pa, the seed average 5.9×10^{-4} Pa (Table 4.1) and the largest in the model, also inside the seed, 7.2×10^{-4} Pa. The element's own growth variable gives 0.49 times the growth and the stress of equation (4.5), which is $\frac{1}{2} \times 1.1/1.125$ for this time step (see the whole-model check above). With the framework's example stiffness ($E = 1000$ MPa, $\nu = 0.3$, blended linearly) the mean stress in the seed is about 4×10^8 times larger and the von Mises stress

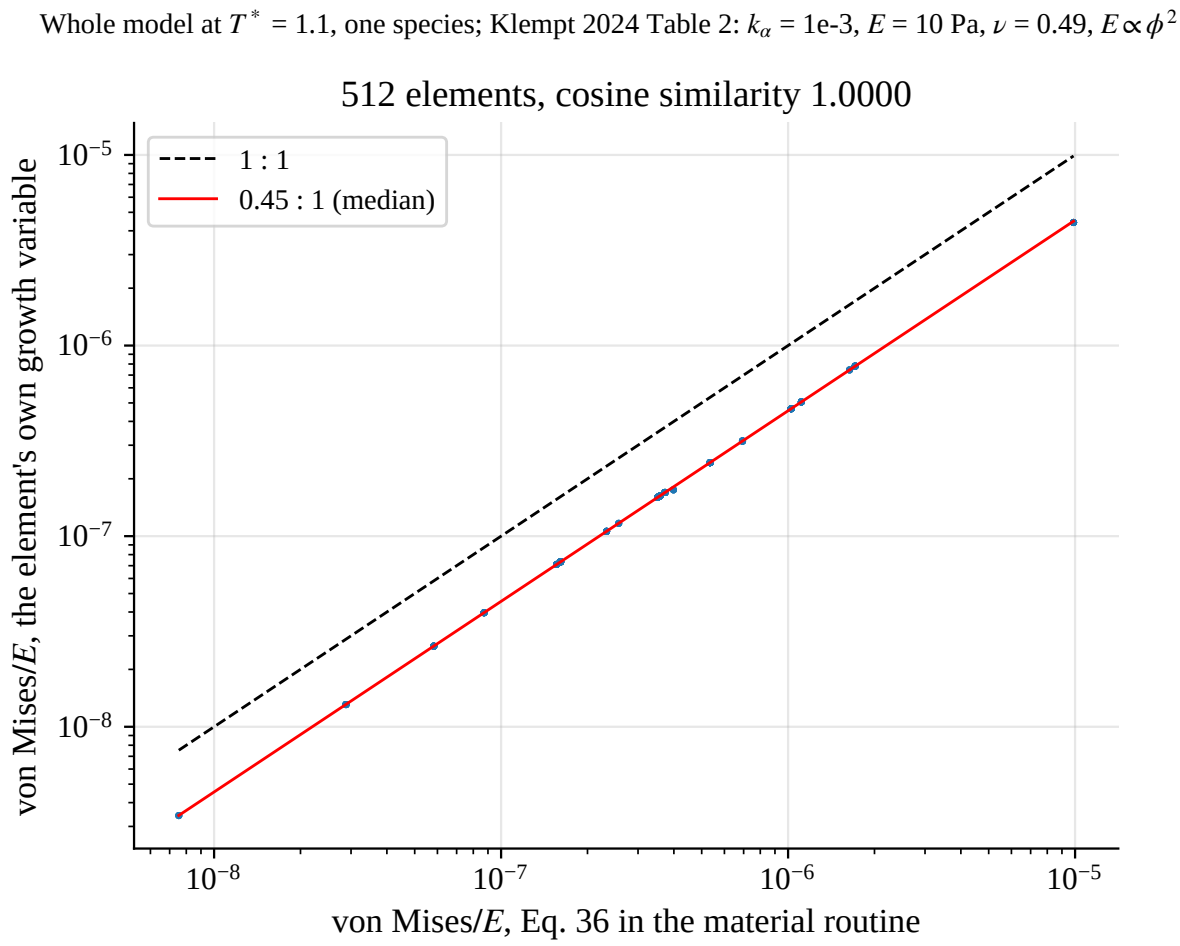


Figure 4.9: Element von Mises stress from equation (4.5) against the element's own growth variable, one point per element.

about 2×10^8 times. Changing ν from 0.49 to 0.45 or 0.499 changes the mean stress in the seed by less than 2 %.

The first author of the paper has since suggested that its stiffness unit may be wrong and that $E = 10$ kPa is more realistic. Growth is the only load, so all stresses in this chapter then scale by 10^3; the comparisons and ratios do not change.

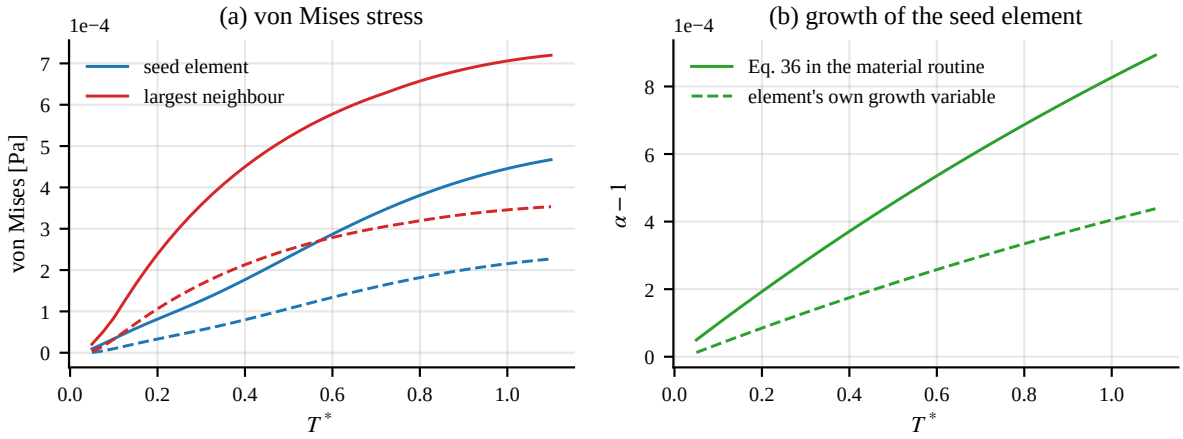


Figure 4.10: (a) Von Mises stress of a monitored element inside the seed and of its most loaded neighbour and (b) growth $\alpha - 1$ of that element over time; 16^3 elements, $\beta = 0.02$ mm$^2/T^*$, stiffness of Klempt et al. 2024. Solid: equation (4.5) in the material routine; dashed: the element's own growth variable.

Mesh and incompressibility. The example model has 8^3 elements, and its seed is 32 of them. I solved the same mechanical problem (seed growth 1.1×10^{-3}, the stiffness above) with a linear hexahedral solver in Python on 8^3, 16^3 and 32^3 elements. The seed averages of von Mises and mean stress on 8^3 are about 1.5 to 1.75 times those on 16^3 and 32^3, which agree with each other; the peaks at the corners of the seed do not converge, because those corners are singular. I therefore quote seed averages and not peaks. With the element's mean-dilatation (B-bar) integration the mean stress hardly changes between $\nu = 0.49$ and 0.499 (-3.8 and -4.1×10^{-5} Pa on 8^3), so there is no volumetric locking; full $2 \times 2 \times 2$ integration would overestimate it by a factor of 3.7 on 8^3 and 2.3 on 16^3 at $\nu = 0.49$.

The ANSYS model uses the same integration (SOLID185, KEYOPT(2) = 0, B-bar). In ANSYS I refined the example model to 16^3 and 24^3 elements, with every boundary condition mapped by position (32^3 exceeds the memory of the workstation in the nutrient solve). Table 4.1 gives the seed averages and the first layer around the seed. The mean stress and the ring of tension agree within about 3 % on the three meshes, and the von Mises stress changes by 3 % from 16^3 to 24^3.

Order of convergence. With the surface points corrected (§4.8), I repeated the seed growth on the three meshes with $\Delta t = 0.1$, 0.05, 0.025 and 0.0125, on 24^3 from $\Delta t = 0.05$

Table 4.1: Seed averages and the first layer of elements around the seed at $T^* = 1.1$, $\beta = 0.02$ mm$^2/T^*$, $\Delta t = 0.025$ [Pa].

	8^3	16^3	24^3
mean stress, seed	-2.52×10^{-4}	-2.53×10^{-4}	-2.45×10^{-4}
mean stress, first layer	$+6.01 \times 10^{-5}$	$+6.04 \times 10^{-5}$	$+5.85 \times 10^{-5}$
von Mises, seed	4.83×10^{-4}	5.87×10^{-4}	6.06×10^{-4}

(Figure 4.11); the seed values agree with Table 4.1 within 0.03 %. Halving the time step halves the change, so the update is first order in time, and the error at $\Delta t = 0.025$ is about 1 % in the seed von Mises stress. A fit through the three meshes alone would give an observed order of about 2.1 for the seed von Mises stress; three meshes cannot check this, and a finer reference shows otherwise. I solved the mechanics with a separate hexahedral code (mean-dilatation integration, the stiffness of Eq. (4.2)) for the reference growth field described next, on meshes up to 48^3, once with one value of ϕ and α per element and once with the values at the $2 \times 2 \times 2$ integration points, as the element carries them. The first series approaches the seed von Mises stress from above, the second from below; on 48^3 they give 6.11 and 6.07×10^{-4} Pa, so the converged value lies between the two, 2 % above the element's value on 24^3 after extrapolation to $\Delta t \to 0$. With the Gauss-point values the code reproduces the element's seed stress within 3.5 % on 8^3 and within 1 % on 16^3 and 24^3; on the coarse mesh the Gauss points smooth the step in α at the surface of the seed inside an element, which lowers the stress. The stress converges only to about first order, because the stiffness jumps at the surface of the seed.

For the growth field itself the error can be measured. Without the front term the seed problem is the linear system of Eqs. (2.5) and (2.7) with the penalty of the element, and I solved it with second-order finite volumes and a fourth-order Runge–Kutta scheme on 48^3, 96^3 and 192^3 cells (the seed is a union of cells on each grid). The seed mean of $\alpha - 1$ converges with an observed order of 1.97, and the Richardson extrapolation of the two finest grids is the reference, with a remaining error below 0.05 %. Averaged over the elements of each NEM mesh and compared element by element (Figure 4.11, right), the element's $\alpha - 1$ after extrapolation to $\Delta t \to 0$ is too large by 4.1 % in the seed mean on 8^3, 1.3 % on 16^3 and 0.7 % on 24^3, an order of about 1.7 in h; the L_2 norm of the element-wise error over the cube is 5.1, 1.9 and 1.1 %. The error is largest near the edge of the seed, where the gradient of ϕ is steepest (5.5 % at 0.4 mm from the centre on 8^3, 0.2 % on 24^3), and smallest at the centre (0.5 to 0.2 %). The domain integral of $\alpha - 1$ is within 0.3 % on every mesh: the explicit update conserves the mass of ϕ up to the source, and the mesh error redistributes the growth between the seed and its surroundings. At $\Delta t = 0.0125$ the errors are about 0.2 % larger than after extrapolation.

The diffusion coefficient sets the size of the stresses (Table 4.2): ϕ that diffuses out of the seed lowers its growth near the surface, and the step in α between the inside and the surface of the seed loads it. From $\beta = 0.005$ to 0.1 mm$^2/T^*$ the mean stress in the seed

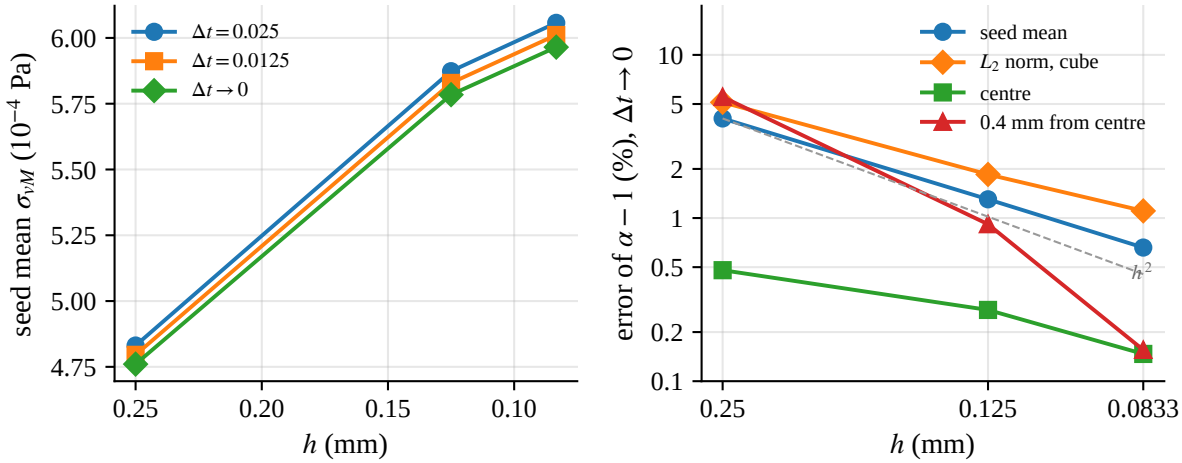


Figure 4.11: Seed growth at $T^* = 1.1$, $\beta = 0.02$ mm$^2/T^*$, surface points corrected. Left: seed mean of the von Mises stress against the element size h for two time steps and the first-order extrapolation $\Delta t \to 0$. Right: error of $\alpha - 1$ of the element against the finite-volume reference solution, averaged over the same elements, after extrapolation to $\Delta t \to 0$: seed mean, L_2 norm over the cube, and the values at the centre of the seed and 0.4 mm from it; the dashed line is second order.

falls by a factor of about four on 8^3 and eight on 16^3. For $\beta \geq 0.02$ the diffusion length $\sqrt{\beta T^*} \geq 0.14$ mm is resolved by the meshes, and 8^3 and 16^3 agree within 11 % at $\beta = 0.02$ and 0.05 and within 16 % at $\beta = 0.1$. For smaller β the layer is thinner than an element and the results depend on the mesh. With the example input, $\beta = 10^{-4}$ mm$^2/T^*$, the layer is 0.01 mm thick: the step in α grows from 0.4 to 1.8 % from 8^3 to 24^3, and the seed von Mises stress (7.5×10^{-5} to 1.5×10^{-4} Pa) does not converge. With $\beta = 0$ the step vanishes and the ratios $16^3/8^3$ are those of the Python study above.

Table 4.2: Mean stress in the seed at $T^* = 1.1$ for different β [Pa]; $\beta = 0.02$ is the value used in this chapter. All runs with ϕ at the surface points updated (§4.8); 16^3 at $\beta = 0.1$ with $\Delta t = 0.0125$.

β [mm$^2/T^*$]	0.005	0.01	0.02	0.05	0.1
8^3	-2.1×10^{-4}	-2.7×10^{-4}	-2.5×10^{-4}	-1.3×10^{-4}	-5.6×10^{-5}
16^3	-3.9×10^{-4}	-3.4×10^{-4}	-2.5×10^{-4}	-1.2×10^{-4}	-4.7×10^{-5}

4.7.1 Two species with nutrient and antibiotic

Set-up. The examples use the coupling of Section 4.6 with the synthetic data of Fritsch et al. [59], case I, two species (A and B, $b = (1, 2)$). All constants are those of the papers: the field and the nutrient from Klempt et al. 2024, Table 2, converted to the 2 mm cube ($\beta = 0.02$ mm$^2/T^*$, $r = 10$ mm$/T^*$, $d = 10^8$ mm$^2/T^*$, $g\ell^2/d = 1$, $E = 10$ Pa, $\nu = 0.49$, void floor $f = 10^{-3}$), and the antibiotic of case 4 of Klempt et al. 2026 ($\alpha_0^* = 100$ from $T^* = 1/3$), with the diffusion and uptake of the nutrient. The threshold is $\phi_{\min} = 0.001$; it is not given by the papers and changes the results by less than 1 % below this value. In the

separate example species A fills the seed of this chapter and species B four small blocks along the edges of the cube; in the mixed example both start in the seed, in equal amounts ($\phi_1 = \phi_2 = 0.275$) or in the proportions of Fritsch et al. (0.25, 0.30). The nutrient is held on the face $y = -1$ mm, the antibiotic on all outer faces. The front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ of Eq. 34 is on in every run.

These runs use a finite-volume code in Python that solves the same equations as the element, with the point model of the ANSYS runs; the coupled point model agrees with the point model alone to 2×10^{-14} in the replay check above. The front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ is advection dominated: its cell Péclet number $rh/(2\beta)$ is about 31 on the 16^3 mesh, and a central gradient oscillates. I discretise it with a second-order upwind scheme (MUSCL with the minmod limiter) and mirror the fields at the walls. The stress is solved on a 16^3 hexahedral mesh with the element's integration; solving the same 64^3 field on 32^3 and 64^3 elements changes the seed von Mises stress by 0.2 %.

Composition. Table 4.3 gives the seed means at $T^* = 1$, and Figures 4.12 and 4.13 the separate and the mixed example with antibiotic. Without antibiotic the living volume fraction of the seed stays near 0.96; the antibiotic lowers it to 0.15 when the species start separately and to 0.04 when they start mixed. Species B, the more sensitive one, is removed first; where the nutrient is low, A dies as well, because the point model compares $c^* = c_0^* c$ with $\alpha^* b_i$. Living volume fraction, shares and nutrient change by less than 3 % between the grids and the time steps used.

Table 4.3: Seed means at $T^* = 1$, Python, 64^3 cells, outer time step 0.025; growth and stress converged values in Table 4.4.

example	$\sum_i \bar{\phi}_i$	$\phi_1/(\phi_1+\phi_2)$	c	$\alpha - 1$	von Mises [Pa]
separate	0.965	0.978	0.536	9.06×10^{-4}	6.56×10^{-4}
separate, antibiotic	0.152	0.951	0.911	8.95×10^{-4}	5.68×10^{-4}
mixed, equal amounts	0.961	0.859	0.608	8.33×10^{-4}	3.50×10^{-4}
mixed, equal amounts, antibiotic	0.038	0.610	0.973	8.30×10^{-4}	3.23×10^{-4}
mixed (0.25, 0.30), antibiotic	0.037	0.510	0.973	8.30×10^{-4}	3.22×10^{-4}

Growth and stress. The growth $\alpha - 1$ hardly depends on the composition: Eq. 36 is unchanged, and the antibiotic lowers the living fraction but not ϕ_{3D}. The composition reaches the stress through the nutrient and the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$, and the mixed seed carries about half the stress of the separate one because its edge is less sharp. Table 4.4 gives the grid series with the three-grid index of Celik et al. [60] (safety factor 1.25) at an outer time step of 0.0025. The growth converges with a GCI of 0.7 to 3.3 % in all examples. The seed stress is set by the jump of α across the edge of the seed, which the front sharpens. In the separate example it converges, with a GCI of 0.7 % without and 8 % with the antibiotic. In the mixed examples it converges slowly, with an observed

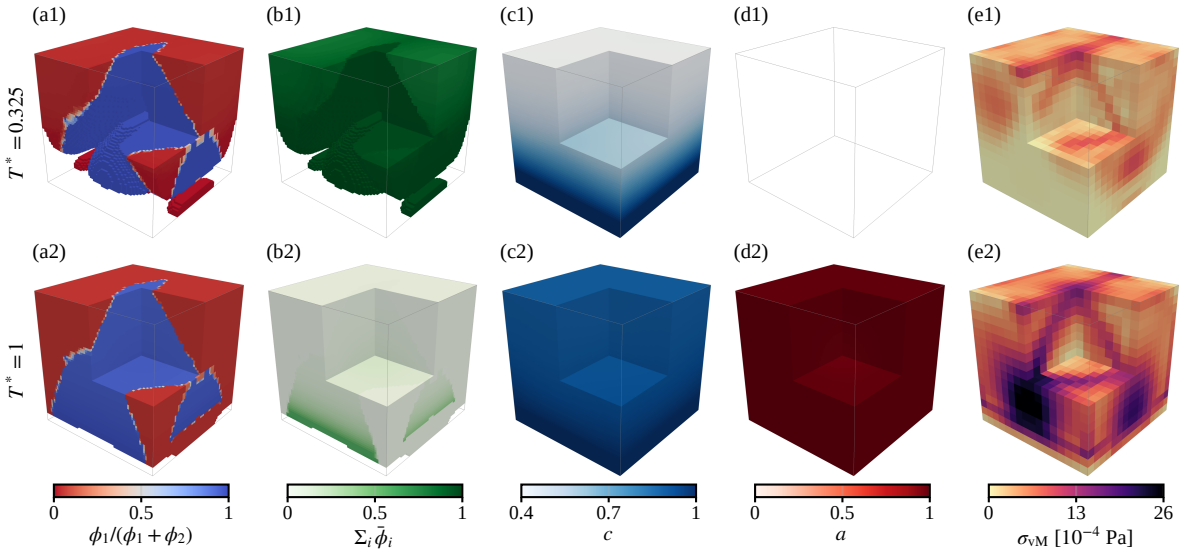


Figure 4.12: Separate example with antibiotic at $T^* = 0.325$ (top) and $T^* = 1$ (bottom), Python, 64^3, one octant removed: (a) share $\phi_1/(\phi_1 + \phi_2)$, (b) living volume fraction $\sum_i \bar{\phi}_i$, (c) nutrient c, (d) antibiotic a, (e) von Mises stress.

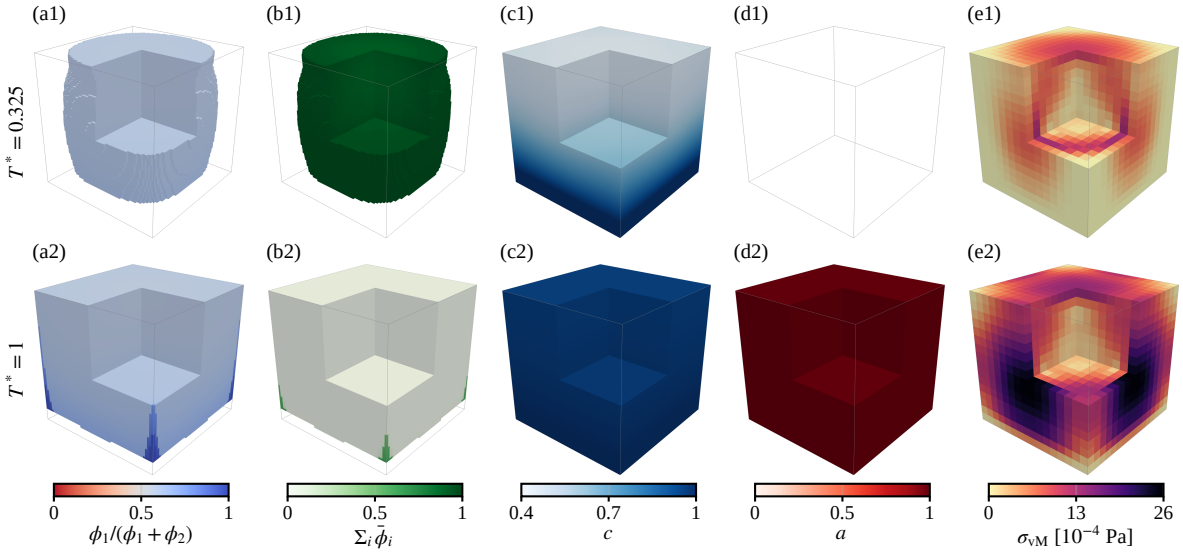


Figure 4.13: Mixed example (equal amounts) with antibiotic, panels as in Figure 4.12. Both species start in the seed; the living volume fraction falls almost everywhere after the antibiotic.

order of about 0.75 and a GCI of 26 to 29 %; with the start fractions of Fritsch et al. the 128^3 grid is not yet in the asymptotic range. There the front carries intermediate values of ϕ_{3D} at a nearly discontinuous edge, and the largest ϕ_{3D} in the seed still rises with the grid (0.60, 0.69, 0.73).

Table 4.4: Seed means at $T^* = 1$, outer time step 0.0025: grid series, observed order p, extrapolated value and GCI. Mixed: equal amounts unless the start fractions (ϕ_1, ϕ_2) are given; antib.: with antibiotic; –: not yet in the asymptotic range.

example	quantity	32^3	64^3	128^3	p	extrapolated	GCI
separate, antib.	$\alpha - 1$ [10^{-4}]	8.453	8.931	9.112	1.4	9.22	1.5 %
separate, antib.	von Mises [10^{-4} Pa]	5.523	6.000	6.284	0.75	6.70	8.3 %
separate	$\alpha - 1$ [10^{-4}]	8.729	9.014	9.110	1.6	9.16	0.7 %
separate	von Mises [10^{-4} Pa]	7.440	6.730	6.585	2.3	6.55	0.7 %
mixed, antib.	$\alpha - 1$ [10^{-4}]	7.42	8.04	8.32	1.2	8.54	3.3 %
mixed, antib.	von Mises [10^{-4} Pa]	2.00	2.82	3.30	0.76	3.99	26 %
mixed	$\alpha - 1$ [10^{-4}]	7.46	8.08	8.35	1.2	8.56	3.1 %
mixed	von Mises [10^{-4} Pa]	2.13	3.08	3.65	0.74	4.50	29 %
mixed (0.25, 0.30), antib.	$\alpha - 1$ [10^{-4}]	7.42	8.04	8.31	1.2	8.51	3.1 %
mixed (0.25, 0.30), antib.	von Mises [10^{-4} Pa]	2.01	2.96	4.01	–	–	–

Time step. The point model advances once per outer step of the field. The composition and $\alpha - 1$ change by less than 3 and 0.3 % between outer steps of 0.02 and 0.0025 (with an integer number of inner steps). The stress does not: in the separate example the step 0.025 of Table 4.3 gives about 6 % less than the limit, which first-order extrapolation puts at about 6.07×10^{-4} Pa on 64^3; in the mixed example it gives about 15 % more (3.23 against 2.82×10^{-4} Pa at 0.0025). The inner steps of the point model, the one-increment lag of $\sum_i \bar{\phi}_i$ and the regularisation of the front direction ($\varepsilon^2 = 10^{-6}$ against 10^{-8}, 0.1 %) are not the cause. Without antibiotic the step changes the stress by 2.5 % (separate) and 12 % (mixed). I therefore take the growth and the stress from the grid series at the small step.

Antibiotic over time. With the same total dose given constantly, in pulses or as a sine from $T^* = 1/3$, $\alpha - 1$ and the stress change by less than 1.5 %, while the living volume fraction follows the dose: in the pauses of the pulsed dose it recovers from 0.16 to 0.28 in the separate example. The antibiotic acts on the composition and the living biomass, not on the mechanics.

Start values of newly reached points. With the start values of the nearer seed instead of the neighbours' living composition, the seed values of the cube examples differ by a few percent (living volume fraction 0.147 against 0.152 in the separate example with antibiotic). In the maze of Klempt et al. 2024, Fig. 11, where the biofilm keeps spreading

after the antibiotic, the rule decides whether it recovers: at $T^* = 60$ the living volume fraction is 0.35 with the neighbours' composition and 0.04 with the start state.

The element. The ANSYS element solves the same equations with its own discretisation (Section 4.5). Its growth agrees with the Python values within 4 % after extrapolation. Its stress depends on how the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ is stabilised: with a constant added diffusion $rh/4$ the seed stress on 16^3 is about a quarter of the converged value, and with upwind gradients it is 2.14, 4.56 and 5.15×10^{-4} Pa on 8^3, 16^3 and 24^3, extrapolated 5.8×10^{-4} Pa (GCI 16.5 %). This lies within the band of the second-order schemes in Python. The element runs the point model in every increment of 0.00125, so its outer-step error is small. I take the values of this section from the Python code, which reaches finer grids.

4.7.2 The calibrated five-species model in the element

The coupling of Section 4.6 takes the number of species as a parameter, so the point model calibrated in Chapter 3 can be run in the element without changes to the routine. I used the MAP of the commensal flow condition (CH) from the calibration to the measured data, with $c^* = 25$ and $\alpha^* = 0$, and otherwise the setting of the two-species runs: the same 8^3 model, the growth law (4.5) driven by the field, $\phi_{\mathrm{cap}} = 0.9$, $s = 0.15$ and $\Delta t = 0.025$ up to $T^* = 1$. The point model starts from the calibrated Day-1 state of the condition (*S. oralis* 0.76, *V. dispar* 0.11, *F. nucleatum* 0.10, *A. naeslundii* 0.02, *P. gingivalis* 0.001, living fractions 0.999). This run predates the agreed coupling of Section 4.6: the coupling is one-way, without front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ and antibiotic, so the growth and the stress do not depend on the composition: the seed mean of $\alpha - 1$ is 6.752×10^{-4}, the same to all printed digits as in the two-species runs and as in a run started from equal shares, and with the stiffness of Klempt et al. 2024 ($E = 10$ Pa, $\nu = 0.49$) the seed mean of the von Mises stress is 4.68×10^{-4} Pa.

Figure 4.14 shows the composition at a Gauss point of the seed. *S. oralis* stays the largest share but falls from 0.74 to 0.55 at $T^* = 1$, while *V. dispar* (0.15), *A. naeslundii* (0.13) and *P. gingivalis* (0.08) grow and *F. nucleatum* stays near 0.09. The living volume fraction $\sum_i \phi_i\psi_i$ falls from 0.81 to 0.23, mostly through the living part of *S. oralis*. Started from equal shares instead (a run with the preliminary MAP), *A. naeslundii* would become the largest share (0.38), so the composition in the element depends on the start as in the point model alone. These numbers show that the calibrated model runs inside the element; they are not compared with a measured spatial composition. An earlier run with a preliminary CH MAP, which differs only in entries not identified in CH, gave the same growth, stress and nutrient field to all digits and shares within 0.02.

4.8 Limitations

The limitations of the whole work (validation, feedback of the stress, transport of the composition, the front speed and the assumptions) are collected in Section 5.2. The

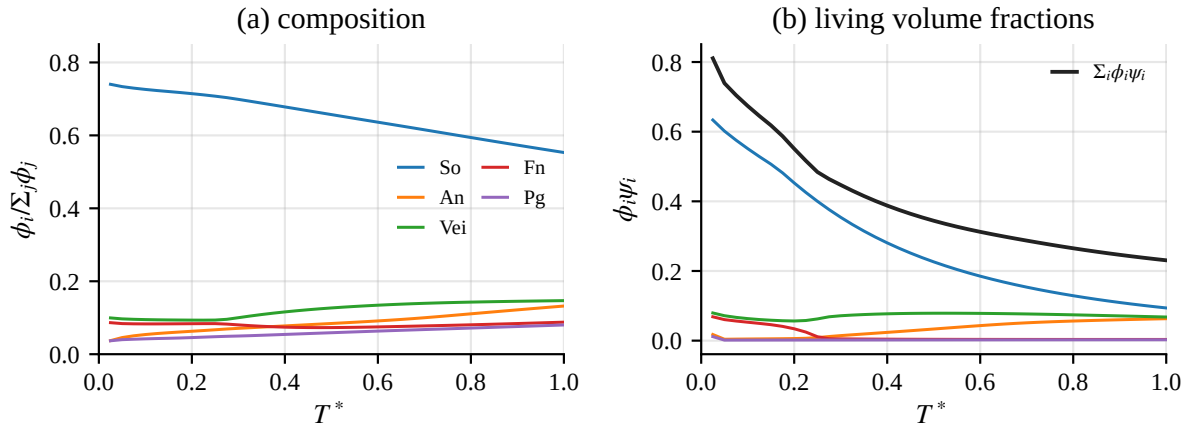


Figure 4.14: The five-species point model calibrated in Chapter 3 (commensal flow condition, MAP) in the element with the earlier one-way coupling of this chapter, 8^3, at a Gauss point of the seed: (a) shares $\phi_i / \sum_j \phi_j$, (b) living volume fractions $\phi_i \psi_i$ and their sum. Start from the calibrated Day-1 state.

following concern the implementation of this chapter.

- *Faces of the cube.* In the element used here, ϕ at the surface points was not updated and stayed 0, so every face acted as $\phi = 0$ where Eq. (4.4) has no flux. With these points set to the mean of their neighbouring Gauss points, the runs with $\beta = 0.02$ mm$^2/T^*$ change by less than 0.13 % in the seed stress and in α and by less than 0.5 % in the composition, so their values are kept. Where ϕ reaches the faces (larger β, longer times) the seed pressure changes by up to 30 %; the 8^3 values of Table 4.2 for $\beta = 0.05$ and 0.1 are those of the corrected runs. The mean is exact only for a constant field; in the seed runs the resulting error stays in the layer of elements at the faces, where $\alpha - 1$ is about 1 % of its maximum.

- *Pressure term.* The spherical term of §4.4.3 is documented, not corrected. It does not change the von Mises stress at a given deformation, and at $\nu = 0.49$ it is 1.3 % of the constrained pressure.

- *Incompressibility.* The paper's exact constraint is approximated by $\nu = 0.49$ with the element's mean-dilatation formulation; the mean stress in the seed changes by less than 2 % between $\nu = 0.45$, 0.49 and 0.499.

- *Mesh.* With $\beta = 0.02$ mm$^2/T^*$ the mean stress in the seed and in the first layer around it agrees within about 3 % on 8^3, 16^3 and 24^3 elements, and the seed von Mises stress within 3 % between 16^3 and 24^3; with the example input $\beta = 10^{-4}$ mm$^2/T^*$ they do not converge. Peak stresses at the seed corners are mesh dependent and not reported.

- *Consumption.* The one-species ANSYS results of this chapter use the consumption $g\,\phi$ as printed (Eq. 35); the two-species examples use $g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,c$, the form $g\,\phi\,c$

confirmed by the authors (Appendix C).

- *Front direction.* The direction $\nabla c/\|\nabla c\|$ of the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ is undefined where $\nabla c = \mathbf{0}$; I regularise it with $\|\nabla c\|^2 + \varepsilon^2$, $\varepsilon^2 = 10^{-6}$, as the authors' implementation does with a small ε (F. Klempt, personal communication, October 2026); $\varepsilon^2 = 10^{-8}$ changes the stress by 0.1 %.

- *Magnitude.* With the published growth rate, $\alpha - 1$ stays below about 10^{-3} up to $T^* = 1$, and with the published stiffness the largest von Mises stress is about 7×10^{-4} Pa (0.7 Pa with $E = 10$ kPa, the value suggested by the paper's first author).

4.9 Earlier one-way computations

The steps beyond the coupling of this chapter (composition into the spreading and the stiffness, stress into growth) are listed in Section 5.3. Before the coupling was agreed, I computed the first of them in Python, outside ANSYS, to see which change the results. These computations use the earlier one-way scheme, in which the composition is scaled to the amount of the field, and cases 3 and 6 of Klempt et al. 2026. The papers give no growth rates or moduli per species; the rates are estimated from the dissipation parameters η_i under a stated assumption, and the moduli are sensitivity parameters.

Step 1: local nutrient. The nutrient is held at the face $y = -1$ mm and consumed in the seed, and each point of the seed runs the point model with $c^* = c_0^*\, c$. Case 3 hardly depends on the nutrient level (share $\phi_1/(\phi_1 + \phi_2)$ 0.63 to 0.66). In case 6 the takeover is slower where the nutrient is low, and the share ranges from 0.03 near the nutrient face to 0.44 inside the seed (Thiele number $\Lambda = \sqrt{g\ell^2/d} = 4$ [61], $\ell = 1$ mm). The same step in ANSYS (8^3, case 6, consumption $6\,T^{*-1}$, chosen in this work) gives a share of 0.11 to 0.48, correlated with the local nutrient (-0.99), and Python gives the same range. Over all runs the share at the end depends on the time link s, the end time and the local nutrient only through the product $s\,T^*c$ (Figure 4.15): the nutrient acts as the clock of the point model. In case 6 the takeover happens between $s\,T^*c \approx 0.03$ and 0.08, so the assumed s decides whether a composition pattern is seen at $T^* = 1$.

Step 2: composition into the spreading. On the reproduced field of Klempt et al. 2024, test case 4.1, the front growth rate becomes the mean of the species rates weighted by their shares, $\bar{r} = \big(\phi_1 r_1 + \phi_2 r_2\big)/(\phi_1 + \phi_2)$, with $r_{1,2} = (1 \pm \delta)\, r$ and r from Table 2.1; for $\delta = 0$ the scheme is the one-way one. The weighting follows Feng et al. [62], whose spreading velocity has the summed species growth as its divergence, and the summed growth sources of Soleimani et al. [63]. The rate contrast has two estimates. In cases 3 and 6 of Klempt et al. 2026 the second species has twice the dissipation of the first, $\eta_1 = 1$, $\eta_2 = 2$; assuming that the front rate of a species scales with $1/\eta_i$ gives $r_1/r_2 = 2$ and $\delta = 1/3$. Feng et al. use maximum growth rates of 3×10^{-5} and 8×10^{-5} s^{-1} for their two

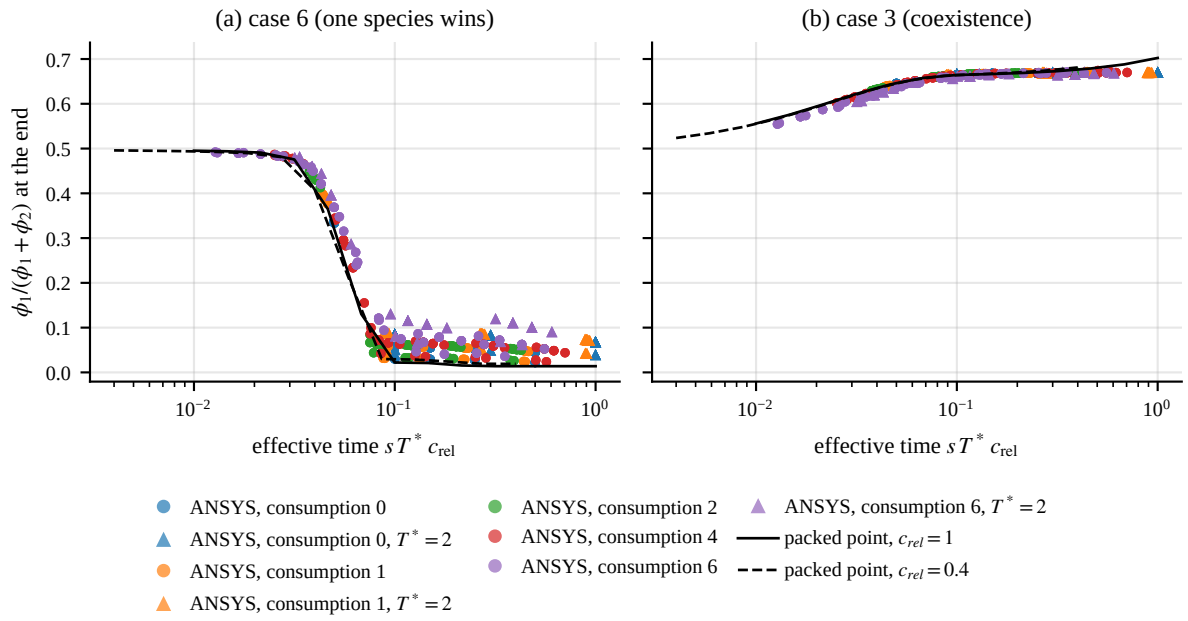


Figure 4.15: $\phi_1/(\phi_1+\phi_2)$ at the end of every seed element of the 8^3 ANSYS runs ($\beta = 0.02$ mm$^2/T^*$; consumption 0 to $6\,T^{*-1}$, $s = 0.05$ to 1, $T^* = 1$ and 2) against $s\,T^*c$, with c the local nutrient. Lines: one packed point in Python at $T^* = 1$. (a) Case 6 and (b) case 3 of Klempt et al. 2026.

species, the first measured and the second estimated, a ratio of 2.7 and $\delta \approx 0.45$. With $\delta = 1/3$ the mean ϕ at $T^* = 1$ rises from 0.704 to 0.735 in case 3, where the faster species is the majority, and falls to 0.648 in case 6, where the slower species takes over; locally ϕ changes by up to 0.47 (Figure 4.16). With $\delta \approx 0.45$ from the rates of Feng et al. the two values are 0.746 and 0.626, with $\delta = 0.5$ they are 0.750 and 0.618. The composition itself hardly changes. Which species wins then decides how fast the biofilm spreads. The values are converged in the coupling step. On a grid of half the spacing (41^3 nodes, 0.5 μm) the one-way mean ϕ falls from 0.704 to 0.674, and the changes for $\delta = 0.5$, at a coupling step of 0.05, shrink from +6 % and −13 % to +5 % and −8 %: the sign and the order of the effect hold, its size depends on the grid of the reproduced field, for which I keep the paper's 1 μm. In ANSYS this step needs a working front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$.

Step 3: composition into the stiffness. With the seed composition of step 1 and a ratio of the species moduli up to 5 at the same mean modulus, the average stresses in the seed change by at most 0.2 %. The seed is about a thousand times stiffer than the void around it, so the soft surroundings set its stress. This step matters only where the biofilm itself carries the load, such as a contiguous biofilm on a substrate.

Step 4: stress into growth. In the example model the mean stress in the seed, about 2.5×10^{-4} Pa, is below 10^{-4} times the shear modulus, so a stress-dependent growth law would change little with the published parameters, while the mechanical term of Klempt

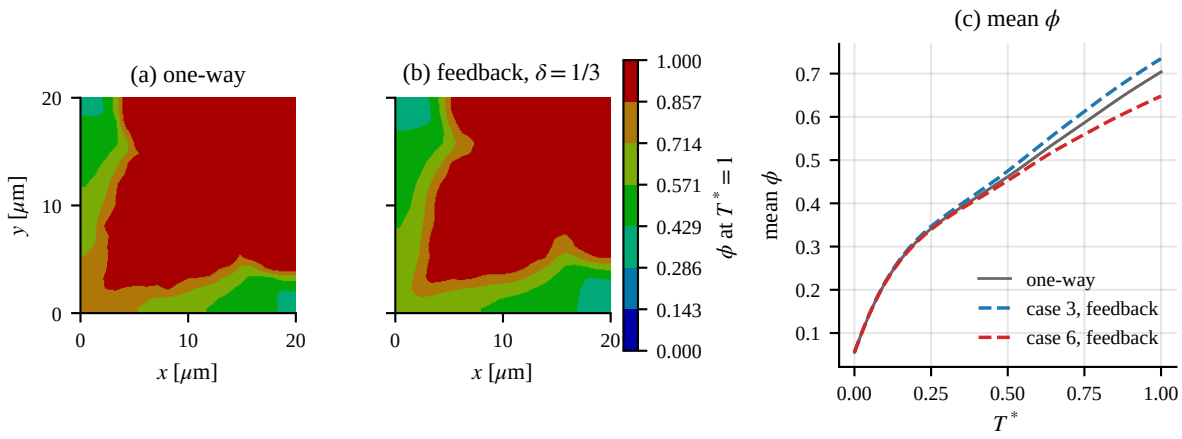


Figure 4.16: Composition acting on the spreading, computed in Python on the reproduced field of Klempt et al. 2024, test case 4.1 (mid-plane $z = 10$ μm), with species growth rates $r_{1,2} = (1 \pm \delta)\, r$, $\delta = 1/3$ from $\eta_1 = 1$, $\eta_2 = 2$ of Klempt et al. 2026 (assumption: front rate $\propto 1/\eta_i$). (a,b) ϕ at $T^* = 1$ for case 6 without and with the feedback (nutrient on the edge at the top right). (c) Mean ϕ over time for cases 3 and 6.

et al. 2024, Eq. 30, kept as written would stop all growth (Appendix C). This step needs a different growth law and realistic stress levels.

Of these steps, step 2 changes the results most; step 1 is now part of the coupling.

5 Conclusion

The task of this thesis was to bring a point model of a multi-species biofilm into an existing finite element framework, so that its composition, its growth and the resulting stresses can be computed in space. Three contributions serve this task.

A calibrated point model (Chapter 3). The 15 interaction coefficients of a five-species point model were estimated with TMCMC from the in-vitro time courses of Heine et al. under four conditions. Compiled with JAX, the forward model runs about 150 times faster on the GPU than on the CPU, which makes posteriors with up to 20,000 particles routine. Every stage was repeated with three seeds and checked against convergence criteria; CS, CH and DS meet them at every final stage; in DH the last stage split across seeds, and DH is reported from the stage before. The data determine 5, 6, 9 and 8 of the 15 entries in the four conditions, and the pH, which was not used in the inference, serves as a check. With a metabolic sign prior, the same type of model also predicts sparse in-vivo data of ten patients out of sample.

A growth law in the finite element framework (Chapter 4). The growth kinematics $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$, $\mathbf{F}_g = \alpha\mathbf{I}$, of Klempt et al. 2024 was implemented as an ANSYS user material and verified against closed-form solutions. It was coupled to the biofilm field of the IKM element through the growth equation $\dot{\alpha} = k_\alpha\phi$ at every Gauss point. Inside ANSYS, the coupled run follows the exact solution of a gradient-free region to 3×10^{-11} in α. With the published parameters the growth in the seeded element reaches $\alpha - 1 = 8.9 \times 10^{-4}$ at $T^* = 1.1$ and the seeded region is compressed; a finer model of the same problem predicts, on average, tension in the surrounding material, consistent with the ring of tension reported by Klempt et al. [32].

The composition at every Gauss point (Chapter 4). The point model of Klempt et al. 2026 runs, unchanged, at every Gauss point. It receives the nutrient and an antibiotic from fields of the element, and its living biomass sets the consumption of both, so that the composition reaches the spreading of the biofilm through the nutrient and the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$. With uniform fields the coupled point model reproduces the point model alone to round-off. In two-species examples with synthetic data an antibiotic lowers the living volume fraction of the seed from 0.97 to 0.15 when the species start separately and to 0.04 when they start mixed, while the growth changes by less than 2 %. The growth converges on refined grids with a GCI below 4 %; the seed stress needs a second-order treatment of the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ and converges more slowly, with a GCI of 8 % in the separate (0.7 % without antibiotic) and 26 to 29 % in the mixed example. The ANSYS element solves the same equations and agrees within these bands.

Together these steps connect a calibrated description of what a biofilm consists of with a finite element description of how it grows and what stresses it carries, for one and two species.

5.1 Discussion

Where the species live. Table 5.1 places this work among the continuum models it builds on. Feng et al. [62] and Soleimani et al. [63] give every species its own field and compute no growth stresses; Klempt et al. [32] compute these stresses with a single biofilm variable. This work keeps the single field for the amount of biofilm and its stresses, and adds the composition as a local state at each Gauss point.

Table 5.1: Continuum models of biofilm growth that this work builds on or compares with.

model	species	space	growth stress	composition
Feng et al. 2021 [62]	two fields	FEM, common velocity	no	from the species fields
Soleimani et al. 2023 [63]	two fields, co-aggregation	ANSYS user element	no	from the species fields
Klempt et al. 2024 [32]	one field	ANSYS user element	yes	none
Klempt et al. 2026 [24]	n fractions at a point	none	no	point model
this work	one field and a point model per Gauss point	ANSYS, IKM element	yes (no feedback)	point model, nutrient and antibiotic fields

The number of transported fields therefore does not grow with the number of species; the species enter as a local state at each Gauss point, and the point model, calibrated on its own, enters unchanged. The price is that the species cannot move relative to each other: the composition moves with the biofilm only through the start values of newly reached points, which take the living composition of their neighbours, and varies in space through the local nutrient and antibiotic. Segregation of species by different spreading is outside what this scheme can show.

Calibrated and chosen parameters. The interaction coefficients of Chapter 3 come from measured time courses; the pH, which did not enter the inference, follows the measured rise in DH through a regression on the predicted fractions ($R^2 = 0.71$) (Section 3.4). The coupled examples of Chapter 4 use synthetic two-species data (case I of Fritsch et al. [59]) with the antibiotic of Klempt et al. [24], since the calibration data contain no antibiotic. The calibrated five-species point model also runs in the element with the same coupling (Section 4.7.2).

What the stresses mean. With the stiffness of Klempt et al. [32] the stresses are of order 10^{-4} Pa and scale with the stiffness. What carries over is their pattern: the seeded region is compressed and the surrounding material is under tension, as in the comparison of growth models by Klempt et al. [33] and in the ring of tension of the 2024 paper. Measured stiffnesses of oral biofilms lie in the kPa range, for example 0.6 to 41 kPa by atomic force microscopy, depending on the sucrose supply [64]; as the stresses scale linearly with E, a stiffness of 10 kPa gives stresses of order 0.1 Pa. A statement about the magnitude in an oral biofilm needs that stiffness together with a measured stress, which this work does not have.

The numerical scheme. The field of the element is advanced explicitly, with a stability limit that depends on the mesh (Appendix D). The front term $-r\frac{c}{k+c}\,\mathbf{n}_c\cdot\nabla\phi_{3D}$ is advection dominated, so the stress needs a second-order scheme (Section 4.7.1). An implicit treatment of the field would remove the step limit; whether it is also more accurate for the same cost is the comparison planned at Keio.

Symmetric interactions. The variational derivation makes the interaction matrix symmetric. A symmetric matrix cannot describe one species feeding another without the reverse, which the metabolic literature reports for several of the pairs. In this work the symmetric model reproduces the in-vitro time courses and the rise of the pH in DH; whether a directed term is needed for in-vivo data is open.

5.2 Limitations

- *Verification only.* The finite element results are verified against closed forms and against the published models. No measured stress of a growing oral biofilm was available to validate them.
- *No stress feedback.* The composition reaches the spreading only through the nutrient; it does not act on the stiffness, and the stress does not act back on growth, following Klempt et al. 2024, who drop the mechanical term in the growth equation.
- *Front term $-r\frac{c}{k+c}\,\mathbf{n}_c\cdot\nabla\phi_{3D}$.* The front term $-r\frac{c}{k+c}\,\mathbf{n}_c\cdot\nabla\phi_{3D}$ needs a second-order upwind treatment, and the seed stress converges slowly when the species start mixed. Its speed depends on the nutrient only, so dead biofilm keeps spreading after an antibiotic. The stiffness counts dead biomass as well.
- *Assumptions.* The interpolation of the nutrient and the antibiotic in time, the start values of points the biofilm reaches later and the threshold for starting the point model are not given by the papers. All are reported with their sensitivity.
- *The source model.* The test cases of Klempt et al. 2024 are reproduced only with stated departures from the printed equations (Appendix C), and the paper gives two forms of the law for α. I follow its Eq. 36.

- *Mesh and magnitude.* For a seed with a sharp edge, the example model's 8^3 mesh overestimates the averaged stresses of the seed by a factor of 1.5 to 1.75. With $\beta = 0.02$ mm$^2/T^*$ the seed mean stress on 8^3 agrees with 16^3 within 1 %, and the seed von Mises stress is about 18 % lower. Peaks at the corners of the seed are mesh dependent. On 24^3 elements the growth $\alpha - 1$ of the seed is 0.7 % above a finite-volume reference solution of the same problem, and the error of the seed stress is estimated at about 2 % from three meshes. With the published parameters the stresses are of order 10^{-4} Pa.

- *Data.* The calibration rests on five species under four conditions in vitro and ten patients in vivo. The commensal and dysbiotic consortia used different *P. gingivalis* strains, so part of the difference between the inferred matrices reflects the strains. The symmetric interaction matrix cannot represent directed effects.

5.3 Outlook

The work continues at Keio University. The steps follow from the limitations, in order of how well the literature supports them.

1. *More species.* The five-species point model calibrated in Chapter 3 and the bridge from ANSYS to Python already exist; with the scheme of Chapter 4 the field stays a single variable, so only the state of the point model and its parameters need to be widened. With the point model calibrated for CH in the element (Section 4.7.2), the stress does not depend on the composition as long as the field sets the amount of biofilm. When α grows instead with the living biomass $\sum_i \phi_i \psi_i$ of the point model, at a rate chosen (not taken from a paper) so that the mean growth of the seed at $T^* = 1$ equals that of Eq. 36 of Klempt et al. 2024, the mean von Mises stress of the seed is 7 % lower. Growth from the living biomass follows classical multispecies biofilm models, in which only the active biomass grows and inactive cells only occupy volume [65]; it lets lysis or an antibiotic in the point model lower the growth.

2. *The front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ in the element.* A second-order stabilisation that stays stable on fine meshes would let the element give the converged stress directly; upwind gradients are a first step.

3. *Composition-dependent spreading.* The point model now sees the local nutrient and antibiotic, and its living biomass sets their consumption. Letting the spreading of the field depend on the composition would close the loop further. First computations in Python (Section 4.9) show that both change the result: with a local nutrient the composition varies across the seed (now also in ANSYS), and with species-dependent spreading the amount of biofilm at $T^* = 1$ changes by up to 8 %.

4. *Composition into stiffness and stress into growth.* Species-specific stiffness and stress-dependent growth would let the composition reach the stress directly and the stress act back. Both need parameters that are not yet available.

5. *Uncertainty into the stress.* The posterior of Chapter 3 could be carried through the element to the stress; time-separated stochastic mechanics [66] does this at a much lower cost than Monte Carlo sampling.

6. *A test of the inferred network.* The calibrated matrices predict that removing *F. nucleatum* from the dysbiotic flow-reactor consortium reduces the late increase of *P. gingivalis* to about a quarter; a co-culture without it would test this.

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A AGORA2 Representative Strains and Oral-Fluid Medium

Table A.1 lists the AGORA2 v2.01 genome-scale model chosen as the representative for each guild in the pFBA and MICOM community-FBA layers of the metabolic sign prior of Section 3.6. The *Other* guild has no representative, so guild pairs involving it receive no FBA-based prior.

Guild	AGORA2 representative strain
Actinobacteria	*Actinomyces naeslundii* Howell 279
Bacilli	*Streptococcus gordonii* CH1
Negativicutes	*Veillonella parvula* Te3 DSM 2008
Bacteroidia	*Prevotella melaninogenica* ATCC 25845
Fusobacteriia	*Fusobacterium nucleatum* ATCC 25586
Clostridia	*Parvimonas micra* ATCC 33270
Coriobacteriia	*Atopobium parvulum* DSM 20469
Gammaproteobacteria	*Haemophilus parainfluenzae* T3T1
Betaproteobacteria	*Eikenella corrodens* ATCC 23834
Other	no AGORA2 representative (no pFBA or MICOM prior)

Table A.1: AGORA2 representative strains per guild used for the pFBA and MICOM community-FBA layers.

Oral-fluid medium. The shared medium was assembled in two steps. First, quantified metabolite concentrations in unstimulated whole saliva were drawn from the literature and converted to exchange bounds ($\mathrm{mmol\,gDW^{-1}\,h^{-1}}$) assuming a biofilm density of $\sim 2\,\mathrm{gDW\,L^{-1}}$ and a reference growth rate of $0.1\,\mathrm{h^{-1}}$: sugars (glucose, fructose, sucrose) and lactate, all twenty amino acids plus ornithine, B-vitamins, trace metals, purines, and inorganic salts. Concentrations were set at the blood-plasma scale ($\sim$100$\times$ realistic saliva) so that every model remains feasible; a realistic-concentration variant produced non-specific competition (growth suppressed by more than 50% in most pairs) and was not used. Second, AGORA2-specific cofactors absent from the saliva list were added at minimal bounds where required for feasibility (protoheme/siroheme, menaquinones, *meso*-diaminopimelate, long-chain fatty acids, 4-hydroxybenzoate, adenosylcobalamin, glutathione, polyamines, cytidine). The O_2 exchange of the strictly anaerobic guilds was shut off before optimisation.

B TMCMC Implementation Details

The four-stage GPU-accelerated TMCMC procedure of Chapter 3 (Sections 3.2.3 and 3.2.4) is summarised in Algorithm 2. The GPU speed-up (170× per tempering step and 147× end to end on the matched benchmark) and the wall-clock times are in Table 3.2; the fit errors and log-likelihoods are in Table 3.3.

Algorithm 2 Four-stage GPU-accelerated TMCMC

Require: Data $\mathbf{y}_{\text{obs}}$ (viable and total composition, live fraction), N_p particles, prior box B, GPU device
Ensure: MAP $\hat{\boldsymbol{\theta}}$ and posterior samples of every stage
▷ **One TMCMC run on a box B with likelihood $p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})$**
draw $\{\boldsymbol{\theta}_j\}$ in B; $\beta_0 \leftarrow 0$
while $\beta_m < 1$ **do**
  find β_{m+1} such that the coefficient of variation of the weights equals its target, by bisection
  weights $w_j \leftarrow p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j)^{\beta_{m+1}-\beta_m}$; resample $\propto w_j$
  estimate the weighted covariance $\hat{\boldsymbol{\Sigma}}^{(m)}$
  for $k = 1$ **to** n_{mut} **do**
    propose a random-walk step scaled by $\hat{\boldsymbol{\Sigma}}^{(m)}$ or a differential-evolution step, alternately
    reject proposals outside B without solving the model
    evaluate all N_p likelihoods at once via `jax.vmap` on the GPU; accept or reject (Metropolis–Hastings)
  end for
end while
▷ **Stage (i):** composition terms only, ψ_i fixed to the data, full box
▷ **Stage (ii):** as (i), box narrowed to the stage-(i) posterior mean ±4 standard deviations
▷ **Stage (iii):** full likelihood, ψ_i evolved, full (widened) box; half of the particles around the stage-(ii) MAP, half from the prior
▷ **Stage (iv):** as (iii), box narrowed to the stage-(iii) posterior mean ±4 standard deviations
if more than 20 % of the posterior mass of an entry lies in the outer 5 % of the box on one side, move that bound outwards and repeat the stage
run every stage with three seeds; accept it only if criteria (a)–(e) of Section 3.2.4 hold, otherwise repeat it with more particles or mutation steps
return $\hat{\boldsymbol{\theta}}$, posterior samples

Settings. Table B.1 lists the particles and mutation steps of every stage; the gauge is $c^* = 25$.

The acceptance criteria, the definition of an identified entry and the treatment of entries with several posterior modes are given in Section 3.2.4.

Stage	CS	CH	DS	DH
(i)	$1{,}000 \times 40$	$1{,}000 \times 40$	$1{,}000 \times 40$	$1{,}000 \times 40$
(ii)	$2{,}000 \times 120$	$2{,}000 \times 150$	$3{,}000 \times 100$	$2{,}000 \times 80$
(iii)	$2{,}000 \times 160$	$2{,}000 \times 150$	$18{,}000 \times 150$	$\mathbf{20{,}000 \times 120}$
(iv)	$\mathbf{5{,}000 \times 150}$	$\mathbf{5{,}000 \times 150}$	$\mathbf{18{,}000 \times 150}$	$20{,}000 \times 150$

Table B.1: Particles N_p times mutation steps n_{mut} per seed. Bold: the final stage of each condition; DH stage (iv) did not meet criterion (e) and is not used.

C Reproduction of the Test Cases of Klempt et al. 2024

Before coupling the field model of Klempt et al. [32] into the finite element framework, I solved its two test cases independently in Python, to see whether the equations and parameters as printed give the published results. They do not in every respect, and this appendix records what is needed to reproduce them. None of the results of Chapter 4 depends on this reproduction; it informs the background of Section 2.2 and the questions to the authors.

C.1 Set-up

The domain is the paper's cube of 20 μm edge length on a regular grid of 21^3 nodes. Equations (2.5)–(2.7) are integrated with explicit time steps and upwinded gradients in the front term; ϕ is limited to $[0, 1]$. All parameters are those of Table 2.1. The two test cases are:

- *4.1*: a spherical seed of 5 μm radius in the centre, the nutrient held at $c = 1$ in one corner of the cube. Read from the paper's Fig. 3 and Table 3, the source is a block of about 2 μm at the corner; the strip of the schematic Fig. 2 fits worse.
- *4.2*: a disk of 5 μm diameter just above a bottom face held at $c = 1$, with the consumption $g = 10^8$ ("high" nutrient level) and $g = 10^{10}$ ("low").

The published curves of the mean ϕ and mean c (Figs. 4 and 7) were read off the PDF by colour and compared with the reproduction by their RMS difference over $T^* \in [0, 1]$.

C.2 The printed equations

Solved as printed, the front term of Eq. (2.5) moves the seed towards the nutrient rather than enlarging it: the factor $\mathbf{n}_{\nabla\phi} \cdot \mathbf{n}_{\nabla c}$ is positive on the face of the colony that looks towards the nutrient and negative on the other, so one face grows while the other recedes. The mean ϕ of Fig. 4 is not reached with any numerical variant tried (seed held fixed, no limiter, a second-order upwind scheme), nor with any time scale. Two further observations point the same way: in case 4.2 "high" the front advances at about 150 μm per T^*, three times the largest speed $r\,c/(k + c)$ that Table 2 allows, and in Table 4 of the paper the biofilm grows away from the nutrient face, while the printed term drives it towards it.

The consumption term gives a clean check. With the consumption $g\,\phi\,c$, proportional to the nutrient, the steady nutrient level of case 4.2 "high" is $\tanh(2)/2 = 0.482$, against 0.489 in the paper. The printed form $g\,\phi$ gives 0.139.

C.3 What reproduces the curves

The following changes, each stated as an assumption, reproduce all three curves without changing a value of Table 2:

1. growth on every face of the colony, with the front term $\|\nabla_X \phi\| \, \frac{r\,c}{k+c}\big[w + (1-w)\,|\mathbf{n}_{\nabla\phi} \cdot \mathbf{n}_{\nabla c}|\big]$ and an isotropic share $w = 0.5$ (the text of the paper says that growth perpendicular to the nutrient gradient is smaller, not zero);
2. consumption $g\,\phi\,c$;
3. a time scale s_T per simulation, applied to r, β and k_α: the paper normalises time by a reference time "chosen based on the conditions for growth at hand", which is not given.

Table C.1 lists the resulting differences. For w between 0.35 and 0.7 the worst of the three stays between 0.029 and 0.032, so the curves do not determine w precisely.

Table C.1: RMS difference between the reproduction and the curves of Klempt et al. 2024 (mean ϕ and mean c, averaged), with the changes listed in the text.

test case	time scale s_T	RMS, $w = 0.5$	RMS, $w = 0$
4.1 (Fig. 4)	1	0.013	0.022
4.2 high (Fig. 7)	5	0.021	0.038
4.2 low (Fig. 7)	3	0.030	0.070

The shapes agree in kind (Figures C.1 and C.2). In case 4.1 the colony becomes an egg drawn out towards the nutrient corner and reaches it, as in the paper. In case 4.2 the early dome and the final layer of the "low" case look like the paper's; in the "high" case the reproduction spreads to the side walls earlier, where the paper's colony rises with straight sides. In the variant without the isotropic share ($w = 0$), varying the regularisation β and the upwind scheme did not change the shapes, so the remaining differences lie in the equations or the set-up rather than in the numerics.

C.4 Mechanics and units

The reproduction has no mechanics. Two bounds show that the published curves can nevertheless be compared with it. The material is incompressible, so the current volume is α^3 times the reference volume; with $\alpha - 1 \leq 0.01$ at $T^* = 1$ the volume-weighted means of ϕ and c differ from the plain ones by at most 0.001. The mechanical term of the ϕ equation, which the paper drops, would be of order 10^4 to 10^6 per T^* after division by the viscosity $\eta_\phi = 10^{-10}$ (the paper gives no unit), against a growth term of about 500 per T^*; kept, it would stop the growth that every figure shows.

The pressure plotted in Table 3 of the paper (of order 10^{-4} MPa) is consistent with $\mu = 3.3557$ MPa, as stated in the text, and not with the 3.3557 Pa of Table 2, which would give pressures a million times smaller. This is an inference from the plot. The calculations of Chapter 4 use $E = 10$ Pa as in Table 2; their stresses scale linearly with E.

C.5 Questions to the authors and their answers

I sent these questions to the first author on 3 and 5 October 2026 and received his answers on 5 October. They settle some points and leave others open:

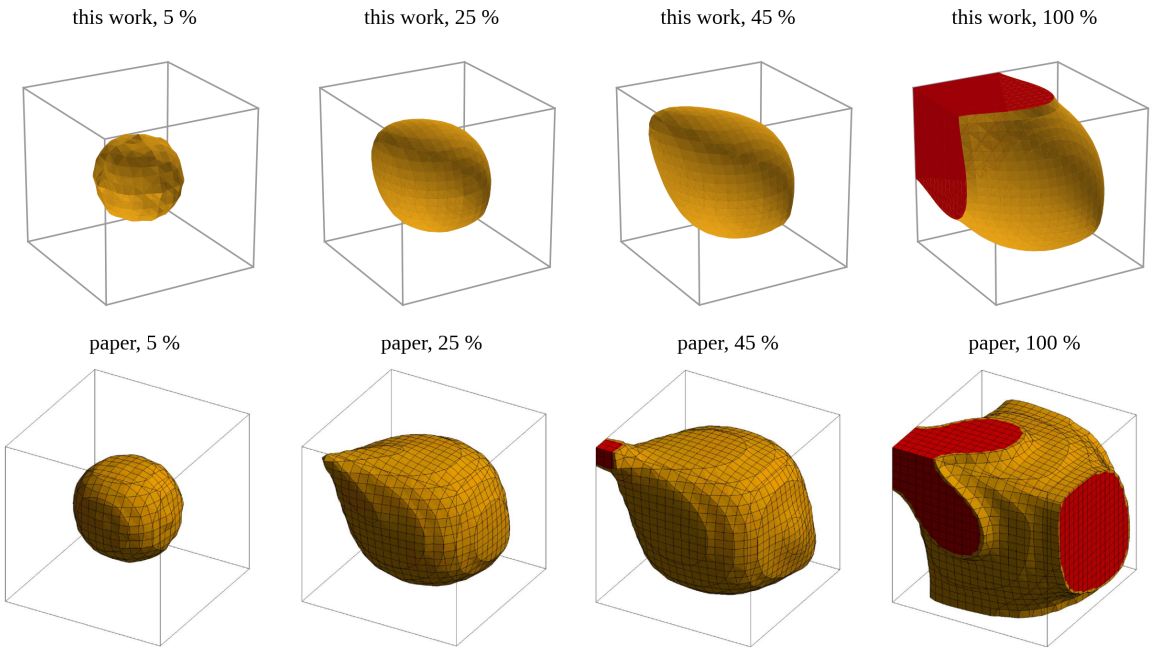


Figure C.1: Test case 4.1: isosurface $\phi = 0.8$ at 5, 25, 45 and 100 % of the simulated time. Top: reproduction with the changes listed in the text ($w = 0.5$, $s_T = 1$). Bottom: Fig. 3 of Klempt et al. 2024 (CC BY 4.0).

- *Front term.* His implementation uses Eq. (2.5) as printed: where the nutrient gradient behind the colony points towards it, the colony does not grow on that side. The isotropic share w of the reproduction is therefore my change, needed to reach the published curves with the explicit scheme used here. Whether the finite element solution of the paper reaches them without it is open.

- *Direction where the nutrient is uniform.* The implementation uses $\nabla_X c/(\|\nabla_X c\| + \varepsilon)$ with a small ε, of order 10^{-8} to 10^{-12} (F. Klempt, personal communication, October 2026).

- *Consumption.* He recalled $g\,\phi$, as printed, but was not certain. The steady nutrient level of case 4.2 “high” (0.489 in the paper) follows from $g\,\phi\,c$ and not from $g\,\phi$ (0.482 against 0.139, see above). Dr. Soleimani, who computed the published examples, confirmed on 6 October 2026 that $g\,\phi\,c$ is correct. The consumption term of Eq. 35 is therefore $g\,\phi\,c$, as in the reproduction.

- *Time.* Every simulation ran over a time of 1 ($T^* = 1$). The time scales $s_T = 5$ and 3 that the reproduction needs for case 4.2 are therefore not explained by different reference times.

- *Nutrient in case 4.1.* Only the corner, as assumed here.

- *Diffusion coefficient β.* Chosen to give sensible results; it depends on the mesh. The value used in Chapter 4, Table 2 converted to the 2 mm cube, is therefore an

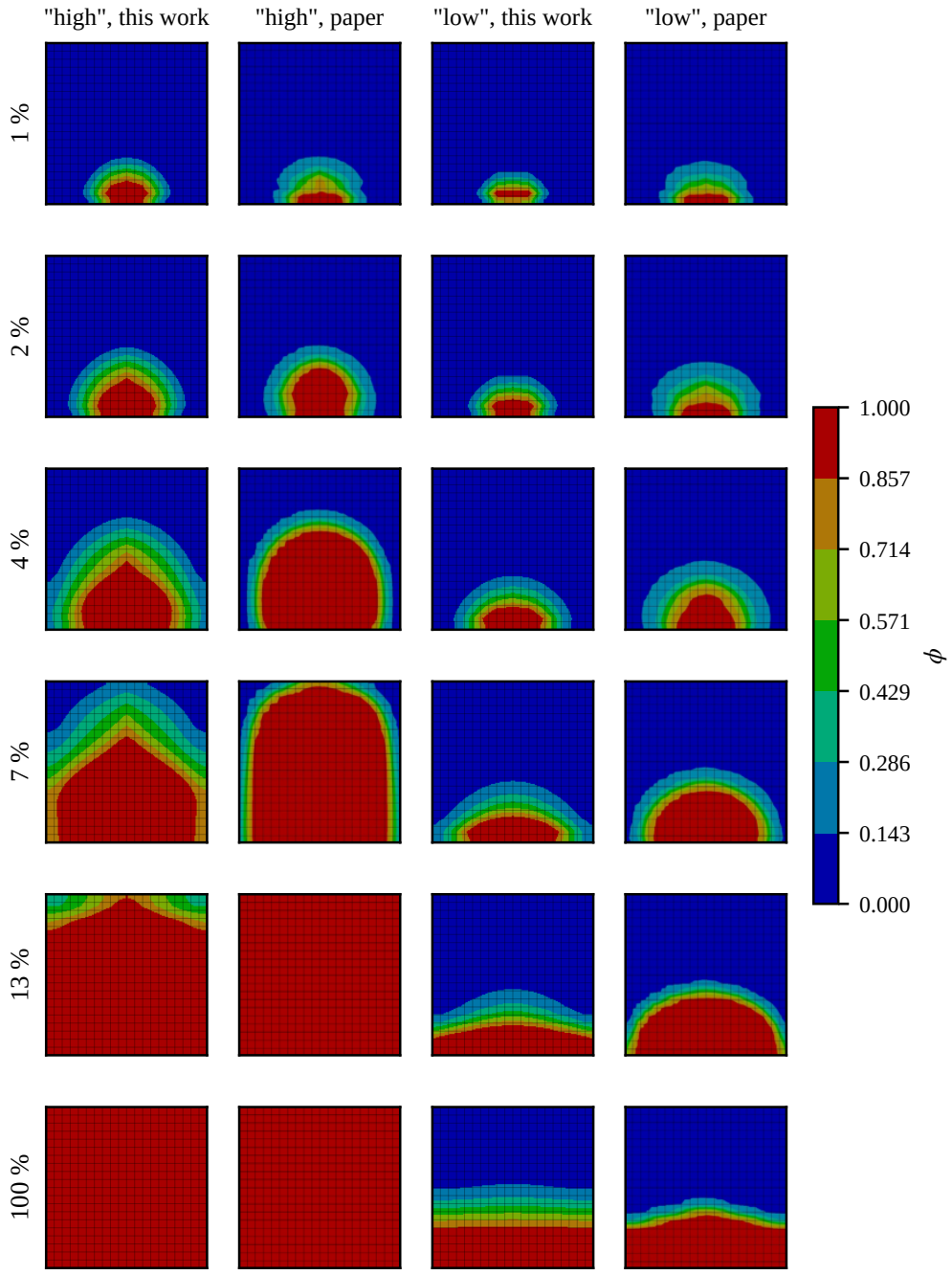


Figure C.2: Test case 4.2: vertical mid-plane, nutrient face at the bottom, reproduction next to Table 4 of Klempt et al. 2024 (CC BY 4.0), in the paper's colour bands. Reproduction with the changes listed in the text: $w = 0.5$, consumption $g\,\phi\,c$, time scales $s_T = 5$ ("high") and 3 ("low"), as in Table C.1.

assumption.

- *Stiffness.* The unit in the paper may be wrong; he suggests $E = 10$ kPa ($\mu \approx 3.3$ kPa). This differs from the inference from the plot of Table 3 above ($\mu = 3.3557$ MPa). The stresses of Chapter 4 scale linearly with E, so either value rescales them without changing any comparison.

Whether α follows Eq. 36 or the update of Table 1 (Section 2.2) remains to be asked. The first author has offered his implementation; comparing it with the reproduction is part of the continuation of this work. His implementation is ongoing work, part of a dissertation in preparation, and is not reproduced here.

D Stability of the Explicit Field Update

The element advances its field explicitly, $\phi_{n+1} = \phi_n + \Delta t\,(\beta L\phi_n + \ldots)$, where L is the Neighbored Element Method approximation of the Laplacian [57, 58]: at every Gauss point a second-order Taylor polynomial is fitted by weighted least squares through the values at the N nearest Gauss points, with the weight $w = \exp[-\frac{1}{2}(4d/\beta^*)^2]$ of the distance d, and the Laplacian is the sum of the three unmixed second derivatives of the fit. In the runs of Chapter 4 the element uses $N = 30$ and $\beta^* = 0.2$ mm. For explicit Euler the update is stable if $\beta\Delta t\,\rho \leq 2$, with ρ the spectral radius of $-L$; written with $\lambda = \beta\Delta t/h^2$ this is $\lambda \leq \lambda_{\max} = 2/(\rho h^2)$. For the three-point stencil in one dimension $\rho = 4/h^2$ and $\lambda_{\max} = 1/2$, which is the condition $h^2/(2\Delta t) \geq \beta$ of Rudolf et al. [58]; for the five-point stencil in two dimensions $\lambda_{\max} = 1/4$ and for the seven-point stencil in three dimensions $1/6$.

Lattice of the Gauss points. On a uniform hexahedral mesh of size h the $2 \times 2 \times 2$ Gauss points of SOLID185 lie, per axis, at $h(\frac{1}{2} \pm \frac{1}{2\sqrt{3}})$ inside each element, so their spacing alternates between $h/\sqrt{3} \approx 0.58h$ inside an element and $h(1 - 1/\sqrt{3}) \approx 0.42h$ across the element boundary. The eight points of an element are equivalent under the reflections of the cube, so the stencil of one of them gives the other seven by mirroring, and the operator L is periodic with the period h of the mesh with eight points per cell.

Spectral radius. I computed the stencil of the WLS fit on this lattice with the deck values and formed the 8×8 Bloch symbol $\hat{L}(\mathbf{k})$ of L over the Brillouin zone of the cell lattice; ρ is the largest modulus of its eigenvalues over $\mathbf{k}$. Three points follow from the result (Table D.1).

First, the limit of the NEM stencil differs from $1/6$ even without the weighting: with $\beta^* \to \infty$ it is 0.18 to 0.19, because the lattice is not uniform and the stencil reaches beyond the nearest neighbours.

Second, the weighting makes the limit depend on the mesh. Because β^* is a length in millimetres, the weight of a neighbour at distance d depends on d/β^*, and on a coarse mesh the fit is dominated by the few nearest points, which gives a stencil with large coefficients and a small $\lambda_{\max}$: 0.062 on 8^3 elements, 0.139 on 16^3 and 0.162 on 24^3 for one choice of 30 neighbours, 0.149 and 0.173 for the symmetric set of 32.

Third, the choice of the neighbours matters. The 30th neighbour lies in a shell of seven points at the distance h, of which the fit takes five; which five follows from the sort order inside the element and cannot be read off the deck. Over 40 random choices of the five, the limit stays at 0.062 on 8^3, where the weight of that shell is negligible, but lies between 0.122 and 0.140 on 16^3 and between 0.122 and 0.162 on 24^3 (shaded in Figure D.1). On the finite cube, with the neighbours taken in the order of a k-d tree, the limit is 0.143

on both meshes. The limit approaches the unweighted value as the mesh is refined. A direct time stepping of the stencil on a periodic block of elements grows by the factor $|1 - 2\lambda/\lambda_{\max}|$ per step, as the analysis predicts.

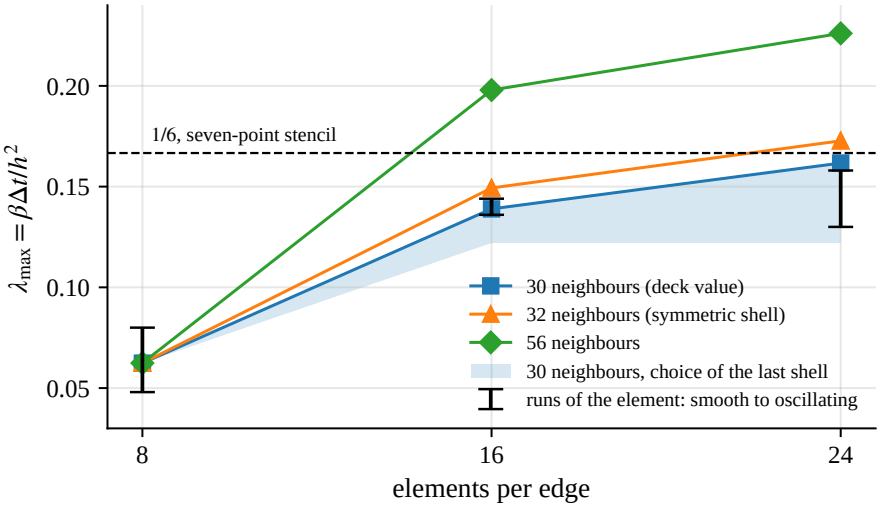


Figure D.1: Stability limit of the explicit field update from the Bloch analysis, against the number of elements per edge, for 30, 32 and 56 neighbours with $\beta^* = 0.2$ mm; shaded: the range over the choice of five of the seven equidistant points of the last shell for 30 neighbours; bars: the runs of the element to $T^* = 5$, from the largest smooth to the smallest stopped λ.

Comparison with the runs. Runs to $T^* = 5$ bracket the limit at $0.048 < \lambda_{\max} < 0.080$ on 8^3 and $0.136 < \lambda_{\max} < 0.144$ on 16^3, and $0.130 < \lambda_{\max} < 0.158$ on 24^3; all three agree with the computed values, on 16^3 and 24^3 within the band of the neighbour choice. The unstable mode is the alternation of sign between neighbouring points, which is what the failing runs show in $\alpha - 1$.

With 32 neighbours set in the deck, the run on 16^3 stays smooth to $T^* = 5$ at $\lambda = 0.144$ and stops at 0.155, around the computed 0.149. On 24^3 the runs at 0.164 and 0.180 both stay smooth to $T^* = 5$, so the limit there lies at or above 0.180, at most 4 % above the computed 0.173 and clearly above the 0.158 at which the run with 30 neighbours stops. The accuracy does not change with the number of neighbours: at $\Delta t = 0.0125$ the error of the seed mean against the finite-volume reference (Figure 4.11) is +1.58 % with 32 and +1.47 % with 30 neighbours on 16^3, and +0.98 % and +0.83 % on 24^3. The results of this work use the deck value 30.

On 8^3 the runs at $\beta = 0.1$ mm$^2/T^*$ to $T^* = 1.1$ stay smooth up to $\lambda = 0.088$ and stop at 0.096, above the computed limit of 0.062. The spectral radius of the operator on the finite 8^3 cube, where each point fits its stencil over the 30 nearest points that exist, gives the same limit, 0.063, so the boundary does not explain the difference. The initial field does: ϕ is constant in each element, and its component along the unstable mode is 8×10^{-7} of its norm. A run to $T^* = 1.1$ takes only 18 to 22 steps on 8^3, and the growth

factor per step, $|1 - 2\lambda/\lambda_{\max}|$, is 1.5 to 2.1 between $\lambda = 0.08$ and 0.096, so the mode reaches the size of ϕ only near the end of the run. Marching the same seed problem with the finite-cube operator reproduces this. I therefore predicted that longer runs show the limit, and two runs to $T^* = 5$ confirm it: at $\lambda = 0.080$ the elements collapse at $T^* = 1.65$, at $\lambda = 0.048$ the run stays smooth to the end. The limit on 8^3 lies between 0.048 and 0.080, in agreement with 0.062. A run judged stable must therefore be long enough for the unstable mode to grow. The same holds on 24^3, where the run at $\lambda = 0.158$ is smooth to $T^* = 1.1$ and stops before $T^* = 5$; marching the seed problem with the finite-cube operator gives the same picture, with $\alpha - 1$ turning negative at $T^* = 1.04$ for $\lambda = 0.158$ and at $T^* = 1.56$ for 0.150.

Table D.1: Stability limit $\lambda_{\max} = \beta\Delta t/h^2$ of the explicit update with the NEM stencil on the lattice of the Gauss points, from the Bloch analysis; $\beta^* = 0.2$ mm as in the element, and the unweighted limit; second row: the range over the choice of the neighbours in the last shell. Last row: the largest smooth and the smallest stopped λ of the runs to $T^* = 5$.

	8^3	16^3	24^3	$\beta^* \to \infty$
$N = 30$, one choice	0.062	0.139	0.162	0.185
$N = 30$, range over the choice	0.062	0.122–0.140	0.122–0.162	
$N = 32$ neighbours	0.062	0.149	0.173	0.180
runs to $T^* = 5$: smooth / stopped	0.048 / 0.080	0.136 / 0.144	0.130 / 0.158	
runs to $T^* = 5$, $N = 32$		0.144 / 0.155	0.180 / –	

Selbstständigkeitserklärung

Hiermit versichere ich, dass ich die vorliegende Masterarbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Dies schließt die Verwendung von elektronischen Medien sowie text- oder anderen inhaltsgenerierenden IT-Werkzeugen wie ChatGPT ein. Alle Stellen der Arbeit, die wörtlich oder sinngemäß aus anderen Quellen übernommen wurden, sind als solche kenntlich gemacht. Die Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

Use of AI tools. I used an AI assistant (Claude, Anthropic) for programming support and for drafting and language editing of parts of the text. All results, the choice of methods and the interpretation are my own; I checked every passage and take full responsibility for the content.

Hannover, _____ Unterschrift: _____